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Offline Reinforcement Learning for Human-Guided Human-Machine Interaction with Private Information

(Authors' names blinded for peer review)

Motivated by the human-machine interaction such as recommending videos for improving customer engagement, we study human-guided human-machine interaction for decision making with private information. We model this interaction as a two-player turn-based game, where one player (Bob, a human) guides the other player (Alice, a machine) towards a common goal. Specifically, we focus on offline reinforcement learning (RL) in this game, where the goal is to find a policy pair for Alice and Bob that maximizes their expected total rewards based on an offline dataset collected a priori. The offline setting presents two challenges: (i) We cannot collect Bob's private information, leading to a confounding bias when using standard RL methods, and (ii) a distributional mismatch between the behavior policy used to collect data and the desired optimal policy we aim to learn. To tackle the confounding bias, we treat Bob's previous action as an instrumental variable for Alice's current decision making so as to adjust for the unmeasured confounding. We establish a novel identification result and propose a new off-policy evaluation (OPE) method for evaluating policy pairs in this two-player turn-based game. To tackle the distributional mismatch, we leverage the idea of pessimism and use our OPE method to develop an off-policy policy learning algorithm for finding a desirable policy pair for both Alice and Bob. Moreover, we prove that under some technical assumptions, the policy pair obtained through our method converges to the optimal one at a satisfactory rate. Finally, we conduct a simulation study to demonstrate the performance of the proposed method.

Key words: Two-player turn-based game with private information, instrumental variables, offline reinforcement learning

1. Introduction

Data driven decision making plays a vital role for improving business decision making. In recent years, there is a surging interest in studying how human feedback can improve AI's decision making, which is essential for improving the use of machine learning for solving real-world problems (Hoc 2000, Gorecky et al. 2014). Motivated by this, in this paper, we study a two-player turn-based cooperative game as an example of human-machine interaction with a particular focus on the human guided setting. Leveraging the pre-collected offline data consisting of decision making

process between human and AI, our goal is to develop a data-driven method to learn an optimal policy when some private information of one particular player (i.e., human) cannot be accessed during the training procedure.

Our primary motivation is to develop an efficient recommendation system using observational data. Take a sequential video recommendation as an example. At each decision point, the AI recommendation system suggests a video based on a customer's characteristics and its own movie catalog or internal popularity scores of all movies. The customer then chooses whether to play one of the recommended videos based on her current preference and mood. After a selection is made, an immediate reward, such as watch time or engagement, is recorded. This interaction between the AI system and the customer continues until the customer leaves. Our objective is to create an algorithm that can effectively leverage offline data from these interactions to enhance long-term outcomes, such as total engagement time. Note that for a growth company, total engagement can be interpreted as the surrogate for long-term revenue and profit. Such an important data-driven business decision-making problem can be framed as a two-player turn-based cooperative game, assuming the customer's objective is aligned with the recommendation platform's. However, in the observational data, the AI system may not be able to access the customer's private information such as the mood, and similarly, the customer cannot make a decision based on the system's private information such as the movie catalog and internal popularity scores of all movies. The absence of private customer information presents a fundamental challenge in training the recommendation system. Therefore, it is crucial to study this two-player, turn-based cooperative game in scenarios where private information is missing. Our work is further motivated by the strategic response of humans to algorithms. In various business activities, algorithms play a key role in decision-making processes, such as auctions, pricing, and trading. For instance, hedge funds rely on machines to collect market data and provide real-time trading suggestions, which traders then act upon. The outcomes of these decisions feed back into the machine, enhancing its learning capability. Hence, it is vital to improve the algorithm's ability to discover significant patterns from big data, leading to better decision-making. This aligns with the two-player, turn-based game concept mentioned earlier.

1.1. Brief Problem Statement and Challenges

In this paper, we formulate a two-player game as a reinforcement learning (RL) problem, where human as a complex agent may also have private information that we fail to deliver to the machine. The forbidden use of the private information is the main obstacle for a successful human-machine interaction. Specifically, there are two main sources that the private information affects the game structure: (i) actions taken by the players; (ii) rewards collected and the next states received after

taking actions. Such a game with private information has been widely considered in practice: See examples in economics (Laffont and Martimort 2009), social science (Sabater and Sierra 2005), robotics/control (Farinelli et al. 2004, Yan et al. 2013, Mertikopoulos and Zhou 2019), cognitive science (Sun et al. 2006), and human-machine negotiation (Lewis et al. 2017), etc.

We model the aforementioned scenario as a two-player turn-based game, where we call two players as Alice (representing a machine) and Bob (representing a human). In such a game, Alice first takes an action, which may potentially depend on the current state, her private information, and Bob's previous action. Then Alice collects her reward and reveals her public information to Bob and transits to the next state. After observing the new state, Bob takes an action, which may potentially depend on the new current state, his private information, and Alice's previous action. Similarly, Bob collects his reward and reveals his public information to Alice and transits to the next state. The goal for Alice and Bob is then to maximize the sum of their expected total rewards. In other words, we (as Alice) aim to find a policy pair for Alice and Bob to maximize their rewards together. We remark that such a turn-based game exactly characterizes a typical human-guided human-machine interaction, where the Bob (as a human) takes an action to provide feedback to Alice's previous action.

In this work, we aim to use offline RL techniques for solving such a two-player turn-based game from the standpoint of Alice, where the goal is to learn the optimal policy pair of the principal based on an offline dataset collected a priori. The challenge in the offline setting is that Alice, as a machine, though can access her own private information, cannot access the private information of Bob, as a human. As a result, Bob's private information is the so-called unobserved confounder for understanding Bob's action and also Alice's in this two-player turn-based game. If the existence of unmeasured confounding is not carefully factored in the design of offline learning algorithms, directly applying standard RL methods could potentially lead to biased and inconsistent estimation of value functions and, consequently learn sub-optimal policies. Meanwhile, as our offline data follow a fixed behavior policy and we cannot interact with the environment to further collect more data, it may result in a distributional mismatch between the behavior policy and the optimal one. This is the common challenge behind offline RL (Levine et al. 2020).

1.2. Main Results and Contributions

To address the above two challenges, we *theoretically* study the offline RL problem in such a two-player turn-based game using the instrumental variable (IV) method (Angrist and Imbens 1995), which has been widely used in the literature of causal inference (e.g., Pearl (2009), Hernán and Robins (2010)) to identify the causal effect of a treatment under unmeasured confounding. In this work, we innovatively treat Bob's previous action as an instrumental variable (IV) for Alice's

current decision making so as to adjust the unmeasured confounding. Such an instrumental variable arises naturally since (i) given all the information of the current stage, Bob's previous action does not depend on his current private information; (ii) Bob's previous action will affect Alice's current action. Condition (i) is usually referred to as IV independence and Condition (ii) is referred to as IV relevance, which are two commonly required conditions for a valid IV. Moreover, a valid IV additionally requires that there is no direct effect of Bob's previous action on the reward obtained by Alice at the current decision-making unless through Alice's action itself. This is called IV exclusion restriction in the causal inference literature (Wang and Tchetgen Tchetgen 2018). However, such a requirement could be easily violated in our two-player turn-based game. For example, a customer's selection (i.e., Bob's action) of the suggested videos would directly affect the customer satisfaction (Alice's reward). Motivated by these, we treat Bob's previous action as *an invalid IV*, where the IV exclusion restriction fails to hold. Relying on the orthogonal conditions on the reward and action's model structure to some extent, we establish a novel identification result using offline data for evaluating policies and learning an optimal one in our two-player turn-based game with private information.

Our new identification results lead to an estimation problem of solving a system of nonlinear integral equations for evaluating policies in two-player turn-based game. We propose to use a sieve minimum distance estimator, which is shown to achieve the known minimax optimal rate for the non-parametric mean instrumental variable regression (Chen and Pouzo 2012). In terms of policy learning, to mitigate the distributional mismatch, instead of directly plugging in the sieve minimum distance estimator, we propose to use conservative estimators as proxies of the action-value functions and maximize such conservative action-value functions to learn an optimal policy pair. Such a method is also termed pessimism in related literature (Jin et al. 2021, Lu et al. 2022, Fu et al. 2022). Theoretically, we provide a finite sample regret analysis of our offline learning algorithm, showing that the learned policy pair converges to the optimal one asymptotically as the sample size of the offline data increases under only the partial coverage assumption, which does not require our offline data to explore the state-action space fully.

Main Contributions. Our contribution is threefold. First, to the best of our knowledge, we are among the first in the literature to study offline policy learning in a two-player turn-based game under unmeasured confounding through the lens of causal inference. This serves as a first step to bridge the gap between causal inference and the cooperative game.

Second, to address the unmeasured confounding (i.e., private information of Bob) during the game, we leverage the structure of the turn-based game and propose to using the other player's action in the previous step as a potential IV for adjusting the confounding bias resulted from

the missing private information of Bob. Since we allow the previous action to have a direct effect on the current reward, standard IV approaches in the literature of causal inference cannot be applied. Motivated by the literature on the causal inference with an invalid IV (e.g., [Lewbel 2012, 2018](#), [Tchetgen et al. 2021](#)), we first study the standard causal inference setting and generalize the existing identification results of invalid IVs, which serves as a foundation for solving our two-player turn-based game. Specifically, in the causal inference under unmeasured confounding, where an IV can have a direct effect on the outcome, the identification is often restricted to the marginal causal effects of the treatment and IV. Nevertheless, in our problem setting, at each decision-point, there may exist an interaction effect between Alice's and Bob's actions on the reward, making existing results hard to apply. Motivated by this, we consider a more general setting where there is an interaction effect between the treatment and IV on the outcome, and establish a novel non-parametric identification result for identifying both marginal and interaction effects of the action and IV. Our result relies on the orthogonal restriction between the effect of unmeasured confounders on the treatment and that on the outcome up to the second-order information, which could be of independent interest. We then adapt these novel identification results in our two-player turn-based game setting and propose a sieve minimum distance estimator for off-policy evaluation, which is shown to achieve the known minimax optimal rate for the standard non-parametric instrumental variable regression.

Last but not least, by our off-policy evaluation procedure and by leveraging the recent idea of using pessimism in the policy learning, we propose a policy pair estimator for both human and machine, which is shown to converge to the optimal policy pair under mild assumptions such as the partial coverage of our offline data. The proposed algorithm addresses the issue of the distributional mismatch. To the best of our knowledge, this is the first established regret result for offline policy learning in a two-player turn-based game under unmeasured confounding.

Related Work. Our work is related to the line of works that study RL under the presence of unobserved confounders. [Chen and Zhang \(2021\)](#) utilize a time-varying instrumental variable to establish a novel estimation framework for dynamic treatment regime. Also, to incorporate the observational data into the finite-horizon RL, [Wang et al. \(2021\)](#) propose an online method termed deconfounded optimistic value iteration. To ensure the identifiability through offline data, they impose a backdoor criterion ([Pearl 2009](#), [Peters et al. 2017](#)) when confounders are partially observed, and a frontdoor criterion when confounders are unobserved. Our work is also closely related to [Liao et al. \(2021\)](#) and [Fu et al. \(2022\)](#). Specifically, [Liao et al. \(2021\)](#) propose an IV-aided value iteration algorithm to study confounded Markov decision processes (MDPs) in the offline setting. It is worth mentioning that they only consider a parametric model, where the transition

dynamics is a linear function of some known feature map. To ensure identifiability, they assume that the unobserved confounders are Gaussian noise. In contrast, we consider a nonparametric saturated model without such restrictive assumptions, which is thus more general and applicable. On the other hand, [Fu et al. \(2022\)](#) establish point identification results with the help of valid IVs in an infinite-horizon confounded MDP. As discussed before, standard IV approaches cannot solve our problem as there exists a direct effect of IV on the outcome/reward. Besides the aforementioned literature, there is a growing body of literature using causal inference tools to study off-policy evaluation in confounded MDPs, see [Kallus and Zhou \(2020\)](#), [Namkoong et al. \(2020\)](#), [Bennett et al. \(2021\)](#), [Shi et al. \(2021\)](#), [Lu et al. \(2022\)](#), [Miao et al. \(2022\)](#) and the references therein. Also we highlight that the aforementioned works all focus on the single-agent setting, while our work focus on the two-player setting.

Our work is also related to the line of works that study two-player games. The majority of this line of works focus on the two-player zero-sum stochastic games. See [Littman \(2001\)](#), [Pérolat et al. \(2015, 2018\)](#), [Srinivasan et al. \(2018\)](#), [Guo et al. \(2021\)](#), [Lagoudakis and Parr \(2012\)](#), [Pérolat et al. \(2016a,b\)](#), [Luo et al. \(2019\)](#), [Jin et al. \(2020\)](#), [Cai et al. \(2020\)](#), [Zhong et al. \(2021\)](#), [Hambly et al. \(2021\)](#) and the reference therein. We remark that all of these works consider the settings where the agents have complete information, and thus do not face the issue of unobserved confounders as in our work.

The rest of our paper is organized as follows. In §2, we describe the offline reinforcement learning problem for the two-player turn-based game. In §3, we study a causal inference problem under unmeasured confounding, where an invalid IV is used to identify the causal effects of both treatment/action and IV on the outcome. In §4, we adapt the identification result developed in §3 to a single-stage two-player game and propose an offline data-driven algorithm to compute an optimal policy pair for Alice and Bob. We then generalize our method for solving a multi-stage game in §5. In §6, we present a numerical study on our proposed method. Lastly, we conclude the paper in §7. All technical proofs can be found in the Appendix.

Notations. We denote by $[H] = \{1, 2, \dots, H\}$ and $[H]_{1/2} = \{3/2, 5/2, \dots, H + 1/2\}$. We denote by $\sigma_{\min}(\Sigma)$ the minimum singular value for any matrix Σ .

2. Problem Formulation

We consider a data-driven two-player turn-based game with private information, where two players called Alice and Bob play the game sequentially based on the observed states and rewards and their private information. Let h be some positive integer. We assume that starting from $h = 1$, at the h -th step of the game, Alice takes an action A_h following a policy π_h^A , which possibly depends on the current state S_h , her private information U_h , and Bob's previous action $B_{h-1/2}$. After taking

such an action, Alice receives a reward R_h^A and the (next) state $S_{h+1/2}$ is revealed to both players. In the meanwhile, Alice and Bob receive their new private information $U_{h+1/2}$ and $V_{h+1/2}$, respectively. Then at the $(h + 1/2)$ -th step, Bob takes an action $B_{h+1/2}$ following a policy $\pi_{h+1/2}^B$ which possibly depends on the current state $S_{h+1/2}$, his private information $V_{h+1/2}$, and Alice's previous action A_h . After taking such an action, Bob receives a reward $R_{h+1/2}^B$ and the state S_{h+1} is revealed to both players. In the meanwhile, Alice and Bob receive their new private information U_{h+1} and V_{h+1} , respectively. We consider a MDP with a finite horizon H (i.e., $h = 1, \dots, H$) to model the trajectory $\{S_h, U_h, V_h, A_h, R_h^A, S_{h+1/2}, U_{h+1/2}, V_{h+1/2}, B_{h+1/2}, R_{h+1/2}^B\}_{h \in [H]}$, where the MDP terminates once the state S_{H+1} is revealed. For convenience, we assume that Bob reveals an action $B_{1/2}$ following $\pi_{1/2}^B$ before the state S_1 is revealed. Moreover, for the simplicity of presentation, we only consider binary actions, i.e., $\mathcal{A} = \mathcal{B} = \{0, 1\}$, where $\mathcal{A}$ and $\mathcal{B}$ are action spaces for Alice and Bob, respectively. Further, we denote by $\mathcal{S}$, $\mathcal{U}$ and $\mathcal{V}$ the state space and private information spaces, respectively. **We remark that to extend our current proposal such as identification results proposed below to the multiple action setting, one only needs to use the one-hot encoding approach. In our recommendation system example, an entertainment service provider may have thousands of movie to suggest. To handle this, one can first learn a low-dimensional embedding of the original action space $\mathcal{A}$ and then imposed our model assumptions on the embedding space and apply our method for learning an effective recommendation policy (e.g., Chandak et al. 2019).**

In our video recommendation example, Alice represents the recommendation system while Bob is the customer. In each turn, Alice selects one out of two different genres of video A_h to recommend to Bob, based on her available movie catalog or internal popularity scores of all movies U_h which is hidden, Bob's previous reaction $B_{h-1/2}$, and current state S_h , which includes observed customer's features such as demographic information, subscription data and history of watching movies. Bob then decides whether to play the recommended video $B_{h+1/2}$, based on the genre of the video A_h , his current mood $V_{h+1/2}$, and current state $S_{h+1/2}$. This implies Bob's (behavior) policy space can depend on his private information, such as his mood. Note that the available movie catalog or internal popularity scores of all movies U_h and Bob's current mood $V_{h+1/2}$ are not revealed to Bob and Alice, respectively. Finally, both Alice's and Bob's rewards (i.e., R_h^A and $R_{h+1/2}^B$) can be measured by the customer engagement time in such a movie. Our goal is to learn a policy based on the data collected by the recommendation system to maximize users' experience, as reflected in engagement time. It is remarked that for a growth company, total engagement of customers is a surrogate for long-term revenue and profit, thus an objective of the company to optimize. In this example, one can actually set R_h^A as zero and the cumulative reward of $R_{h+1/2}^B$ as the common objective for both Alice and Bob. However, to keep its generality, we still consider one can observed R_h^A after Alice takes her action.

In this paper, we consider a fully cooperative game of Alice and Bob. Specifically, if all information is available at each time-step, then the goal of the two-player turn-based game considered here is to find a policy pair that maximizes the sum of Alice's and Bob's expected total rewards, as we will formally introduce in §2.1. As discussed in the §1, we consider the scenario such as training a recommendation system with the help of customers' feedback, where human interacts with the machine to achieve a common goal. In such an application, Bob plays a role of a human, while Alice plays a role of a machine. Our goal is to leverage the offline data collected from the past experience to help Alice and Bob learn a good policy pair that maximizes the total expected rewards. The challenge behind this task is the existence of the private information owned by Bob that hinders Alice from learning a desirable policy. Without accessing such private information, the offline data may not ensure the identification of the policy value and thus cannot guarantee to learn a good policy. In this paper, we develop a novel IV approach to address such a challenge. Finally, we assume that the private information V_h and $V_{h+1/2}$ for $h \in [H]$ are memoryless so that given other observed variables at the same decision point, private information is independent of the past variables. Memoryless assumption has been widely used in the literature (e.g., [Kallus and Zhou 2018](#), [Shi et al. 2022](#), [Fu et al. 2022](#)). In this case, the observed sequence by Alice can also be modelled by MDP. In the following subsections, we introduce the mathematical framework in more details.

2.1. Value Functions

We use the policy value to evaluate the performance of any policies. Specifically, for any policy pair $\pi = (\pi^A, \pi^B)$, and the timestamp $h \in [H]$, we denote by $Q_h^{A,\pi}(S_h, A_h, B_{h-1/2}, U_h, V_h)$ and $Q_{h+1/2}^{A,\pi}(S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2})$ the action-value functions of Alice for the expected cumulative rewards starting at times h and $h + 1/2$, i.e.,

$$Q_h^{A,\pi}(S_h, A_h, B_{h-1/2}, U_h, V_h) = \mathbb{E}_\pi \left[\sum_{j=h}^H R_j^A \middle| S_h, A_h, B_{h-1/2}, U_h, V_h \right],$$

$$Q_{h+1/2}^{A,\pi}(S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2}) = \mathbb{E}_\pi \left[\sum_{j=h+1}^H R_j^A \middle| S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2} \right],$$

where the expectation $\mathbb{E}_\pi[\cdot]$ are taken following the policy pair $\pi = (\pi^A, \pi^B)$. Similarly, we define the action-value functions of Bob at times h and $h + 1/2$ as $Q_h^{B,\pi}(S_h, A_h, B_{h-1/2}, U_h, V_h)$ and $Q_{h+1/2}^{B,\pi}(S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2})$ respectively. Finally, we denote by

$$J^A(\pi^A, \pi^B) = \mathbb{E}_\pi [Q_1^{A,\pi}(S_1, A_1, B_{1/2}, U_1, V_1)], \quad J^B(\pi^A, \pi^B) = \mathbb{E}_\pi [Q_1^{B,\pi}(S_1, A_1, B_{1/2}, U_1, V_1)],$$

the expected total rewards (i.e., policy value) for Alice and Bob, respectively.

In a fully cooperative turn-based game of Alice and Bob, we aim to solve the following optimization problem together,

$$\max_{\pi^A \in \Pi^A, \pi^B \in \Pi^B} J(\pi^A, \pi^B), \quad \text{where } J(\pi^A, \pi^B) = J^A(\pi^A, \pi^B) + J^B(\pi^A, \pi^B), \quad (1)$$

where Π^A and Π^B are some pre-specified policy classes. Let $\Pi = (\pi^A, \pi^B)$. Since we are on the side of Alice, we consider Alice's policy space Π^A consisting of policies $\pi^A = \{\pi_h^A\}_{h \in [H]}$ that depends on the current state S_h , Bob's previous action $B_{h-1/2}$, and Alice's private information U_h ; while the policy space of Bob Π^B consists of policies $\pi^B = \{\pi_{h+1/2}^B\}_{h \in [H]}$ that only depends on the current state $S_{h+1/2}$ and Alice's previous action A_h . Recall that we only construct or execute policies using the knowledge of Alice without accessing Bob's private information. See Figure 1 for a graphical illustration of the two-player turn-based game described in §2, where we aim to solve in (1). In some scenarios, Bob and Alice may not have a common utility function to maximize. For example, if the company is a value company, then the objective of the company includes maximizing the revenue/profit, which is not aligned with the customer's objective. In this case, one can modify the optimization problem (1) as a Stackelberg game, i.e., $\max_{\pi^A \in \Pi^A} J^A(\pi^A, \tilde{\pi}^B)$, where $\tilde{\pi}^B \in \arg \max_{\pi^B \in \Pi^B} J^B(\pi^A, \pi^B)$. We believe our proposed identification and estimation procedure below can still be used in this scenario because they are used to estimate the value of some fixed policies or perform off-policy evaluation using historic data. Therefore it does not related to what objective function to optimize. However, the optimization problem is substantially different and indeed more difficult than ours. One possible way is to generalize the algorithm proposed by (Cheng et al. 2022), which needs further investigation.

2.2. Data Collection Process

We aim to leverage the offline data collected from the previous experience to solve Problem (1). In this subsection, we describe how the offline data are collected in a two-player turn-based game as follows. For any $h \in [H]$, at the h -th step, Alice observes the state S_h and her private information U_h , and then takes an action A_h following a behavior policy b_h^A and receives a reward R_h^A . At the $(h + 1/2)$ -th step, Bob observes the state $S_{h+1/2}$ and his hidden state $V_{h+1/2}$, and then takes an action $B_{h+1/2}$ following a behavior policy $b_{h+1/2}^B$ and receives and reveals a reward $R_{h+1/2}^B$. The game terminates once the state S_{H+1} is revealed to both players. In this way, we can collect data as the form of $\mathcal{D} = \{S_h, U_h, A_h, R_h^A, S_{h+1/2}, B_{h+1/2}, R_{h+1/2}^B\}_{h \in [H]} \cup \{B_{1/2}, S_{H+1}\}$ from Alice's position. We assume that our offline data consist of n independent and identically distributed copies/trajectories of the form in $\mathcal{D}$. For i -th trajectory, we observe $\{S_h^i, U_h^i, A_h^i, R_h^{A,i}, S_{h+1/2}^i, B_{h+1/2}^i, R_{h+1/2}^{B,i}\}_{h \in [H]} \cup \{B_{1/2}^i, S_{H+1}^i\}$, which represents one example of previous decision process. For the video recommendation, the i -th trajectory is referred to as the history of video recommendation for one particular

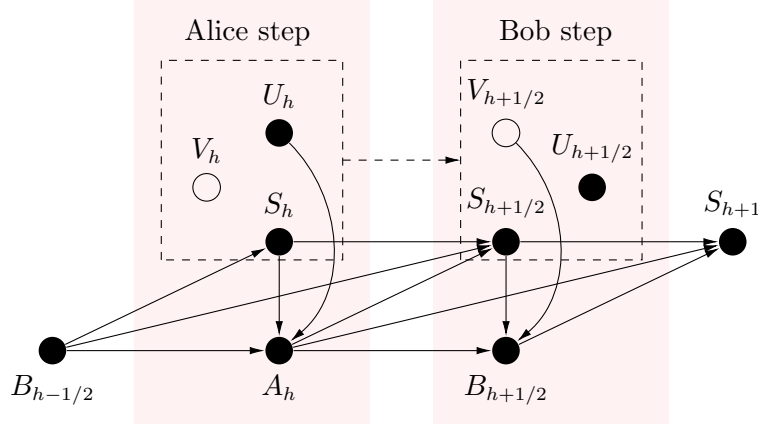


Figure 1 A graphical illustration of the offline data generated by a two-player turn-based game described in §2 satisfying Assumption 1. Here $\{U_h, V_h\}$ are private information, which satisfy the independence assumption, i.e.,

$$B_{h-1/2} \perp\!\!\!\perp V_h \mid (U_h, S_h) \text{ and } A_h \perp\!\!\!\perp V_{h+1/2} \mid (U_{h+1/2}, S_{h+1/2}).$$

Also the trajectory $\{S_h, U_h, V_h, A_h, S_{h+1/2}, U_{h+1/2}, V_{h+1/2}, B_{h+1/2}\}_{h \in [H]}$ satisfies the Markov property. In the meanwhile, the dependence of $(S_{h+1/2}, U_{h+1/2}, V_{h+1/2})$ on (S_h, U_h, V_h) is characterized by the dashed arrow in the figure for the simplicity of the presentation.

customer generated by the existing recommendation algorithm and $B_{h+1/2}^i$ reflects the customer's choice across different decision points. Here for simplicity we assume that all trajectories have the same length of horizon. However, this is not needed for implementing our method introduced below. Our goal is to use such offline data to find an optimal pair of policies that maximizes (1).

We consider the following model for such a two-player turn-based game. Specifically, we impose the structure of the action and reward for each player and the structure of state transitions at the h -th step as follows.

Alice's model. At the h -th step, the *saturated* model for Alice takes the following form,

$$\begin{aligned} \mathbb{E}[A_h \mid S_h, B_{h-1/2}, U_h, V_h] &= \alpha_z^A(U_h, V_h, S_h) B_{h-1/2} + \alpha_{u,s}^A(U_h, V_h, S_h), \\ \mathbb{E}[R_h^A \mid S_h, A_h, B_{h-1/2}, U_h, V_h] &= \beta_a^A(U_h, V_h, S_h) A_h + \beta_z^A(U_h, V_h, S_h) B_{h-1/2} \\ &\quad + \beta_{az}^A(U_h, V_h, S_h) A_h B_{h-1/2} + \beta_{u,s}^A(U_h, V_h, S_h). \end{aligned} \quad (2)$$

Here in the structural formulation of Alice's action, the function $\alpha_z^A(U_h, V_h, S_h)$ measures the impact of Bob's previous action $B_{h-1/2}$ on Alice's policy, while $\alpha_{u,s}^A(U_h, V_h, S_h)$ measures other effects from the state and all related private information. We remark that the functions α_z^A and $\alpha_{u,s}^A$ are determined by Alice's behavior policy b_h^A . In the meanwhile, in the structural formulation of Alice's reward, $\beta_a^A(U_h, V_h, S_h)$ measures the impact completely from Alice's action, $\beta_z^A(U_h, V_h, S_h)$ measures the impact completely from Bob's previous action, $\beta_{az}^A(U_h, V_h, S_h)$ measures the impact from the interaction of Bob's previous and Alice's actions, while the last term $\beta_{u,s}^A(U_h, V_h, S_h)$ measures the random effect from the state and all related private information. In our applications such as

training a video recommendation system, it is important to consider the interaction effect between Alice's action and Bob's action. For example, a customer's overall satisfaction is clearly affected by whether she likes the recommended video and plays it or not, which can be quantified by the interaction effect to some extent. We remark that because the action spaces are binary, all models can be written in the form of Model (2). Thus our Model (2) is *saturated* and *general*.

State Transition from h to $h + 1/2$. We assume that

$$\begin{aligned} \mathbb{E} [f(S_{h+1/2}, U_{h+1/2}, V_{h+1/2}) | S_h, A_h, B_{h-1/2}, U_h, V_h] \\ = \theta_h^f(U_h, V_h, S_h)A_h + \gamma_h^f(U_h, V_h, S_h)B_{h-1/2} + \omega_h^f(U_h, V_h, S_h)A_hB_{h-1/2} + \iota_h^f(U_h, V_h, S_h) \end{aligned} \quad (3)$$

for any function $f: \mathcal{S} \times \mathcal{U} \times \mathcal{V} \rightarrow \mathbb{R}$. In such a formulation, $\theta_h^f(U_h, V_h, S_h)$ measures the impact on the future state-transition completely from Alice's action, $\gamma_h^f(U_h, V_h, S_h)$ measures the impact completely from Bob's previous action, $\omega_h^f(U_h, V_h, S_h)$ measures the impact from the interaction of Bob's previous and Alice's actions, while the last term $\iota_h^f(U_h, V_h, S_h)$ measures the random effect from the state and all related private information. Again we emphasize that the state-transition Model (3) is *saturated* as our action spaces are all binary. Similar as Alice's model, we consider the following Bob's model.

Bob's model. At the $(h + 1/2)$ -th step, the model for Bob takes the following form,

$$\begin{aligned} \mathbb{E} [B_{h+1/2} | S_{h+1/2}, A_h, U_{h+1/2}, V_{h+1/2}] \\ = \alpha_z^B(U_{h+1/2}, V_{h+1/2}, S_{h+1/2})A_h + \alpha_{u,s}^B(U_{h+1/2}, V_{h+1/2}, S_{h+1/2}), \\ \mathbb{E} [R_{h+1/2}^B | S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2}] \\ = \beta_a^B(U_{h+1/2}, V_{h+1/2}, S_{h+1/2})B_{h+1/2} + \beta_z^B(U_{h+1/2}, V_{h+1/2}, S_{h+1/2})A_h \\ + \beta_{az}^B(U_{h+1/2}, V_{h+1/2}, S_{h+1/2})A_hB_{h+1/2} + \beta_{u,s}^B(U_{h+1/2}, V_{h+1/2}, S_{h+1/2}). \end{aligned} \quad (4)$$

The interpretation for Bob's reward and action model is the same as Alice's and thus is omitted. We remark that the functions α_z^B and $\alpha_{u,s}^B$ are determined by Bob's adopted policy $b_{h+1/2}^B$. Similarly, Bob's transition is modelled as follows.

Transition from $h + 1/2$ to $h + 1$. We have

$$\begin{aligned} \mathbb{E} [f(S_{h+1}, U_{h+1}, V_{h+1}) | S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2}] \\ = \theta_{h+1/2}^f(U_{h+1/2}, V_{h+1/2}, S_{h+1/2})A_h + \gamma_{h+1/2}^f(U_{h+1/2}, V_{h+1/2}, S_{h+1/2})B_{h+1/2} \\ + \omega_{h+1/2}^f(U_{h+1/2}, V_{h+1/2}, S_{h+1/2})A_hB_{h+1/2} + \iota_{h+1/2}^f(U_{h+1/2}, V_{h+1/2}, S_{h+1/2}) \end{aligned} \quad (5)$$

for any function $f: \mathcal{S} \times \mathcal{U} \times \mathcal{V} \rightarrow \mathbb{R}$.

We want to remark again that because the action spaces are binary, all models can be written in the form of Model (2)-(5). Therefore, (2)-(5) are not model assumptions and our model defined

above is saturated. In Equations (3) and (5), we use any function f to characterize the transition models. One can view such a function f as a summary/utility function of the future transition. By letting f be some indicator function, one can recover the transition kernel function. This implies that using any function f in Equations (3) and (5) are equivalent to specifying the transition models.

The challenge of this two-player turn-based game is that as Alice, she cannot access the private information of Bob, which will prevent us from using the offline data to identify the value function. In this case, standard policy learning methods will in general induce the so-called confounding bias. To address this challenge, we treat Bob's action $B_{h-1/2}$ at $(h-1/2)$ -th step as an IV for Alice h -th step's decision making in order to remove the effect of unobserved confounding caused by V_h . When considering the decision-making for Bob from Alice's side, we will use Alice h -step's action as an IV for Bob's $(h+1/2)$ -th step's decision. In particular, we impose the following assumptions for the collected offline data.

ASSUMPTION 1 (Identification). *For any $h \in [H]$, the following statements hold for the offline data generated by the behavior policy $b = (b^A, b^B)$.*

- (a) $A_h \not\perp\!\!\!\perp B_{h-1/2} \mid (U_h, S_h)$.
- (b) $B_{h-1/2} \perp\!\!\!\perp V_h \mid (U_h, S_h)$.
- (c) $B_{h+1/2} \not\perp\!\!\!\perp A_h \mid (U_{h+1/2}, S_{h+1/2})$.
- (d) $A_h \perp\!\!\!\perp V_{h+1/2} \mid (U_{h+1/2}, S_{h+1/2})$.
- (e) $\mathbb{E}_{V_h}[\beta_{u,s}^A(U_h, V_h, S_h)] = \mathbb{E}_{V_{h+1/2}}[\beta_{u,s}^A(U_{h+1/2}, V_{h+1/2}, S_{h+1/2})] = 0$.
- (f) $\mathbb{E}_{V_h}[\iota_h^f(U_h, V_h, S_h)] = \mathbb{E}_{V_{h+1/2}}[\iota_h^f(U_{h+1/2}, V_{h+1/2}, S_{h+1/2})] = 0, \forall f$,

where $\mathbb{E}_{V_{h+1/2}}$ and $\mathbb{E}_{V_h}$ denote the expectations with respect to V_h and $V_{h+1/2}$ given other variables respectively.

Assumption 1 (a)–(b) are two well-known core IV conditions in the literature of causal inference (Angrist and Imbens 1995, Angrist et al. 1996). Specifically, Assumption 1 (a) ensures that Bob's action $B_{h-1/2}$, which is regarded as an IV for Alice's action A_h , is correlated with the action A_h of Alice given the observed state variables U_h and S_h , which is mild. In our previous video recommendation example, the system's next recommendation is correlated with whether the user likes the previously recommended video or not. This is reasonable as the recommendation system will adjust its current suggestion especially if the user does not choose to watch the previously suggested video. Assumption 1 (b) states that the causal effect of $B_{h-1/2}$ on the future transition cannot be confounded by V_h , which ensures that the observed process satisfies the Markovian property. Under such an assumption, the model becomes so-called confounded MDP (Shi et al. 2022, Kallus and Zhou 2020, Fu et al. 2022). In the context of the video recommendation, Assumption 1 (b) implies

that the user's previous reaction to the recommended video does not affect their current (private) mood. This assumption could be violated if the user watches the movie in the previous stage and decides to find another movie to watch in a short period. **In practice, Assumption 1 (b) cannot be verified and tested due to the unobserved confounders.**

Assumption 1 (c)–(d) are similar conditions, where we consider A_h as an instrumental variable for adjusting the causal effect of $B_{h+1/2}$ on future transitions without accessing $V_{h+1/2}$. The last two assumptions 1 (e)–(f) ensure that the baseline effects in the reward and transition models at each decision point considered in (2)–(5) will not be affected by the early actions. In other words, actions at each decision do not have a carry-over effect on the future baseline effect. In our video recommendation example, it basically means that the recommendation made by the machine and whether the customer will play the video or not will not affect exogenous variables such as demographic and subscription information of the user. These two conditions are used to guarantee that the backward induction is valid during our learning procedure. See Figure 1 for a graphical illustration of the offline data collected from the two-player turn-based game satisfying Assumption 1. Assumption 1 is imposed so that one can leverage the offline data and instrumental variables to understand the true causal effects of user's watching behavior and the recommendation system on the long-term reward when there exists unmeasured confounding. Most existing approaches rely on the unconfoundedness assumption, which is restrictive and leads to a suboptimal recommendation system. Relaxing the standard identification condition can greatly enhance the applicability of our method, improving the current recommendation system, compared with existing approaches. **However, our model on the data generating process has several limitations. First of all, Assumption 1 is not testable by our data, despite that some of them are typical assumptions in the existing literature of causal inference. Moreover, Assumptions 1 (e)–(f) are relatively strong regarding with the carry-over effect of the baseline covariates. One possible way to relax these assumptions is to consider an alternative model called confounded partially observable Markov decision process and generalize it to model the offline data with two players, which allows more general carry-over effect. Recent developments along this direction include (Shi et al. 2021, Miao et al. 2022, Lu et al. 2022, Hong et al. 2023).**

Under the model structure specified in (2)–(5), immediately we have the following property for the action-value functions of Alice. Similar results also hold for Bob.

PROPOSITION 1. *For any policy $\pi = (\pi^A, \pi^B) \in \Pi^A \times \Pi^B$ and $h \in [H]$, there exist functions $\tilde{\theta}_h^A$, $\tilde{\theta}_{h+1/2}^A$, $\tilde{\gamma}_h^A$, $\tilde{\gamma}_{h+1/2}^A$, $\tilde{\omega}_h^A$, and $\tilde{\omega}_{h+1/2}^A$ such that*

$$\begin{aligned} Q_h^{A,\pi}(S_h, A_h, B_{h-1/2}, U_h, V_h) \\ = \tilde{\theta}_h^A(S_h, U_h, V_h; \pi) \cdot A_h + \tilde{\gamma}_h^A(S_h, U_h, V_h; \pi) \cdot B_{h-1/2} \end{aligned}$$

$$\begin{aligned}
& + \tilde{\omega}_h^A(S_h, U_h, V_h; \pi) \cdot A_h B_{h-1/2} + \tilde{\zeta}_h^A(S_h, U_h, V_h; \pi), \\
Q_{h+1/2}^{A,\pi}(S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2}) \\
& = \tilde{\theta}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}, V_{h+1/2}; \pi) \cdot A_h + \tilde{\gamma}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}, V_{h+1/2}; \pi) \cdot B_{h+1/2} \\
& + \tilde{\omega}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}, V_{h+1/2}; \pi) \cdot A_h B_{h+1/2} + \tilde{\zeta}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}, V_{h+1/2}; \pi).
\end{aligned}$$

Proposition 1 implies that the action-value functions can always be decomposed into three components as the effects from Alice's action, Bob's action, and their interaction. It can be seen that our data generating process can be easily generalized to the multiple-action setting. Since V_h is not observed, one cannot use the offline data to identify $Q_h^{A,\pi}$ and $Q_{h+1/2}^{A,\pi}$, and ultimately $J^A(\pi^A, \pi^B)$. Under Assumption 1 (e), we can define

$$\begin{aligned}
Q_h^{A,\pi}(S_h, A_h, B_{h-1/2}, U_h) &= \tilde{\theta}_h^A(S_h, U_h; \pi) \cdot A_h + \tilde{\gamma}_h^A(S_h, U_h; \pi) \cdot B_{h-1/2} + \tilde{\omega}_h^A(S_h, U_h; \pi) \cdot A_h B_{h-1/2}, \\
Q_{h+1/2}^{A,\pi}(S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}) &= \tilde{\theta}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h + \tilde{\gamma}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) \cdot B_{h+1/2} \\
&+ \tilde{\omega}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h B_{h+1/2},
\end{aligned}$$

where,

$$(\tilde{\theta}_h^A(S_h, U_h; \pi), \tilde{\gamma}_h^A(S_h, U_h; \pi), \tilde{\omega}_h^A(S_h, U_h; \pi))^\top = \mathbb{E}_{V_h} \left[(\tilde{\theta}_h^A(S_h, U_h, V_h; \pi), \tilde{\gamma}_h^A(S_h, U_h, V_h; \pi), \tilde{\omega}_h^A(S_h, U_h, V_h; \pi))^\top \right]$$

and

$$\begin{aligned}
& (\tilde{\theta}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi), \tilde{\gamma}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi), \tilde{\omega}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi))^\top \\
& = \mathbb{E}_{V_{h+1/2}} \left[(\tilde{\theta}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}, V_{h+1/2}; \pi), \tilde{\gamma}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}, V_{h+1/2}; \pi), \tilde{\omega}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}, V_{h+1/2}; \pi))^\top \right].
\end{aligned}$$

As long as we are able to use the offline data to identify $Q_h^{A,\pi}(S_h, A_h, B_{h-1/2}, U_h)$ and $Q_{h+1/2}^{A,\pi}(S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2})$, we can obtain that

$$J^A(\pi^A, \pi^B) = \mathbb{E}[Q_1^{A,\pi}(S_h, A_h, B_{h-1/2}, U_h)].$$

Similar notations and results hold for Bob's state-action value function. In the following two sections, we leverage instrumental variables and establish a novel identification result for identifying the policy value of both players.

3. Identification Results under Instrumental Variables: A Detour

Since our identification results in the two-player turned-based game rely on the instrumental variable approach, in this subsection, we take a detour and introduce a new instrumental variable approach for identifying the effect of both the action and IV in the standard causal inference setting under unmeasured confounding. Such an approach serves as the foundation of our proposal

for learning the optimal in-class policy pair in the two-player turned-based game without accessing Bob's private information, which will be introduced later in §4.

Consider a setting where the outcome $Y(\pi)$ is generated by some observed state (S, U) , unobserved state V , a binary action $A \in \mathcal{A}$ and a binary instrumental variable $B \in \mathcal{B}$, where $\pi \in \Pi$ is some generic policy. Here we let $Y(\pi)$ depend on π so that it can be interpreted as the future rewards under some target policy π . We impose the following structural equation models for the outcome $Y(\pi)$ and the action A . Specifically, for unknown functions $\boldsymbol{\vartheta}^*(S, U, V; \pi) = (\vartheta_a^*(S, U, V; \pi), \vartheta_z^*(S, U, V; \pi), \vartheta_{az}^*(S, U, V; \pi))$, the following two conditional expectation models always hold,

$$\begin{aligned} \mathbb{E}[Y(\pi) | S, U, V, A, B] &= \vartheta_a^*(S, U, V; \pi)A + \vartheta_z^*(S, U, V; \pi)B + \vartheta_{az}^*(S, U, V; \pi)AB + \vartheta_{us}^*(S, U, V; \pi), \\ \mathbb{E}[A | S, B, U, V] &= \alpha_z^*(S, U, V)B + \alpha_{u,s}^*(S, U, V), \end{aligned} \quad (6)$$

where one can treat $Y(\pi)$ as the cumulative reward under π , A as the action of Alice, B as the action taken by Bob in the previous step, (S, U) as the information accessed by the Alice and V as the private information of Bob in the current step. For notational simplicity, we often write $Y, \boldsymbol{\vartheta}^*(S, U, V) = (\vartheta_a^*(S, U, V), \vartheta_z^*(S, U, V), \vartheta_{az}^*(S, U, V))$ for $Y(\pi), \boldsymbol{\vartheta}^*(S, U, V; \pi) = (\vartheta_a^*(S, U, V; \pi), \vartheta_z^*(S, U, V; \pi), \vartheta_{az}^*(S, U, V; \pi))$ respectively when there is no confusion. In the standard causal inference setting, existing works such as Tchetgen et al. (2021) did not model the interaction effect $\vartheta_{az}^*(S, U, V)$. In other words, Tchetgen et al. (2021) let $\vartheta_{az}^*(S, U, V) \equiv 0$. As discussed in the Alice's model, motivated by our applications, it is essential to model such an effect. This implies that our model is more general than the existing literature. In this section, we aim to establish identification results for $\boldsymbol{\vartheta}^*(S, U) = (\vartheta_a^*(S, U), \vartheta_z^*(S, U), \vartheta_{az}^*(S, U))$ so that one can leverage the offline data, which are i.i.d copies of $\{Y, S, U, A, B\}$ (i.e., $\{Y^i, S^i, U^i, A^i, B^i\}_{i \in [n]}$), to learn the effects of the action A and the IV B on the outcome Y . Here $\boldsymbol{\vartheta}^*(S, U) = \mathbb{E}_V[\boldsymbol{\vartheta}^*(S, U, V)]$. Once $\boldsymbol{\vartheta}^*(S, U)$ is identified, one can figure out the best actions for A and B given observed information so as the expected outcome of Y is maximized. Since this problem can be regarded as a special case of the two-player turn-based game, we impose the following assumptions similar as Assumption 1, together with some orthogonal conditions.

ASSUMPTION 2 (Identification for A Basic Setting). *The following statements hold for the random sample $\{Y, S, U, A, B\}$ and $\pi \in \Pi$.*

- (a) $B \not\perp\!\!\!\perp A | (S, U)$.
- (b) $B \perp\!\!\!\perp V | (S, U)$.
- (c) *The following orthogonality conditions hold.*

$$\text{Cov}(\vartheta_a^*(S, U, V), \alpha_z^*(S, U, V) | S, U) = \text{Cov}(\vartheta_a^*(S, U, V), \alpha_{u,s}^*(S, U, V) | S, U) = 0 \quad (7)$$

$$\text{Cov}(\vartheta_z^*(S, U, V), \alpha_z^*(S, U, V) | S, U) = \text{Cov}(\vartheta_z^*(S, U, V), \alpha_{u,s}^*(S, U, V) | S, U) = 0 \quad (8)$$

$$\text{Cov}(\alpha_z^*(S, U, V), \vartheta_{az}^*(S, U, V) | S, U) = \text{Cov}(\alpha_z^*(S, U, V), \vartheta_{us}^*(S, U, V) | S, U) = 0 \quad (9)$$

$$\text{Cov}(\vartheta_{az}^*(S, U, V), \alpha_{u,s}^*(S, U, V) | S, U) = 0. \quad (10)$$

Again, Assumption 2 (a)-(b) impose IV-relevant and independence assumptions on $\{Y, S, U, A, B\}$, which are standard in the existing literature (e.g., Wang and Tchetgen Tchetgen 2018). Since the IV B (i.e., Bob's action) may have an effect on the outcome Y , we did not impose the exclusion restriction on B by only allowing it to affect Y directly. Instead, we impose the orthogonal conditions in Assumption 2 (c) to relax the IV exclusion restriction. One *sufficient* condition for Assumption 2 (c) is that there is no common effect modifiers resulted from V in the aforementioned conditional expectation models related to Y and A (see Figure 2 for a graphical illustration of such a case). We remark that identification conditions in Assumption 2 serves as the foundation for those imposed in Assumption 1. In other words, it can be interpreted as the identification condition for single-stage decision making problem when there exists unmeasured confounding and an invalid instrumental variable. In the sequential setting, basically, at each decision point, either Alice or Bob's action can be understood as an invalid instrumental variable. Assumption 2 is used to identify the effect of Bob or Alice's action on the immediate reward and future transitions. Our identification condition is motivated by Tchetgen et al. (2021), where in their outcome model of Y , only the marginal effects of A and B are considered but not their interaction, i.e., $\vartheta_{az}^*(S, U, V) \equiv 0$ in their model. In order to identify the conditional average effect of A on Y , we impose two additional orthogonality conditions related to $\vartheta_{az}^*(S, U, V)$ so that the interaction effect is being characterized, compared with those in Tchetgen et al. (2021). Therefore, our models, identification conditions and results are more general than theirs. The limitation of Assumption 2 is similar to Assumption 1 as the orthogonal conditions are not testable, while it is weaker than the existing literature. In the following, we present our identification results so that the offline data can identify $\vartheta^*(S, U)$.

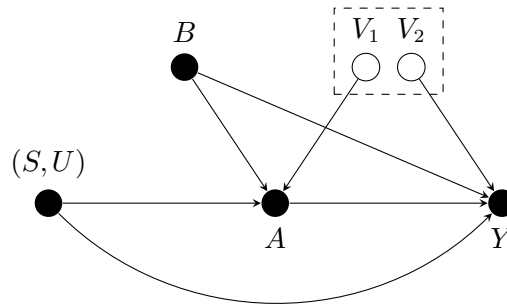


Figure 2 A graphical illustration of one sufficient condition for Assumption 2 (c), where $V = (V_1, V_2)$ and V_1 and V_2 are independent.

THEOREM 1. *Under Assumption 2, if the following three linear equations with respect to $\vartheta^*(S, U)$ have a unique solution, then $\vartheta^*(S, U)$ can be non-parametrically identified by our offline data.*

$$\begin{aligned}
& \mathbb{E}[(B - \mathbb{E}[B | S, U]) \cdot (A - \mathbb{E}[A | B, S, U]) \cdot Y | S, U] \\
&= \mathbb{E}[(B - \mathbb{E}[B | S, U]) \cdot (A - \mathbb{E}[A | B, S, U]) \cdot A | S, U] \cdot \vartheta_a^*(S, U) \\
&\quad + \mathbb{E}[B \cdot (B - \mathbb{E}[B | S, U]) \cdot (A - \mathbb{E}[A | B, S, U]) \cdot A | S, U] \cdot \vartheta_{az}^*(S, U), \\
& \mathbb{E}[(B - \mathbb{E}[B | S, U]) \cdot Y | S, U] \\
&= \mathbb{E}[A \cdot (B - \mathbb{E}[B | S, U]) | S, U] \cdot \vartheta_a^*(S, U) \\
&\quad + \mathbb{E}[B \cdot (B - \mathbb{E}[B | S, U]) | S, U] \cdot \vartheta_z^*(S, U) \\
&\quad + \mathbb{E}[AB \cdot (B - \mathbb{E}[B | S, U]) | S, U] \cdot \vartheta_{az}^*(S, U), \\
& \text{Cov}(B \cdot (B - \mathbb{E}[B | S, U]), (A - \mathbb{E}[A | B, S, U]) \cdot Y | S, U) \\
&= \text{Cov}(B \cdot (B - \mathbb{E}[B | S, U]), A \cdot (A - \mathbb{E}[A | B, S, U]) | S, U) \cdot \vartheta_a^*(S, U) \\
&\quad + \text{Cov}(B \cdot (B - \mathbb{E}[B | S, U]), AB \cdot (A - \mathbb{E}[A | B, S, U]) | S, U) \cdot \vartheta_{az}^*(S, U).
\end{aligned}$$

Proof. The proof can be found in §B.1. ■

The uniqueness in Theorem 1 is satisfied as long as that

$$\begin{aligned}
& \mathbb{E}[(B - \mathbb{E}[B | S, U]) \cdot (A - \mathbb{E}[A | B, S, U]) \cdot A | S, U] \times \text{Cov}(B \cdot (B - \mathbb{E}[B | S, U]), AB \cdot (A - \mathbb{E}[A | B, S, U]) | S, U) \\
& \neq \mathbb{E}[B \cdot (B - \mathbb{E}[B | S, U]) \cdot (A - \mathbb{E}[A | B, S, U]) \cdot A | S, U] \times \text{Cov}(B \cdot (B - \mathbb{E}[B | S, U]), A \cdot (A - \mathbb{E}[A | B, S, U]) | S, U),
\end{aligned}$$

and $\mathbb{E}[AB \cdot (B - \mathbb{E}[B | S, U]) | S, U] \neq 0$. By Theorem 1, we can identify ϑ_a^* , ϑ_z^* , and ϑ_{az}^* non-parametrically by solving a system of linear equations. In the following, we introduce our estimator for ϑ^* based on these identification results. To simplify the presentation, we assume that there exists an oracle that gives the following nuisance parameters,

$$\begin{aligned}
f_1(S, U) &= \mathbb{E}[B | S, U], \quad f_2(S, U, B) = \mathbb{E}[A | S, U, B], \quad f_3(S, U) = \mathbb{E}[(A - f_2(S, U, B))Y | S, U], \\
f_4(S, U) &= \mathbb{E}[A(A - f_2(S, U, B)) | S, U], \quad f_5(S, U) = \mathbb{E}[AB(A - f_2(S, U, B)) | S, U].
\end{aligned} \tag{11}$$

We remark that such nuisance parameters can be estimated via supervised learning without oracle. See §B for such a setting. Further, we define

$$\begin{aligned}
w_1(Y, A, B, S, U) &= \rho_1(Y, S, U, B) - \rho_2(S, U, B)\vartheta_a^*(S, U) - \rho_3(S, U, B)\vartheta_{az}^*(S, U), \\
w_2(Y, A, B, S, U) &= \rho_4(Y, S, U, B) - \rho_2(S, U, B, f_1)\vartheta_a^*(S, U) - \rho_6(S, U, B)\vartheta_z^*(S, U) - \rho_7(S, U, B)\vartheta_{az}^*(S, U), \\
w_3(Y, A, B, S, U) &= \rho_8(Y, S, U, B) - (1 - f_1(S, U))f_1(S, U)f_3(S, U) \\
&\quad - (\rho_9(S, U, B) - (1 - f_1(S, U))f_1(S, U)f_4(S, U))\vartheta_a^*(S, U) \\
&\quad - (\rho_{10}(S, U, B) - (1 - f_1(S, U))f_1(S, U)f_5(S, U))\vartheta_{az}^*(S, U),
\end{aligned} \tag{12}$$

where $\{\rho_i\}_{i \in [10]}$ are defined in (56) of the appendix. We denote by $W(Y, S, U, A, B, \boldsymbol{\vartheta}^*) = (w_1, w_2, w_3)$, where we recall that $\boldsymbol{\vartheta}^*(S, U) \in \mathbb{R}^3$. Observing the linear system in (12), we can write

$$W = \Phi(\mathbf{f})\boldsymbol{\vartheta}^* + \alpha(\mathbf{f}), \quad (13)$$

where $\Phi(\mathbf{f}) \in \mathbb{R}^{3 \times 3}$, $\alpha(\mathbf{f}) \in \mathbb{R}^3$, and $\mathbf{f} = (f_1, f_2, \dots, f_5)$. Then by the identification results in Theorem 1, we have the following conditional moment restriction,

$$\mathbb{E}[W(Y, S, U, A, B, \boldsymbol{\vartheta}^*) | S, U] = 0. \quad (14)$$

We aim to recover $\boldsymbol{\vartheta}^*$ via such a conditional moment restriction. In general, we need to estimate both $\boldsymbol{\vartheta}^*$ and $\mathbf{f}$, where (14) is a nonlinear conditional moment restriction model with respect to $\boldsymbol{\vartheta}^*$ and $\mathbf{f}$. Although we assume that we have accessed the oracle $\mathbf{f}$, we still consider this problem as solving a nonlinear conditional moment restriction model. We follow Chen and Pouzo (2012) and propose to use a sieve minimum distance (SMD) estimator for $\boldsymbol{\vartheta}^*$. We remark that such a nonlinear conditional moment restriction has been studied in the econometrics literature. See Chen and Pouzo (2012) and the reference therein. We first introduce the definition of the sieve space $\mathcal{H}_n$,

$$\mathcal{H}_n = \left\{ \boldsymbol{\vartheta} \in \mathcal{H} : \boldsymbol{\vartheta}(\cdot) = \sum_{k=1}^{k(n)} a_k q_k(\cdot) \right\}, \quad (15)$$

where $k(n) < \infty$, $k(n) \rightarrow \infty$ as $n \rightarrow \infty$, and $\{q_k\}_{k=1}^\infty$ is a sequence of known basis functions of a Banach space (i.e., $\mathcal{H}$) such as wavelet basis functions. Then the SMD estimator for $\boldsymbol{\vartheta}^*$ takes the following form,

$$\begin{aligned} \hat{\boldsymbol{\vartheta}}_n &\in \arg \min_{\boldsymbol{\vartheta} \in \mathcal{H}_n} \hat{L}(\boldsymbol{\vartheta}), \\ \text{where } \hat{L}(\boldsymbol{\vartheta}) &= \frac{1}{n} \sum_{i=1}^n \hat{m}(Y^i, S^i, U^i, A^i, B^i; \boldsymbol{\vartheta})^\top \hat{m}(Y^i, S^i, U^i, A^i, B^i; \boldsymbol{\vartheta}). \end{aligned} \quad (16)$$

Here $\hat{m}(Y^i, S^i, U^i, A^i, B^i; \boldsymbol{\vartheta})$ is a consistent estimator of $\mathbb{E}[W(Y, S, U, A, B, \boldsymbol{\vartheta}) | S, U]$ under some mild conditions. For example, one can use a non-parametric regression with linear sieve to obtain $\hat{m}(Y^i, S^i, U^i, A^i, B^i; \boldsymbol{\vartheta})$. Then the optimization problem in (16) is convex and can be solved efficiently. In the meanwhile, we denote by $m(S, U, \boldsymbol{\vartheta}) = \mathbb{E}[W(Y, S, U, A, B, \boldsymbol{\vartheta}) | S, U]$ for any $\boldsymbol{\vartheta}$.

In the following, based on the result of Chen and Pouzo (2012), we derive the convergence rate of $\hat{\boldsymbol{\vartheta}}_n$. We introduce the following assumptions.

ASSUMPTION 3 (Consistency). *The SMD estimator $\hat{\boldsymbol{\vartheta}}_n$ is a consistent estimator of $\boldsymbol{\vartheta}^*$ uniformly for any π .*

As we will show in Theorem 5 in §B of the appendix, such an assumption of consistency can be achieved under mild conditions. The main difference between our result and those in (Chen and Pouzo 2012) lies that we require our results hold uniformly over the policy class Π as the outcome Y depends on the policy π . Therefore, our proof needs to take the complexity of Π into consideration. With Assumption 3, we can restrict our space $\mathcal{H}$ to a shrinking neighborhood around the ground truth $\boldsymbol{\vartheta}^*$ and define

$$\mathcal{H}_s = \{\boldsymbol{\vartheta} \in \mathcal{H} : \|\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*\| \leq \delta, \|\boldsymbol{\vartheta}\| \leq M\}, \quad \mathcal{H}_{sn} = \mathcal{H}_s \cap \mathcal{H}_n$$

for a positive constant M and a sufficiently small positive δ such that $\mathbb{P}(\hat{\boldsymbol{\vartheta}}_n \notin \mathcal{H}_s) < \delta$. $\|\bullet\|$ refers to the L_2 norm with respect to some underlying probability measure. Thus, we restrict the sieve space $\mathcal{H}_n$ in (16) to be $\mathcal{H}_{sn}$ hereafter. Meanwhile, we define the path-wise derivative in the direction $[\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*]$ evaluated at $\boldsymbol{\vartheta}^*$ as follows,

$$\frac{dm(S, U, \boldsymbol{\vartheta}^*)}{d\boldsymbol{\vartheta}}[\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*] = \left. \frac{d\mathbb{E}[W(Y, S, A, B, U, (1 - \tau)\boldsymbol{\vartheta}^* + \tau\boldsymbol{\vartheta}) | S, U]}{d\tau} \right|_{\tau=0}.$$

We also define the following notation of pseudometric for any $\boldsymbol{\vartheta}_1, \boldsymbol{\vartheta}_2 \in \mathcal{H}_s$,

$$\|\boldsymbol{\vartheta}_1 - \boldsymbol{\vartheta}_2\|_{ps} = \sqrt{\mathbb{E} \left[\left\| \frac{dm(S, U, \boldsymbol{\vartheta}^*)}{d\boldsymbol{\vartheta}}[\boldsymbol{\vartheta}_1 - \boldsymbol{\vartheta}_2] \right\|_{\ell_2}^2 \right]}.$$

We introduce the following assumptions for the shrinking space $\mathcal{H}_s$.

ASSUMPTION 4 (Local Curvature). *The following statements hold uniformly for any π .*

- $\mathcal{H}_s$ and $\mathcal{H}_{sn}$ are convex, and $m(S, U, \boldsymbol{\vartheta})$ is continuously path-wise differentiable with respect to $\boldsymbol{\vartheta} \in \mathcal{H}_s$.

- There exists a constant $c > 0$ such that $\|\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*\|_{ps} \leq c \cdot \|\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*\|$ for any $\boldsymbol{\vartheta} \in \mathcal{H}_s$.

- There exist finite constants $c_1, c_2 > 0$ such that $\|\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*\|_{ps}^2 \leq c_1 \mathbb{E}[\|m(S, U, \boldsymbol{\vartheta})\|^2]$ for any $\boldsymbol{\vartheta} \in \mathcal{H}_{sn}$; and $c_2 \mathbb{E}[\|m(S, U, \Gamma_n \boldsymbol{\vartheta}^*)\|^2] \leq \|\Gamma_n \boldsymbol{\vartheta}^* - \boldsymbol{\vartheta}^*\|_{ps}^2$ for some $\Gamma_n \boldsymbol{\vartheta}^* \in \mathcal{H}_{k(n)}$ such that $\|\Gamma_n \boldsymbol{\vartheta}^* - \boldsymbol{\vartheta}^*\| = o(1)$.

The first and the second statements in Assumption 4 imply that the pseudometric $\|\cdot\|_{ps}$ is well-defined in $\mathcal{H}_s$ and is weaker than $\|\cdot\|$, which is the L_2 norm. The last statement in Assumption 4 implies that the pseudometric $\|\cdot\|_{ps}$ is Lipschitz continuous with respect to the function $\mathbb{E}[\|m(S, U, \boldsymbol{\vartheta})\|^2]$ for any $\boldsymbol{\vartheta} \in \mathcal{H}_{sn}$. Assumption 4 is imposed so that a fast convergence rate in terms of the pseudometric can be obtained.

ASSUMPTION 5 (Sieve Approximation Error). *We have $\|\boldsymbol{\vartheta}^* - \sum_{j=1}^{k(n)} \langle \boldsymbol{\vartheta}^*, q_j \rangle q_j\| = O(\nu_{k(n)}^{-\alpha})$ uniformly for any π with a finite $\alpha > 0$ and a positive sequence $\{\nu_j\}_{j=1}^\infty$ that strictly increases and $\nu_j = \Theta(j^{1/d})$, where d is the dimension of S and U .*

ASSUMPTION 6 (**Sieve Link Condition**). *There are finite constants $c, C > 0$ such that for any π*

- $\|\boldsymbol{\vartheta}\|_{ps}^2 \geq c \sum_{j=1}^{\infty} \varphi(\nu_j^{-2}) \cdot |\langle \boldsymbol{\vartheta}, q_j \rangle|^2$ for all $\boldsymbol{\vartheta} \in \mathcal{H}_{sn}$;
- $\|\Gamma_n \boldsymbol{\vartheta}^* - \boldsymbol{\vartheta}^*\|_{ps}^2 \leq C \sum_{j=1}^{\infty} \varphi(\nu_j^{-2}) \cdot |\langle \Gamma_n \boldsymbol{\vartheta}^* - \boldsymbol{\vartheta}^*, q_j \rangle|^2$, where $\Gamma_n \boldsymbol{\vartheta}^*$ is defined in Assumption 4;
- $\varphi(\tau) = \tau^\varsigma$ for some $\varsigma \geq 0$.

Assumption 5 implies that $\mathcal{H}_n$ is a natural sieve for the approximation of $\boldsymbol{\vartheta}^*$, which is a mild condition on the smoothness of $\boldsymbol{\vartheta}^* \in \mathcal{H}_s$. The constant α usually refers to the smoothness constant of $\mathcal{H}$ and d is the dimension of the input features S and U . In the meanwhile, the first statement in Assumption 6 implies that the norm in the shrinking sieve neighborhood $\mathcal{H}_{sn}$ is upper bounded by the pseudometric; the second statement is a stability condition that is required only for the sieve approximation error; and the last statement characterizes the ill-posedness of solving the conditional moment restriction (14) via a link function φ . Here we consider the mildly ill-posed case (Chen and Reiss 2011) and ς is the corresponding exponent. When ς is large, the hardness of solving (14) increases. Assumptions 5 and 6 are used to establish the link between the error quantified by the pseudometric and that by the L_2 -norm. See Chen and Pouzo (2012) for more details on these assumptions.

With the above assumptions, we have the following results characterizing the convergence of $\widehat{\boldsymbol{\vartheta}}_n$.

THEOREM 2 (Chen and Pouzo (2012)). *Suppose that Assumptions 2–6 hold. It holds for any π that*

$$\left\| \widehat{\boldsymbol{\vartheta}}_n - \boldsymbol{\vartheta}^* \right\| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right),$$

where $\widehat{\boldsymbol{\vartheta}}_n$ is the SMD estimator defined in (16).

Theorem 2 indicates that the convergence rate depends on the smoothness constant α , the dimension of S and U (i.e., d), and the ill-posed constant ς . Specifically, when the smoothness of the true-function increases, our SMD estimator converges faster. In contrast, when the dimension and the ill-posed constant increase, the converge rate becomes slower. Moreover, as shown in Chen and Pouzo (2012), the convergence rate in Theorem 2 is minimax-optimal when the function W defined in (14) is linear. We can thus construct plug-in estimators of ϑ_a^* , ϑ_z^* , and ϑ_{az}^* as $\widehat{\vartheta}_a^*$, $\widehat{\vartheta}_z^*$, and $\widehat{\vartheta}_{az}^*$ from $\widehat{\boldsymbol{\vartheta}}_n$ and obtain the following convergence results.

COROLLARY 1. *Suppose that Assumptions 3–6 hold. It holds for any π that*

$$\left\| \widehat{\vartheta}_a^* - \vartheta_a^* \right\| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right), \quad \left\| \widehat{\vartheta}_z^* - \vartheta_z^* \right\| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right), \quad \left\| \widehat{\vartheta}_{az}^* - \vartheta_{az}^* \right\| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right).$$

Corollary 1 demonstrates that our estimators are valid for the effects of the action A and the IV B on the outcome Y . For example, in our video recommendation example, by leveraging such a result, we can reliably estimate the true causal effect of suggesting a certain video genre to the user and understand how a user's selection will affect the recommendation of the next video. Overall, these results serve as the foundation of our proposal below.

4. Warm-up: Single-Stage Game

In this section, as a warm-up, we consider a single-stage version of the aforementioned two-player turn-based game, where the length of horizon $H = 1$. Under such a setting, the game terminates once the state S_2 is revealed to both players. In order to perform policy learning under the unmeasured confounding, we first need to conduct off-policy evaluation. Specifically, we need to establish identification results so that we are able to estimate the action-value functions $Q_1^{A,\pi}$ and $Q_1^{B,\pi}$ for any policy $\pi = (\pi^A, \pi^B)$ using the offline data without accessing the private information of Bob. In §3, we study a standard causal setting, where we introduce a general identification condition and an estimating procedure using the instrumental variable approach. The established results motivate our proposal in the single-stage game below. In §4.1, we tailor the results of §3 to the single-stage two-player turn-based game and propose our estimators for $Q_1^{A,\pi}$ and $Q_1^{B,\pi}$. Based on these results, in §4.2, we propose our method for offline policy learning and finally in §4.3, we show that our proposed method provably converges to an optimal policy.

4.1. Evaluation of Expected Total Rewards

In this subsection, we extend results from §3 to our single-stage turn-based game. For any policy pair $\pi = (\pi^A, \pi^B)$, we introduce the estimation of $Q_1^{A,\pi}$, $Q_{3/2}^{B,\pi}$, and $Q_1^{B,\pi}$ as follows.

Estimation of $Q_1^{A,\pi}$. Recall that $Q_{3/2}^{A,\pi} = 0$. Then by Alice's model (2) and the definition of $Q_1^{A,\pi}(S_1, A_1, B_{1/2}, U_1)$, we have

$$Q_1^{A,\pi}(S_1, A_1, B_{1/2}, U_1) = \beta_a^A(U_1, S_1)A_1 + \beta_z^A(U_1, S_1)B_{1/2} + \beta_{az}^A(U_1, S_1)A_1B_{1/2}, \quad (17)$$

where we write $\beta_a^A(U_1, S_1) = \mathbb{E}_{V_1}[\beta_a^A(U_1, V_1, S_1)]$, $\beta_z^A(U_1, S_1) = \mathbb{E}_{V_1}[\beta_z^A(U_1, V_1, S_1)]$, and $\beta_{az}^A(U_1, S_1) = \mathbb{E}_{V_1}[\beta_{az}^A(U_1, V_1, S_1)]$ for notational convenience. Thus, to construct an estimator of $Q_1^{A,\pi}$, we only need to estimate the right-hand side of (17). Following from the estimating procedure in §3 by treating A_1 as the action and $B_{1/2}$ as an IV, together with similar orthogonal conditions, we can obtain an estimation of the right-hand side of (17) as follows,

$$\widehat{Q}_1^{A,\pi}(S_1, A_1, B_{1/2}, U_1) = \widehat{\beta}_a^A(U_1, S_1)A_1 + \widehat{\beta}_z^A(U_1, S_1)B_{1/2} + \widehat{\beta}_{az}^A(U_1, S_1)A_1B_{1/2}. \quad (18)$$

Estimation of $Q_{3/2}^{B,\pi}$. Similar to the derivation of (17), we have the following equation,

$$Q_{3/2}^{B,\pi}(S_{3/2}, A_1, B_{3/2}, U_{3/2}) = \beta_a^B(U_{3/2}, S_{3/2})B_{3/2} + \beta_z^B(U_{3/2}, S_{3/2})A_1 + \beta_{az}^B(U_{3/2}, S_{3/2})A_1B_{3/2},$$

where the estimation of the right-hand side can be obtained via the procedure in §3 by treating $B_{3/2}$ as the action and A_1 as the IV, i.e.,

$$\widehat{Q}_{3/2}^{B,\pi}(S_{3/2}, A_1, B_{3/2}, U_{3/2}) = \widehat{\beta}_a^B(U_{3/2}, S_{3/2})B_{3/2} + \widehat{\beta}_z^B(U_{3/2}, S_{3/2})A_1 + \widehat{\beta}_{az}^B(U_{3/2}, S_{3/2})A_1B_{3/2}. \quad (19)$$

Estimation of $Q_1^{B,\pi}$. Note that by the memoryless assumption of $V_{3/2}$ and Assumption 1 (e), we have the following Bellman equation,

$$Q_1^{B,\pi}(S_1, A_1, B_{1/2}, U_1, V_1) = \mathbb{E} \left[Q_{3/2}^{B,\pi}(S_{3/2}, A_1, B_{3/2}, U_{3/2}) \mid S_1, A_1, B_{1/2}, U_1, V_1 \right].$$

Then we can obtain that

$$\begin{aligned} Q_1^{B,\pi}(S_1, A_1, B_{1/2}, U_1, V_1) &= \mathbb{E} \left[Q_{3/2}^{B,\pi}(S_{3/2}, A_1, B_{3/2}, U_{3/2}) \mid S_1, A_1, B_{1/2}, U_1, V_1 \right] \\ &= \mathbb{E} \left[\beta_a^B(U_{3/2}, S_{3/2})B_{3/2} \mid S_1, A_1, B_{1/2}, U_1, V_1 \right] \\ &\quad + \mathbb{E} \left[\beta_z^B(U_{3/2}, S_{3/2}) \mid S_1, A_1, B_{1/2}, U_1, V_1 \right] \cdot A_1 \\ &\quad + \mathbb{E} \left[\beta_{az}^B(U_{3/2}, S_{3/2})B_{3/2} \mid S_1, A_1, B_{1/2}, U_1, V_1 \right] \cdot A_1 \\ &\triangleq (I) + (II) \cdot A_1 + (III) \cdot A_1 \end{aligned} \quad (20)$$

By Assumption 1 (e) and integrating out V_1 given (S_1, U_1) , the model specified in (3) and Bob's reward model, we know that there exist functions $\{\theta_1^{B,j}, \gamma_1^{B,j}, \omega_1^{B,j}\}_{j \in [3]}$ such that

$$\begin{aligned} \mathbb{E}[(I) \mid S_1, U_1] &= \theta_1^{B,1}(S_1, U_1; \pi) \cdot A_1 + \gamma_1^{B,1}(S_1, U_1; \pi) \cdot B_{1/2} + \omega_1^{B,1}(S_1, U_1; \pi) \cdot A_1 B_{1/2}, \\ \mathbb{E}[(II) \mid S_1, U_1] &= \theta_1^{B,2}(S_1, U_1; \pi) \cdot A_1 + \gamma_1^{B,2}(S_1, U_1; \pi) \cdot B_{1/2} + \omega_1^{B,2}(S_1, U_1; \pi) \cdot A_1 B_{1/2}, \\ \mathbb{E}[(III) \mid S_1, U_1] &= \theta_1^{B,3}(S_1, U_1; \pi) \cdot A_1 + \gamma_1^{B,3}(S_1, U_1; \pi) \cdot B_{1/2} + \omega_1^{B,3}(S_1, U_1; \pi) \cdot A_1 B_{1/2}. \end{aligned} \quad (21)$$

Again, following from the estimation procedure in §3 where we treat A_1 as the action and $B_{1/2}$ as the IV, we can obtain the estimators of $\{\theta_1^{B,j}, \gamma_1^{B,j}, \omega_1^{B,j}\}_{j \in [3]}$, which are denoted by $\{\widehat{\theta}_1^{B,j}, \widehat{\gamma}_1^{B,j}, \widehat{\omega}_1^{B,j}\}_{j \in [3]}$. Combining (20)-(21), we obtain the following estimator of $Q_1^{B,\pi}$,

$$\widehat{Q}_1^{B,\pi}(S_1, A_1, B_{1/2}, U_1) = \widehat{\theta}_1^B(S_1, U_1; \pi) \cdot A_1 + \widehat{\gamma}_1^B(S_1, U_1; \pi) \cdot B_{1/2} + \widehat{\omega}_1^B(S_1, U_1; \pi) \cdot A_1 B_{1/2}, \quad (22)$$

where

$$\begin{aligned} \widehat{\theta}_1^B(S_1, U_1; \pi) &= \widehat{\theta}_1^{B,1}(S_1, U_1; \pi) + \widehat{\theta}_1^{B,2}(S_1, U_1; \pi) + \widehat{\theta}_1^{B,3}(S_1, U_1; \pi), \\ \widehat{\gamma}_1^B(S_1, U_1; \pi) &= \widehat{\gamma}_1^{B,1}(S_1, U_1; \pi), \\ \widehat{\omega}_1^B(S_1, U_1; \pi) &= \widehat{\omega}_1^{B,1}(S_1, U_1; \pi) + \widehat{\omega}_1^{B,2}(S_1, U_1; \pi) + \widehat{\omega}_1^{B,3}(S_1, U_1; \pi) \\ &\quad + \widehat{\gamma}_1^{B,2}(S_1, U_1; \pi) + \widehat{\gamma}_1^{B,3}(S_1, U_1; \pi). \end{aligned} \quad (23)$$

See Figure 3 for a graphical illustration of the above construction.

Estimation of J^A and J^B . In summary, to construct $\widehat{Q}_1^A$ and $\widehat{Q}_1^B$ in (18) and (22), respectively, we need to obtain

$$\widehat{\beta}^A = (\widehat{\beta}_a^A, \widehat{\beta}_z^A, \widehat{\beta}_{az}^A), \quad \widehat{\beta}^B = (\widehat{\beta}_a^B, \widehat{\beta}_z^B, \widehat{\beta}_{az}^B), \quad \widehat{\theta}_1^{B,j} = (\widehat{\theta}_1^{B,j}, \widehat{\gamma}_1^{B,j}, \widehat{\omega}_1^{B,j}) \quad \text{for any } j \in [3]$$

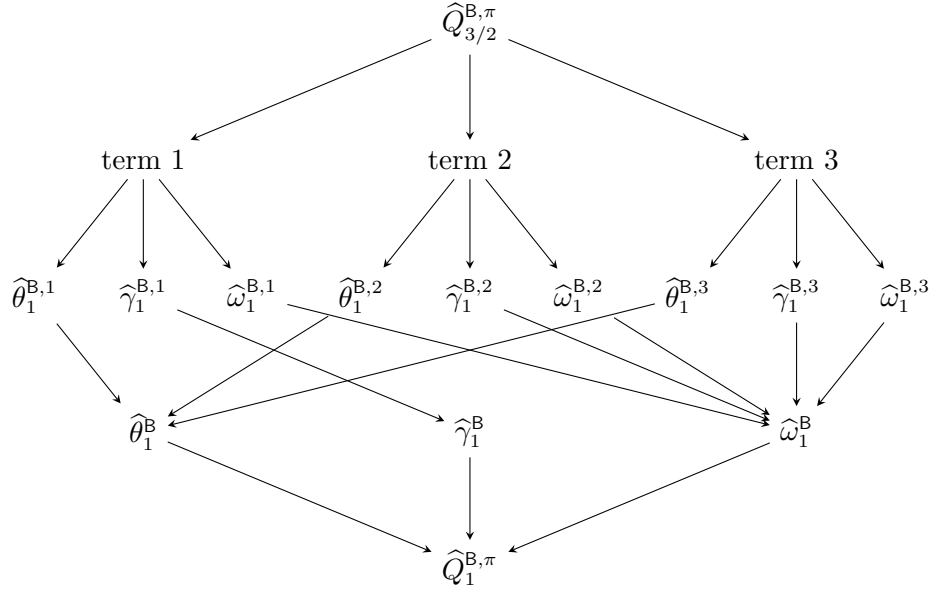


Figure 3 A graphical illustration of the estimation of action-value function $Q_1^{B, \pi}$. In particular, the three items below $\hat{Q}_{3/2}^{B, \pi}$ illustrate that the conditional expectation of $\hat{Q}_{3/2}^{B, \pi}$, which yields $Q_1^{B, \pi}$, can be decomposed into three terms, as shown in (20). The arrows starting from these three terms show that they can each be further decomposed into smaller terms, as shown in (21). Finally, we aggregate these terms according to (22).

as estimators of the following functions,

$$\beta^A = (\beta_a^A, \beta_z^A, \beta_{az}^A), \quad \beta^B = (\beta_a^B, \beta_z^B, \beta_{az}^B), \quad \theta_1^{B,j} = (\theta_1^{B,j}, \gamma_1^{B,j}, \omega_1^{B,j}) \quad \text{for any } j \in [3].$$

Once we obtain these estimators, we can estimate $J^A(\pi^A, \pi^B)$ and $J^B(\pi^A, \pi^B)$ as follows,

$$\hat{J}^A(\pi^A, \pi^B) = \mathbb{E}_\pi \left[\hat{Q}_1^{A, \pi}(S_1, A_1, B_{1/2}, U_1) \right], \quad \hat{J}^B(\pi^A, \pi^B) = \mathbb{E}_\pi \left[\hat{Q}_1^{B, \pi}(S_1, A_1, B_{1/2}, U_1) \right],$$

where the expectation $\mathbb{E}_\pi[\cdot]$ is taken with following the policy $\pi = (\pi^A, \pi^B)$.

4.2. Pessimistic Policy Learning Method

With the evaluation procedure proposed in §4.1 and §3, we present our method to find an optimal policy in this section. Instead of directly maximizing $\hat{J}^A(\pi^A, \pi^B)$ and $\hat{J}^B(\pi^A, \pi^B)$, we use the pessimistic idea in policy learning. See Jin et al. (2021), Fu et al. (2022) for more details. To begin with, in light of (13), we can define similar system of linear equations, which are helpful in constructing estimators for β^A, β^B , and $\theta_1^{B,j}$ for $j \in [3]$.

$$W^{\diamond, r} \triangleq \Phi^{\diamond, r} \beta^\diamond + \alpha^{\diamond, r}, \quad W^{B,j, \pi}(\tilde{\beta}^B) \triangleq \Phi_1^{B,j, \pi}(\tilde{\beta}^B) \theta_1^{B,j, \pi}(\tilde{\beta}^B) + \alpha_1^{B,j, \pi}(\tilde{\beta}^B), \quad (24)$$

where $\Phi^{\diamond, r}, \Phi_1^{B,j, \pi}(\tilde{\beta}^B) \in \mathbb{R}^{3 \times 3}$ and $\alpha^{\diamond, r}, \alpha_1^{B,j, \pi}(\tilde{\beta}^B) \in \mathbb{R}^3$ for any $\tilde{\beta}^B \in \mathcal{H}^3$, $\diamond \in \{A, B\}$, and $j \in [3]$. Here $W^{\diamond, r} = (w_1^{\diamond, r}, w_2^{\diamond, r}, w_3^{\diamond, r})$ and $W^{B,j, \pi}(\tilde{\beta}^B) = (w_1^{B,j, \pi}, w_2^{B,j, \pi}, w_3^{B,j, \pi})$ are similarly constructed as (12) by

treating A_1 as the action and $B_{1/2}$ as the IV or by treating $B_{3/2}$ as the action and A_1 as the IV. According to our identification result, one can show that

$$\mathbb{E}[W^{A,r} | S_1, U_1] = \mathbb{E}[W^{B,r} | S_1, U_1] = \mathbb{E}[W^{B,j,\pi} | S_1, U_1] = 0$$

With (24), we further define the loss functions for the construction of estimators in (18), (19), and (22) as follows,

$$\begin{aligned}\widehat{L}^{A,r}(\tilde{\beta}^A) &= \frac{1}{n} \sum_{i=1}^n \widehat{m}^{i,A,r}(\tilde{\beta}^A)^\top \widehat{m}^{i,A,r}(\tilde{\beta}^A), \\ \widehat{L}^{B,r}(\tilde{\beta}^B) &= \frac{1}{n} \sum_{i=1}^n \widehat{m}^{i,B,r}(\tilde{\beta}^B)^\top \widehat{m}^{i,B,r}(\tilde{\beta}^B), \\ \widehat{L}_1^{B,j,\pi}(\tilde{\theta}_1^{B,j}, \tilde{\beta}^B) &= \frac{1}{n} \sum_{i=1}^n \widehat{m}_1^{i,B,j,\pi}(\tilde{\theta}_1^{B,j}, \tilde{\beta}^B)^\top \widehat{m}_1^{i,B,j,\pi}(\tilde{\theta}_1^{B,j}, \tilde{\beta}^B),\end{aligned}\tag{25}$$

where $\widehat{m}^{i,A,r}(\tilde{\beta}^A)$ is a consistent estimator of $\mathbb{E}[\Phi^{A,r}\tilde{\beta}^A + \alpha^{A,r} | S_1, U_1]$, $\widehat{m}^{i,B,r}(\tilde{\beta}^B)$ is a consistent estimator of $\mathbb{E}[\Phi^{B,r}\tilde{\beta}^B + \alpha^{B,r} | S_{3/2}, U_{3/2}]$, while $\widehat{m}_1^{i,B,j,\pi}(\tilde{\theta}_1^{B,j}, \tilde{\beta}^B)$ is a consistent estimator of $\mathbb{E}[\Phi_1^{B,j,\pi}(\tilde{\beta}^B)\theta_1^{B,j,\pi}(\tilde{\beta}^B) + \alpha_1^{B,j,\pi}(\tilde{\beta}^B) | S_1, U_1]$. By minimizing the loss functions in (25), we can obtain the following estimators needed to construct estimators of action-value functions defined in (18), (19), and (22),

$$\widehat{\beta}^A = (\widehat{\beta}_a^A, \widehat{\beta}_z^A, \widehat{\beta}_{az}^A), \quad \widehat{\beta}^B = (\widehat{\beta}_a^B, \widehat{\beta}_z^B, \widehat{\beta}_{az}^B), \quad \widehat{\theta}_1^{B,j} = (\widehat{\theta}_1^{B,j}, \widehat{\gamma}_1^{B,j}, \widehat{\omega}_1^{B,j}) \quad \text{for any } j \in [3].\tag{26}$$

In this work, instead of directly plugging in the estimators defined in (26), we propose to use conservative estimators as proxies of action-value functions, and maximize such conservative action-value functions to learn an optimal policy. Such a method is also termed pessimism in related literature. Specifically, given the loss functions defined in (25), we define the following confidence regions,

$$\begin{aligned}\text{conf}^{\diamond,r} &= \left\{ \tilde{\beta}^\diamond = (\tilde{\beta}_a^\diamond, \tilde{\beta}_z^\diamond, \tilde{\beta}_{az}^\diamond) : \widehat{L}^{\diamond,r}(\tilde{\beta}^\diamond) - \widehat{L}^{\diamond,r}(\widehat{\beta}^\diamond) \leq \eta^{\diamond,r} \right\}, \\ \text{conf}_1^{B,j,\pi}(\tilde{\beta}^B) &= \left\{ \tilde{\theta}_1^{B,j} = (\tilde{\theta}_1^{B,j}, \tilde{\gamma}_1^{B,j}, \tilde{\omega}_1^{B,j}) : \widehat{L}_1^{B,j,\pi}(\tilde{\theta}_1^{B,j}, \tilde{\beta}^B) - \widehat{L}_1^{B,j,\pi}(\widehat{\theta}_1^{B,j}, \tilde{\beta}^B) \leq \eta_1^{B,j} \right\}, \\ \text{conf}_1^{B,\pi}(\tilde{\beta}^B) &= \left\{ \tilde{\theta}_1^B = (\tilde{\theta}_1^B, \tilde{\gamma}_1^B, \tilde{\omega}_1^B) \text{ satisfying (23): } \tilde{\theta}_1^{B,j} \in \text{conf}_1^{B,j,\pi}(\tilde{\beta}^B) \text{ for any } j \in [3] \right\},\end{aligned}\tag{27}$$

for any $j \in [3]$ and $\diamond \in \{A, B\}$. Here $\eta^{\diamond,r}$ and $\eta_1^{B,j}$ measure the size of the confidence regions, which will be specified later in (31). Then, we construct confidence regions for the action-value functions as follows,

$$\begin{aligned}\text{conf}_1^{A,Q,\pi} &= \left\{ Q_1^A = \tilde{\beta}_a^A A_1 + \tilde{\beta}_z^A B_{1/2} + \tilde{\beta}_{az}^A A_1 B_{1/2} : (\tilde{\beta}_a^A, \tilde{\beta}_z^A, \tilde{\beta}_{az}^A) \in \text{conf}^{A,r} \right\}, \\ \text{conf}_1^{B,Q,\pi} &= \bigcup_{\tilde{\beta}^B \in \text{conf}^{B,r}} \left\{ Q_1^B = \tilde{\theta}_1^B A_1 + \tilde{\gamma}_1^B B_{1/2} + \tilde{\omega}_1^B A_1 B_{1/2} : (\tilde{\theta}_1^B, \tilde{\gamma}_1^B, \tilde{\omega}_1^B) \in \text{conf}_1^{B,\pi}(\tilde{\beta}^B) \right\}.\end{aligned}\tag{28}$$

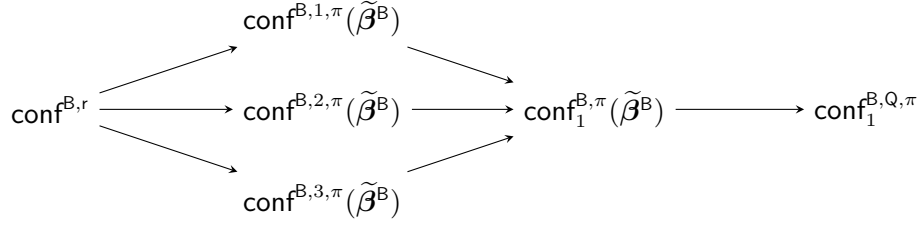


Figure 4 A graphical illustration of the construction of the confidence region $\text{conf}_1^{B,Q,\pi}$.

See Figure 4 for a graphical illustration.

We solve the following optimization problem, which maximizes the sum of the estimated expected total rewards of Alice and Bob,

$$(\hat{\pi}^A, \hat{\pi}^B) \in \arg \max_{\pi^A \in \Pi^A, \pi^B \in \Pi^B} \min_{Q_1^A \in \text{conf}_1^{A,Q,\pi}, Q_1^B \in \text{conf}_1^{B,Q,\pi}} \mathbb{E}_{\pi} \left[\sum_{\diamond \in \{A,B\}} Q_1^{\diamond}(S_1, A_1, B_{1/2}, U_1) \right], \quad (29)$$

where the confidence regions $\text{conf}_1^{A,Q,\pi}$ and $\text{conf}_1^{B,Q,\pi}$ are defined in (28). $(\hat{\pi}^A, \hat{\pi}^B)$ is denoted as our estimated policy pair.

4.3. Theoretical Results

In this section, we characterize the performance of the learned policy pair $(\hat{\pi}^A, \hat{\pi}^B)$ in (29). Specifically, we aim to upper bound the performance gap as follows,

$$\text{Gap}(\hat{\pi}^A, \hat{\pi}^B) = \sup_{\pi^A \in \Pi^A, \pi^B \in \Pi^B} J(\pi^A, \pi^B) - J(\hat{\pi}^A, \hat{\pi}^B), \quad (30)$$

where $J(\cdot, \cdot)$ is the expected total reward defined in (1). For the sizes of the confidence regions in (27), we let

$$\eta^{A,r} = O\left(n^{-\frac{2\alpha}{2\alpha+2\zeta+d}}\right), \quad \eta^{B,r} = O\left(n^{-\frac{2\alpha}{2\alpha+2\zeta+d}}\right), \quad \eta_1^{B,j} = O\left(n^{-\frac{2\alpha}{2\alpha+2\zeta+d}}\right) \quad (31)$$

for any $j \in [3]$.

ASSUMPTION 7. We have $\sigma_{\min}(\mathbb{E}[\Phi_1^{B,j,\pi^*}(\beta^B)^\top \Phi_1^{B,j,\pi^*}(\beta^B)^\top]) \geq c$ and $\sigma_{\min}(\mathbb{E}[\Phi^{\diamond,r\top} \Phi^{\diamond,r}]) \geq c$ for any $\diamond \in \{A, B\}$ and $j \in [3]$, where c is a positive absolute constant.

Assumption 7 only requires that the loss functions in (25) admit a unique minimizer under the distribution induced by the optimal policy pair π^* , which is indeed a weak assumption. For example, as long as statements in Assumption 7 hold under the data generating distribution and π^* is covered by the behavior policy, Assumption 7 holds as well. This is often called partial coverage in the literature of RL.

ASSUMPTION 8. The policy classes π^A and π^B have finite cardinalities.

For simplicity, in Assumption 8 we assume elements in π^A and π^B are finite. One can generalize to the setting where only assuming Vapnik–Chervonenkis indexes of π^A and π^B are finite. See Liao et al. (2022) for more details. Before showing our main theorem, we provide our identification result below.

COROLLARY 2. *Under Assumption 1 and similar orthogonal conditions in Assumption 2, one can use the offline data distribution to identify $Q_1^{A,\pi}(S_1, A_1, B_{1/2}, U_1)$, $Q_1^{B,\pi}(S_1, A_1, B_{1/2}, U_1)$ and eventually $J(\pi^A, \pi^B)$.*

Now we are ready to state our theorem, which upper bounds the performance gap in (30).

THEOREM 3. *Under Assumptions 1, 3–8, and similar sets of conditions as Assumption 2 (c), it holds that*

$$\text{Gap}(\hat{\pi}^A, \hat{\pi}^B) = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right).$$

Proof. We first introduce two supporting lemmas. The following lemma shows that with the sizes of the confidence regions in (31), the true action-value functions $Q_1^{A,\pi}$ and $Q_1^{B,\pi}$ lie in the confidence regions.

LEMMA 1. *Under the assumptions stated in Theorem 2, it holds that $\lim_{n \rightarrow \infty} \mathbb{P}(Q_1^{A,\pi} \in \text{conf}_1^{A,Q,\pi}, Q_1^{B,\pi} \in \text{conf}_1^{B,Q,\pi} \forall \pi \in \Pi) = 1$.*

Proof. See §E.1 for a detailed proof. ■

The following lemma shows that with the sizes of the confidence regions in (31), for any functions in the confidence regions, they are close to the true action-value functions $Q_1^{A,\pi}$ and $Q_1^{B,\pi}$.

LEMMA 2. *Under Assumption 7 and the assumptions stated in Theorem 2, for any $Q_1^A \in \text{conf}_1^{A,Q,\pi^*}$ and $Q_1^B \in \text{conf}_1^{B,Q,\pi^*}$, we have*

$$\left| \mathbb{E}_{\pi^*} \left[Q_1^{A,\pi^*} - Q_1^A \right] \right| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right), \quad \left| \mathbb{E}_{\pi^*} \left[Q_1^{B,\pi^*} - Q_1^B \right] \right| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right).$$

Proof. See §E.2 for a detailed proof. ■

Finally, it holds that

$$\begin{aligned} & J(\pi^{*,A}, \pi^{*,B}) - J(\hat{\pi}^A, \hat{\pi}^B) \\ &= \mathbb{E}_{\pi^*} \left[(Q_1^{A,\pi^*} + Q_1^{B,\pi^*})(S_1, A_1, B_{1/2}, U_1) \right] - \mathbb{E}_{\hat{\pi}} \left[(Q_1^{A,\hat{\pi}} + Q_1^{B,\hat{\pi}})(S_1, A_1, B_{1/2}, U_1) \right] \\ &\leq \mathbb{E}_{\pi^*} \left[(Q_1^{A,\pi^*} + Q_1^{B,\pi^*})(S_1, A_1, B_{1/2}, U_1) \right] - \min_{Q_1^A \in \text{conf}_1^{A,Q,\hat{\pi}}, Q_1^B \in \text{conf}_1^{B,Q,\hat{\pi}}} \mathbb{E}_{\hat{\pi}} \left[(Q_1^A + Q_1^B)(S_1, A_1, B_{1/2}, U_1) \right] \\ &\leq \mathbb{E}_{\pi^*} \left[(Q_1^{A,\pi^*} + Q_1^{B,\pi^*})(S_1, A_1, B_{1/2}, U_1) \right] - \min_{Q_1^A \in \text{conf}_1^{A,Q,\pi^*}, Q_1^B \in \text{conf}_1^{B,Q,\pi^*}} \mathbb{E}_{\pi^*} \left[(Q_1^A + Q_1^B)(S_1, A_1, B_{1/2}, U_1) \right] \\ &\leq \max_{Q_1^A \in \text{conf}_1^{A,Q,\pi^*}, Q_1^B \in \text{conf}_1^{B,Q,\pi^*}} \left| \mathbb{E}_{\pi^*} \left[(Q_1^{A,\pi^*} - Q_1^A + Q_1^{B,\pi^*} - Q_1^B)(S_1, A_1, B_{1/2}, U_1) \right] \right|, \end{aligned}$$

where in the first and second inequalities, we use Lemma 1 and the optimality of $(\hat{\pi}^A, \hat{\pi}^B)$, respectively. Further, by Lemma 2, we have

$$J(\pi^{*,A}, \pi^{*,B}) - J(\hat{\pi}^A, \hat{\pi}^B) = O_p \left(n^{-\frac{\alpha}{2\alpha+2\zeta+d}} \right),$$

which concludes the proof of the theorem. $\blacksquare$

In Theorem 3, with only partial coverage assumption and the help of convergence of the SMD estimators, the upper bound on the performance gap implies that the regret of finding an optimal in-class policy pair converges to 0 as long as the number of trajectories n goes to infinity. For standard policy learning methods without using pessimism, the regret upper bound will rely on the so-called uniform concentratability coefficient, which is much larger than that in our bound.

5. Policy Learning Method for Multi-Stage Game

In a multi-stage game with $H \geq 2$, the idea of policy evaluation is similar to the single-stage game in §4.1. Therefore details are deferred to §A.1 of the appendix. In the meanwhile, similar to the construction of confidence regions as in (28), we define the following confidence regions for the action-value function at the first step,

$$\text{conf}_1^{\diamond, Q, \pi} = \bigcup_{\substack{\tilde{\theta}^\diamond \in \text{conf}_h^{\diamond, \pi}(\tilde{\theta}_{h+1/2}^\diamond), \\ \text{for any } h \in \mathcal{C}^\diamond}} \left\{ Q_1^\diamond : \left(\tilde{\theta}_1^\diamond, \tilde{\gamma}_1^\diamond, \tilde{\omega}_1^\diamond \right) \in \text{conf}_1^{\diamond, \pi}(\tilde{\theta}_{3/2}^\diamond) \right\}, \quad (32)$$

where $Q_1^\diamond = \tilde{\theta}_1^\diamond A_1 + \tilde{\gamma}_1^\diamond B_{1/2} + \tilde{\omega}_1^\diamond A_1 B_{1/2}$ for any $\diamond \in \{A, B\}$, $\mathcal{C}^A = \{3/2, 2, \dots, H - 1/2\}$, $\mathcal{C}^B = \{3/2, 2, \dots, H\}$, and the confidence regions $\text{conf}^{\diamond, r}$ and $\text{conf}_h^{\diamond, \pi}$ are defined in (54) in §A.2 of the appendix. Figure 5 illustrates how $\text{conf}_1^{A, Q, \pi}$ is constructed for an example. We remark that the construct of $\text{conf}_1^{B, Q, \pi}$ can also be visualized in a similar way.

We solve the following optimization problem, which maximizes the sum of the estimated expected total rewards of Alice and Bob,

$$(\hat{\pi}^A, \hat{\pi}^B) \in \arg \max_{\pi^A \in \Pi^A, \pi^B \in \Pi^B} \min_{Q_1^\diamond \in \text{conf}_1^{\diamond, Q, \pi} \forall \diamond \in \{A, B\}} \mathbb{E}_\pi \left[\sum_{\diamond \in \{A, B\}} Q_1^\diamond(S_1, A_1, B_{1/2}, U_1) \right], \quad (33)$$

where the confidence regions $\text{conf}_1^{\diamond, Q, \pi}$ are defined in (32) for any $\diamond \in \{A, B\}$. While our pessimistic policy learning approach can conservatively estimate the optimal policy and prevent extrapolation in out-of-distribution, the above optimization problem is difficult to solve efficiently, especially when the time horizon grows. In our current implementation, we develop a heuristic zero-order type algorithm, which is not efficient when the state and parameter spaces are large, and the length of horizon is large. Moreover, a marketplace might have hundreds of thousands of items, a streaming

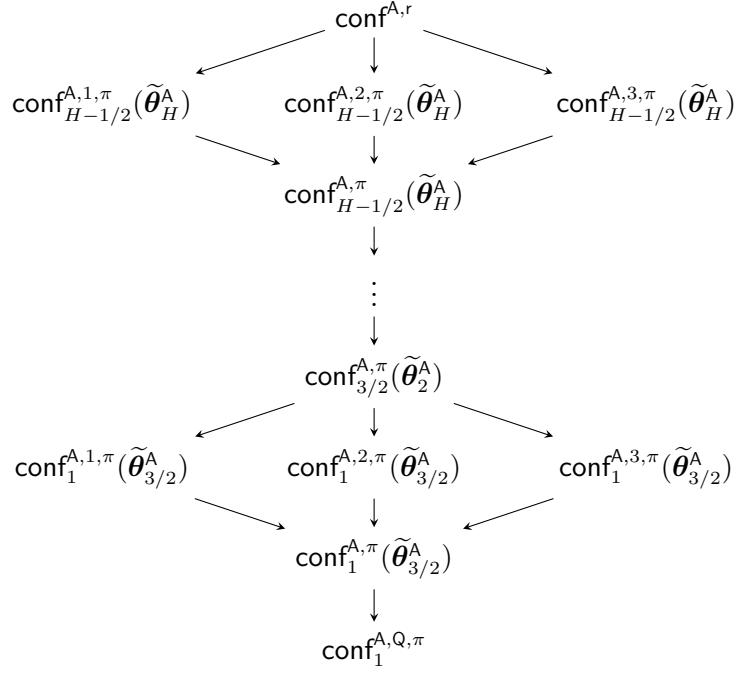


Figure 5 A graphical illustration of the construction of the confidence region $\text{conf}_1^{A,Q,\pi}$ in (32).

service might have millions of titles, and a social network might have billions of posts or ads. Our current methodology cannot handle this level of applications. One possible method is to use the penalization approach to replace the constraint set in Equation (33) and develop a gradient-type algorithm. We leave the development of an efficient algorithm as the future work. We present our algorithm in Algorithm 1. Note that for Lines 6 and 7 of Algorithm 1, under some special cases, we can efficiently solve the problems. See §6 for details.

5.1. Theoretical Results

In this section, we characterize the performance of the learned policy pair $(\hat{\pi}^A, \hat{\pi}^B)$ in (33). Specifically, we aim to upper bound the performance gap defined in (30). For the sizes of the confidence regions in (27), we let

$$\eta^{\diamond,r} = O\left(n^{-\frac{2\alpha}{2\alpha+2\zeta+d}}\right), \quad \eta_h^{\diamond,j} = O\left(H(H-h)^3 \cdot n^{-\frac{2\alpha}{2\alpha+2\zeta+d}}\right) \quad (34)$$

for any $j \in [3]$, $h \in [H] \cup [H]_{1/2}$, and $\diamond \in \{A, B\}$.

The following assumption ensures the loss functions in (54) admit unique minimizers only under the optimal policy pair π^* , which is indeed a weak assumption only requiring partial coverage. We remark that such an assumption is in parallel to Assumption 7 for single-stage games.

ASSUMPTION 9. *We have*

$$\begin{aligned} \sigma_{\min}(\mathbb{E}[\Phi_h^{\diamond,j,\pi^*}(\theta_{h+1/2}^\diamond, \mathbf{f}_h^\diamond)^\top \Phi_h^{\diamond,j,\pi^*}(\theta_{h+1/2}^\diamond, \mathbf{f}_h^\diamond)^\top]) &\geq c, \\ \sigma_{\min}(\mathbb{E}[\Phi^{\diamond,r}(\mathbf{f}^\diamond)^\top \Phi^{\diamond,r}(\mathbf{f}^\diamond)]) &\geq c \end{aligned}$$

Algorithm 1 Off-policy Policy Learning for Multi-Stage Human-Machine Interaction

- 1: Set an initial policy $\hat{\pi}_0$.
- 2: **for** $k = 0, \dots, K - 1$ **do**
- 3: Set $\hat{\theta}_{H+1}^\diamond(\cdot, \cdot; \hat{\pi}_k)$, $\hat{\gamma}_{H+1}^\diamond(\cdot, \cdot; \hat{\pi}_k)$, and $\hat{\omega}_{H+1}^\diamond(\cdot, \cdot; \hat{\pi}_k)$ as zero functions over $\mathcal{S} \times \mathcal{U}$ for any $\diamond \in \{\mathbf{A}, \mathbf{B}\}$.
- 4: Obtain $\hat{\theta}_h^\diamond(\cdot, \cdot; \hat{\pi}_k)$, $\hat{\gamma}_h^\diamond(\cdot, \cdot; \hat{\pi}_k)$, and $\hat{\omega}_h^\diamond(\cdot, \cdot; \hat{\pi}_k)$ for any $h \times \diamond \in ([H] \cup [H]_{1/2}) \times \{\mathbf{A}, \mathbf{B}\}$ by following §A.1 in a backward fashion.
- 5: Construct $\text{conf}^{\diamond, r}$ and $\text{conf}_h^{\diamond, \hat{\pi}_k}(\cdot)$ for any $h \times \diamond \in ([H] \cup [H]_{1/2}) \times \{\mathbf{A}, \mathbf{B}\}$ by following §A.2 in a backward fashion, and set $\text{conf}_1^{\diamond, Q, \hat{\pi}_k}$ as (32).
- 6: Solve the following minimization problem

$$\hat{J}_k = \min_{Q_1^\diamond \in \text{conf}_1^{\diamond, Q, \hat{\pi}_k} \forall \diamond \in \{\mathbf{A}, \mathbf{B}\}} \mathbb{E}_{\hat{\pi}_k} \left[\sum_{\diamond \in \{\mathbf{A}, \mathbf{B}\}} Q_1^\diamond(S_1, A_1, B_{1/2}, U_1) \right],$$

where $(S_1, B_{1/2}, U_1)$ is sampled from the offline data while $A_1 \sim \hat{\pi}_k^{\mathbf{A}}(S_1, U_1, B_{1/2})$.

- 7: Construct a gradient estimator $\hat{\nabla}_k \hat{J}_k$ and set $\hat{\pi}_{k+1} \leftarrow \hat{\pi}_k + \eta_k \hat{\nabla}_k \hat{J}_k$, where η_k is the stepsize.
 - 8: **end for**
-

for any $\diamond \in \{\mathbf{A}, \mathbf{B}\}$ and $j \in [3]$, where c is a positive absolute constant, and the matrices $\Phi_h^{\diamond, j, \pi^*}(\theta_{h+1/2}^\diamond, \mathbf{f}_h^\diamond)$ and $\Phi^{\diamond, r}(\mathbf{f}^\diamond)$ are defined in (51) in §A.2 of the appendix.

Now we are ready to state our theorem, which upper bounds the performance gap in (30).

THEOREM 4. *Under Assumptions 1, 9, and similar sets of conditions as Assumption 2 (c), Assumptions 3–6, it holds that*

$$\text{Gap}(\hat{\pi}^{\mathbf{A}}, \hat{\pi}^{\mathbf{B}}) = O_p \left(H^{2.5} \cdot n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right).$$

Proof. See §F.1 for a detailed proof. ■

Similar to Theorem 3, in Theorem 4, with only partial coverage assumption and the help of convergence of the SMD estimators, the upper bound on the performance gap implies that the regret of finding an optimal in-class policy pair converges to 0 as long as the number of trajectories n goes to infinity in a multi-stage game. Regarding the optimality of upper bound, we conjecture that H^2 and $n^{-\frac{\alpha}{2\alpha+2\varsigma+d}}$ may be sub-optimal because in the standard MDP/Markov Game, the minimax optimal regret should be linear in H and $1/\sqrt{n}$. However the optimal dependency in terms of H and n for solving MDP/Markov Game under unmeasured confounding has not been formally studied in the literature. Since this is beyond the scope of our paper, we leave it for future work.

To summarize this section, we establish a finite-sample regret bound for our proposed algorithm. Our theory establishes that as long as the sample size of our offline data is large enough, our algorithm can asymptotically recover the optimal policy. In other words, in our video recommendation example, by using our algorithm, we are able to improve the machine's recommendation system as long as the offline data are large enough. In the next section, we will use an empirical study to validate the effective performance of our method.

6. Numerical Simulation

To illustrate the efficacy of the proposed method with IVs, we conduct a simulation study. In the following, we introduce the setup of our numerical experiment.

Dynamics and Rewards. For the simplicity of the simulation, we assume that the related reward functions and transition functions take the following linear form, i.e., so-called linear MDP

$$\mathcal{G} = \{g(\cdot, \cdot, \cdot) : \exists \phi \text{ s.t. } g(u, v, s) = \phi^\top \text{concat}(u, v, s), \forall (u, v, s)\}, \quad (35)$$

where the feature vector $\text{concat}(u, v, s)$ is the concatenation of the vectors u , v , and s . With that said, we have $\alpha_z^\diamond, \alpha_{u,s}^\diamond, \beta_a^\diamond, \beta_{a,z}^\diamond, \beta_{u,s}^\diamond, \theta_h^f, \gamma_h^f, \omega_h^f, \iota_h^f \in \mathcal{G}$ for any $(\diamond, h, f) \in \{\mathbf{A}, \mathbf{B}\} \times ([H] \cup [H]_{1/2}) \times \mathbb{R}^{S \times \mathcal{U} \times \mathcal{V}}$. We detail those function in Appendix C.

Behavior Policy. For the simplicity of the simulation, the behavior policies of Alice and Bob are taken as follows with some constant $p = 0.2$. Recall that behavior policy is the probability distribution of choosing actions in the offline data generating process. Consider U_h as the internal popularity scores of top 2 movies. In particular, $U_h > 0$ refers to Movie 1 is more popular than Movie 0 at h decision point. Note that at each decision point, top 2 movies could be different. In addition, we consider $V_{h+1/2} > 0$ or $V_{h+1/2} \leq 0$ interpreted as two different types of mood, e.g., happy or not. We set the behavior policy as follows.

- if $U_h > 0$, i.e., Movie 1 is considered more popular than Movie 0:
 - if $B_{h-1/2} = 0$ i.e., Bob did not watch the previous recommended movie, we recommend Movie $A_h = a$ with probability p_a for any $a \in \{0, 1\}$, where $(p_0, p_1) = (p, 1 - p)$, i.e., with a higher probability of recommending Movie 1, which is more popular at this moment;
 - if $B_{h-1/2} = 1$, we take an opposite approach by recommending $A_h = a$ with probability p_a for any $a \in \{0, 1\}$, where $(p_0, p_1) = (1 - p, p)$;
- if $U_h \leq 0$, we take an opposite action of the scenario that $U_h > 0$.
- if $V_{h+1/2} > 0$, i.e., the customer is happy now,
 - if $A_h = 0$, i.e., Movie 0 is recommended, then the customer will choose to play the movie with probability p , i.e., $B_{h+1/2} = b$ with probability p_b for any $b \in \{0, 1\}$, where $(p_0, p_1) = (p, 1 - p)$;

—if $A_h = 1$ i.e., Movie 1 is recommended, then the customer will choose to play the movie with probability $1 - p$, i.e., $B_{h+1/2} = b$ with probability p_b for any $b \in \{0, 1\}$, where $(p_0, p_1) = (1 - p, p)$.

- if $V_{h+1/2} \leq 0$, i.e., the customer is not happy, the customer will take an opposite action of the scenario that $V_{h+1/2} > 0$.

Simulation Setup. We compare results under the following three settings.

- **iv:** This is the setting with the proposed algorithm in the paper.
- **full_obs:** Under this setting, we assume that private information U_h and V_h for any $h \in [H]_{1/2}$ are revealed to both Alice and Bob, so that Alice and Bob can utilize such information for decision making. Under such a setting, the problem degenerates to a regular two-player game, which can be solved efficiently to optimality.
- **w/o_iv:** Under this setting, we assume that Alice is not aware of the unmeasured confounder V_h for any $h \in [H]_{1/2}$; thus just do regression to solve (6).

Under the aforementioned setting, we assume that $\mathcal{S} = \mathcal{U} = \mathcal{V} = \mathbb{R}$ and we collect $n = 100$ trajectories with the length of horizon $H = 1$ following the behavior policies and the transition model, i.e., the episode terminates once the state S_2 is revealed to both players. Note that by (35), we only need to estimate the associated vectors of those functions in the policy evaluation phase, as the ground truth of the reward and transition functions are linear in $\text{concat}(u, v, s)$. Meanwhile, as v is the unmeasured confounder, we replace the feature vector by $\text{concat}(u, 1, s)$ (but still use $\text{concat}(u, v, s)$ in the setting **full_obs**). We assume there is an oracle to give the nuisance parameters in (11), which is achieved by Monte Carlo method. **We consider $H = 1$ because it includes all the basic components of our method and is enough to show the effective of our method.**

Implementation. We implement our proposed method in a bilevel optimization style, where the inner loop aims to find the pessimistic estimators, while the outer loop aims to maximize the estimated total reward associated with such pessimistic estimators.

For the inner loop, note that following the aforementioned simulation setup, the loss functions defined in (25) simplify to quadratic functions in the underlying parameters associated with $\mathcal{G}$ in (35). In the evaluation procedure, we adopt Adam (Kingma 2014) with a learning rate 0.05 and exponential decay rates (0.9, 0.999) to get the minimizers of the loss functions. To find the pessimistic estimators, we (i) randomly sample 10 sets of auxiliary estimators in the Euclidean balls of radius 0.01 centered at the minimizers, (ii) calculate their associated estimated total reward, and (iii) take the ones with the minimum estimated total reward as the pessimistic estimators.

For the outer loop, to maximize the estimated total reward associated with the obtained pessimistic estimators, we employ policy gradient method (Sutton et al. 1999). Specifically, we calculate the policy gradient associated with the estimated action-value functions, and plug in such

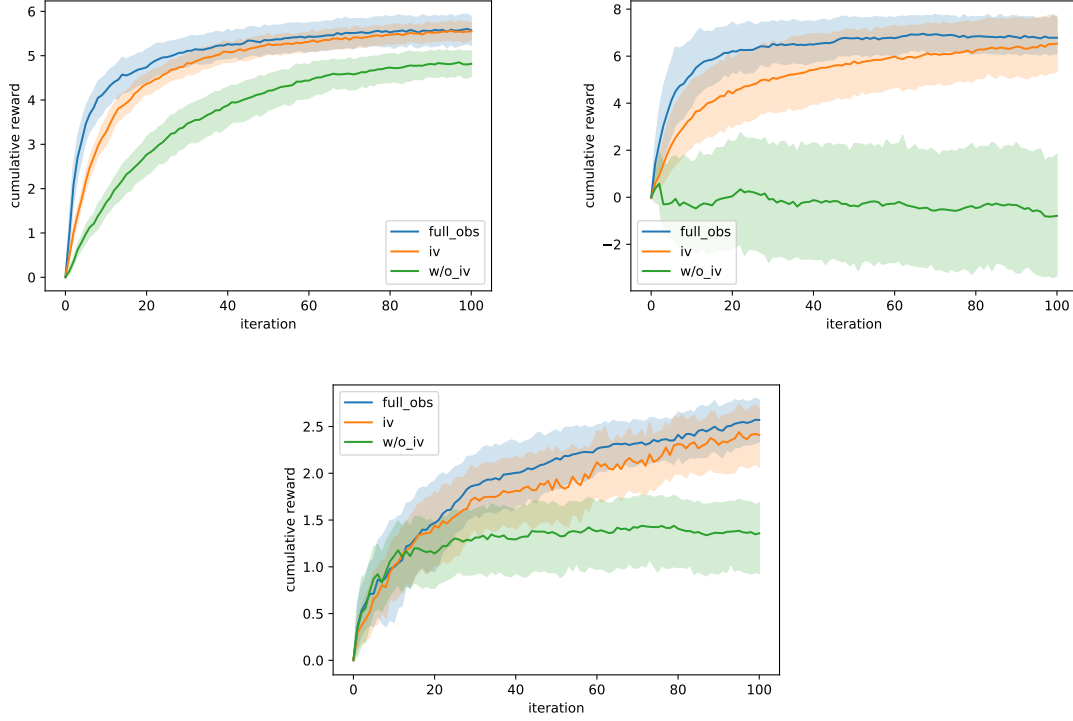


Figure 6 Cumulative rewards against iteration by using `iv`, `full_obs`, and `w/o_iv`. Here the solid lines are means of the cumulative rewards, while the shaded areas represent the corresponding 95% confidence intervals.

calculated policy gradient to an Adam optimizer with a learning rate 0.02 and exponential decay rates (0.9, 0.999).

As our implementation is only for demonstration purpose on the feasibility of the proposed method, it may not be very efficient. For a more practical implementation, we can adopt the pessimistic soft policy iteration method proposed in Xie et al. (2021), and we leave such an optimization as a potential future direction.

Numerical Results. We plot the results in Figure 6 with three sets of ground truth parameters, which are randomly generated; see Appendix C for details. Here in Figure 6, for each set of parameters, the solid lines represent the means, while the shaded areas represent 95% confidence intervals generated by 10 seeds. From Figure 6, we see that settings `iv` and `full_obs` have similar performance, while `w/o_iv` has poor performance. Such a sharp comparison demonstrates the effectiveness of using instrumental variables in our algorithm. Finally, it is remarked that our current simulation does not study how our method performs when there are more features, actions and decision making stages. This is due to the high cost of our optimization algorithm. Since the main focus of this paper is the proposed mathematical framework, identification results and the related theory, we leave the thorough numerical studies as the future work.

7. Conclusion

In this paper, we study human-guided human-machine interaction via a form of two-player turn-based games, where one player (a human) tries to guide another player (a machine) to maximize the sum of their expected total rewards together under the presence of private information of the machine. Specifically, we consider an offline setting where the data are pre-collected. Two key challenges are presented in solving this problem: (i) confounding biases induced by the private information, which cannot be accessed by the human; (ii) distributional shift in offline setting. To solve (i), we innovatively use machine's action as a natural instrumental variable, and establish a general non-parametric identification result for the causal effect of both action and IV, based on which we propose an estimator for off-policy evaluation. To tackle (ii), we leverage the idea of pessimism and propose an optimal policy pair estimator for both human and machine. Theoretically, a rate of convergence to an optimal policy pair is rigorously established.

For future research, it would be valuable to extend our framework to multi-agent cooperative games. For example, a recommendation system may need to suggest movies to multiple users simultaneously at each stage. In this scenario, different users could have varying utility functions based on their decision to watch or not. Developing an efficient algorithm to enhance the recommendation system under these conditions presents a significant challenge. One potential approach is to consider a general Stackelberg game, where the upper level optimization is to maximize the recommendation system's utility function involving users' actions, and the lower level solves an optimization problem to get users' actions based on their utility function. Furthermore, our method could be applied to other contexts, such as cooperation between a company's marketing and sales departments. For instance, a data science team might aim to improve an AI system (Alice) that suggests marketing campaigns, in conjunction with a pricing strategy employed by the sales team (Bob), to collectively enhance customer satisfaction and profitability. To achieve this shared objective, it is crucial to understand the causal effect of the marketing campaign on the final outcome. However, this effect may be confounded by hidden factors, such as unrecorded communication between the sales team and customers. Our proposed framework and method could help address this challenge. Finally, it would be interesting to apply our method in solving the real-world video recommendation problem. Specifically, we first need to collect historical data of video recommendation to different customers and obtain as many static and time-varying features as possible. Secondly, since our method allows non-parametric models, we could use either linear sieve or neural network models in our algorithm to estimate some unknown functions. Once we obtain an estimated policy by implementing the proposed method, we can deploy it to the real-world environment and conduct A/B testing to verify the effectiveness of the estimated policy, compared with the previous recommendation policy.

Appendix A: Detailed Method for Multi-Stage Game

A.1. Evaluation of Expected Total Rewards

For any policy pair $\pi = (\pi^A, \pi^B)$, we estimate $J^A(\pi^A, \pi^B)$ and $J^B(\pi^A, \pi^B)$ in a backward fashion. We introduce the estimation procedure of $Q_h^{A,\pi}$, $Q_{h+1/2}^{A,\pi}$, $Q_h^{B,\pi}$, and $Q_{h+1/2}^{B,\pi}$ for any $h \in [H]$ as follows.

Estimation of $Q_h^{A,\pi}$. For any $h \in [H]$, with an estimator $\hat{Q}_{h+1/2}^A(\cdot, \cdot, \cdot, \cdot; \pi)$, we estimate $Q_h^{A,\pi}(\cdot, \cdot, \cdot, \cdot)$ as follows. Note that by the memoryless assumption of $V_{h+1/2}$, the following Bellman equation holds,

$$\begin{aligned} Q_h^{A,\pi}(S_h, A_h, B_{h-1/2}, U_h, V_h) &= \mathbb{E}_\pi [R_h^A + Q_{h+1/2}^{A,\pi}(S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2})] \\ &= \mathbb{E} [R_h^A] + \mathbb{E} [\hat{\theta}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi)] \cdot A_h \\ &\quad + \mathbb{E} [\hat{\gamma}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) \cdot \pi_{h+1/2}^B(S_{h+1/2})] \\ &\quad + \mathbb{E} [\hat{\omega}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) \cdot \pi_{h+1/2}^B(S_{h+1/2})] \cdot A_h. \end{aligned} \quad (36)$$

Note that the above expectations are taken conditioning on S_h , A_h , $B_{h-1/2}$, U_h , and V_h . Thus, to construct an estimator of $Q_h^{A,\pi}$, we only need to estimate the right-hand side of (36). Following from (17), we can estimate the first term $\mathbb{E}[R_h^A]$ on the right-hand side of (36) as

$$\hat{\beta}_a^A(S_h, U_h)A_h + \hat{\beta}_z^A(S_h, U_h)B_{h-1/2} + \hat{\beta}_{az}^A(S_h, U_h)A_hB_{h-1/2}. \quad (37)$$

Note that the conditional expectation of $\beta_{u,s}^A(U_h, V_h, S_h)$ is zero by Assumption 1, thus we have the above formulation. For the remaining three terms on the right-hand side of (36), since we have estimators $\hat{\theta}_{h+1/2}^A$, $\hat{\gamma}_{h+1/2}^A$, and $\hat{\omega}_{h+1/2}^A$ for $\theta_{h+1/2}^A$, $\gamma_{h+1/2}^A$, and $\omega_{h+1/2}^A$, respectively from the estimator $\hat{Q}_{h+1/2}^A$, we only need to estimate the following three terms,

$$\begin{aligned} (I) &= \mathbb{E} [\hat{\theta}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) | S_h, A_h, B_{h-1/2}, U_h, V_h], \\ (II) &= \mathbb{E} [\hat{\gamma}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) \cdot \pi_{h+1/2}^B(S_{h+1/2}) | S_h, A_h, B_{h-1/2}, U_h, V_h], \\ (III) &= \mathbb{E} [\hat{\omega}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) \cdot \pi_{h+1/2}^B(S_{h+1/2}) | S_h, A_h, B_{h-1/2}, U_h, V_h]. \end{aligned}$$

By Assumption 1 (e) and the model specified in (3), we know that there exist functions $\{\theta_h^{A,j}, \gamma_h^{A,j}, \omega_h^{A,j}\}_{j \in [3]}$ such that

$$\begin{aligned} (I) &= \theta_h^{A,1}(S_h, U_h, V_h; \pi) \cdot A_h + \gamma_h^{A,1}(S_h, U_h, V_h; \pi) \cdot B_{h-1/2} + \omega_h^{A,1}(S_h, U_h, V_h; \pi) \cdot A_hB_{h-1/2}, \\ (II) &= \theta_h^{A,2}(S_h, U_h, V_h; \pi) \cdot A_h + \gamma_h^{A,2}(S_h, U_h, V_h; \pi) \cdot B_{h-1/2} + \omega_h^{A,2}(S_h, U_h, V_h; \pi) \cdot A_hB_{h-1/2}, \\ (III) &= \theta_h^{A,3}(S_h, U_h, V_h; \pi) \cdot A_h + \gamma_h^{A,3}(S_h, U_h, V_h; \pi) \cdot B_{h-1/2} + \omega_h^{A,3}(S_h, U_h, V_h; \pi) \cdot A_hB_{h-1/2}, \end{aligned} \quad (38)$$

Following from a similar idea as in (21) by integrating out V_h in (38), we construct the following estimators for the above terms,

$$\begin{aligned} &\hat{\theta}_h^{A,1}(S_h, U_h; \pi) \cdot A_h + \hat{\gamma}_h^{A,1}(S_h, U_h; \pi) \cdot B_{h-1/2} + \hat{\omega}_h^{A,1}(S_h, U_h; \pi) \cdot A_hB_{h-1/2}, \\ &\hat{\theta}_h^{A,2}(S_h, U_h; \pi) \cdot A_h + \hat{\gamma}_h^{A,2}(S_h, U_h; \pi) \cdot B_{h-1/2} + \hat{\omega}_h^{A,2}(S_h, U_h; \pi) \cdot A_hB_{h-1/2}, \\ &\hat{\theta}_h^{A,3}(S_h, U_h; \pi) \cdot A_h + \hat{\gamma}_h^{A,3}(S_h, U_h; \pi) \cdot B_{h-1/2} + \hat{\omega}_h^{A,3}(S_h, U_h; \pi) \cdot A_hB_{h-1/2}, \end{aligned} \quad (39)$$

respectively. Combining (36), (37), and (39), we obtain the following estimator of $Q_h^{A,\pi}$,

$$\begin{aligned} \hat{Q}_h^A(S_h, A_h, B_{h-1/2}, U_h; \pi) \\ = \hat{\theta}_h^A(S_h, U_h; \pi) \cdot A_h + \hat{\gamma}_h^A(S_h, U_h; \pi) \cdot B_{h-1/2} + \hat{\omega}_h^A(S_h, U_h; \pi) \cdot A_hB_{h-1/2}, \end{aligned} \quad (40)$$

where

$$\begin{aligned}\hat{\theta}_h^A(S_h, U_h; \pi) &= \hat{\beta}_a^A(U_h, S_h) + \hat{\theta}_h^{A,1}(S_h, U_h; \pi) + \hat{\theta}_h^{A,2}(S_h, U_h; \pi) + \hat{\theta}_h^{A,3}(S_h, U_h; \pi), \\ \hat{\gamma}_h^A(S_h, U_h; \pi) &= \hat{\beta}_z^A(U_h, S_h) + \hat{\gamma}_h^{A,2}(S_h, U_h; \pi), \\ \hat{\omega}_h^A(S_h, U_h; \pi) &= \hat{\beta}_{az}^A(U_h, S_h) + \hat{\omega}_h^{A,1}(S_h, U_h; \pi) + \hat{\omega}_h^{A,2}(S_h, U_h; \pi) + \hat{\omega}_h^{A,3}(S_h, U_h; \pi) \\ &\quad + \hat{\gamma}_h^{A,1}(S_h, U_h; \pi) + \hat{\gamma}_h^{A,3}(S_h, U_h; \pi).\end{aligned}\tag{41}$$

Estimation of $Q_{h+1/2}^{A,\pi}$. For any $h \in [H]$, we estimate $Q_{h+1/2}^{A,\pi}(\cdot, \cdot, \cdot, \cdot; \pi)$ as follows. Since we have estimators $\hat{\theta}_{h+1}^A$, $\hat{\gamma}_{h+1}^A$, and $\hat{\omega}_{h+1}^A$ of $\hat{\theta}_{h+1}^A$, $\hat{\gamma}_{h+1}^A$, and $\hat{\omega}_{h+1}^A$, respectively from the estimator $\hat{Q}_{h+1}^A$, we first estimate the following three terms,

$$\begin{aligned}\mathbb{E} \left[\hat{\theta}_{h+1}^A(S_{h+1}, U_{h+1}^A; \pi) \cdot \pi_{h+1}^A(S_{h+1}, U_{h+1}^A) \mid S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2} \right], \\ \mathbb{E} \left[\hat{\gamma}_{h+1}^A(S_{h+1}, U_{h+1}^A; \pi) \mid S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2} \right], \\ \mathbb{E} \left[\hat{\omega}_{h+1}^A(S_{h+1}, U_{h+1}^A; \pi) \cdot \pi_{h+1}^A(S_{h+1}, U_{h+1}^A) \mid S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2} \right]\end{aligned}$$

by

$$\begin{aligned}\hat{\theta}_{h+1/2}^{A,1}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h + \hat{\gamma}_{h+1/2}^{A,1}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot B_{h+1/2} \\ + \hat{\omega}_{h+1/2}^{A,1}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h B_{h+1/2}, \\ \hat{\theta}_{h+1/2}^{A,2}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h + \hat{\gamma}_{h+1/2}^{A,2}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot B_{h+1/2} \\ + \hat{\omega}_{h+1/2}^{A,2}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h B_{h+1/2}, \\ \hat{\theta}_{h+1/2}^{A,3}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h + \hat{\gamma}_{h+1/2}^{A,3}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot B_{h+1/2} \\ + \hat{\omega}_{h+1/2}^{A,3}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h B_{h+1/2},\end{aligned}\tag{42}$$

respectively. We remark that the above estimators in (42) are constructed following from a similar idea as in (39). Similar as in (40), we obtain the following estimator of $Q_{h+1/2}^{A,\pi}$,

$$\begin{aligned}\hat{Q}_{h+1/2}^A(S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}; \pi) \\ = \hat{\theta}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h + \hat{\gamma}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) \cdot B_{h+1/2} \\ + \hat{\omega}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h B_{h+1/2}.\end{aligned}\tag{43}$$

where

$$\begin{aligned}\hat{\theta}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) &= \hat{\theta}_{h+1/2}^{A,1}(S_{h+1/2}, U_{h+1/2}; \pi), \\ \hat{\gamma}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) &= \hat{\gamma}_{h+1/2}^{A,1}(S_{h+1/2}, U_{h+1/2}; \pi) + \hat{\gamma}_{h+1/2}^{A,2}(S_{h+1/2}, U_{h+1/2}; \pi) \\ &\quad + \hat{\gamma}_{h+1/2}^{A,3}(S_{h+1/2}, U_{h+1/2}; \pi), \\ \hat{\omega}_{h+1/2}^A(S_{h+1/2}, U_{h+1/2}; \pi) &= \hat{\omega}_{h+1/2}^{A,1}(S_{h+1/2}, U_{h+1/2}; \pi) + \hat{\omega}_{h+1/2}^{A,2}(S_{h+1/2}, U_{h+1/2}; \pi) \\ &\quad + \hat{\omega}_{h+1/2}^{A,3}(S_{h+1/2}, U_{h+1/2}; \pi) + \hat{\theta}_{h+1/2}^{A,2}(S_{h+1/2}, U_{h+1/2}; \pi) \\ &\quad + \hat{\theta}_{h+1/2}^{A,3}(S_{h+1/2}, U_{h+1/2}; \pi).\end{aligned}\tag{44}$$

Estimation of $Q_h^{B,\pi}$. For any $h \in [H]$, with an estimator $\hat{Q}_{h+1/2}^B(\cdot, \cdot, \cdot, \cdot; \pi)$, we estimate $Q_h^{B,\pi}(\cdot, \cdot, \cdot, \cdot)$ as follows. Since we have estimators $\hat{\theta}_{h+1/2}^B$, $\hat{\gamma}_{h+1/2}^B$, and $\hat{\omega}_{h+1/2}^B$ for $\hat{\theta}_{h+1/2}^B$, $\hat{\gamma}_{h+1/2}^B$, and $\hat{\omega}_{h+1/2}^B$, respectively from the estimator $\hat{Q}_{h+1/2}^B$, we first estimate the following three terms,

$$\mathbb{E} \left[\hat{\theta}_{h+1/2}^B(S_{h+1/2}, U_{h+1/2}; \pi) \mid S_h, A_h, B_{h-1/2}, U_h, V_h \right],$$

$$\begin{aligned} & \mathbb{E} [\hat{\gamma}_{h+1/2}^B(S_{h+1/2}, U_{h+1/2}; \pi) \cdot \pi_{h+1/2}^B(S_{h+1/2}) \mid S_h, A_h, B_{h-1/2}, U_h, V_h], \\ & \mathbb{E} [\hat{\omega}_{h+1/2}^B(S_{h+1/2}, U_{h+1/2}; \pi) \cdot \pi_{h+1/2}^B(S_{h+1/2}) \mid S_h, A_h, B_{h-1/2}, U_h, V_h] \end{aligned}$$

by

$$\begin{aligned} & \hat{\theta}_h^{B,1}(S_h, U_h; \pi) \cdot A_h + \hat{\gamma}_h^{B,1}(S_h, U_h; \pi) \cdot B_{h-1/2} + \hat{\omega}_h^{B,1}(S_h, U_h; \pi) \cdot A_h B_{h-1/2}, \\ & \hat{\theta}_h^{B,2}(S_h, U_h; \pi) \cdot A_h + \hat{\gamma}_h^{B,2}(S_h, U_h; \pi) \cdot B_{h-1/2} + \hat{\omega}_h^{B,2}(S_h, U_h; \pi) \cdot A_h B_{h-1/2}, \\ & \hat{\theta}_h^{B,3}(S_h, U_h; \pi) \cdot A_h + \hat{\gamma}_h^{B,3}(S_h, U_h; \pi) \cdot B_{h-1/2} + \hat{\omega}_h^{B,3}(S_h, U_h; \pi) \cdot A_h B_{h-1/2}, \end{aligned} \quad (45)$$

respectively. We remark that the above estimators in (45) are constructed following from a similar idea as in (39). Similar as in (40), we obtain the following estimator of $Q_h^{B,\pi}$,

$$\begin{aligned} & \hat{Q}_h^B(S_h, A_h, B_{h-1/2}, U_h; \pi) \\ & = \hat{\theta}_h^B(S_h, U_h; \pi) \cdot A_h + \hat{\gamma}_h^B(S_h, U_h; \pi) \cdot B_{h-1/2} + \hat{\omega}_h^B(S_h, U_h; \pi) \cdot A_h B_{h-1/2}, \end{aligned} \quad (46)$$

where

$$\begin{aligned} & \hat{\theta}_h^B(S_h, U_h; \pi) = \hat{\theta}_h^{B,1}(S_h, U_h; \pi) + \hat{\theta}_h^{B,2}(S_h, U_h; \pi) + \hat{\theta}_h^{B,3}(S_h, U_h; \pi), \\ & \hat{\gamma}_h^B(S_h, U_h; \pi) = \hat{\gamma}_h^{B,2}(S_h, U_h; \pi), \\ & \hat{\omega}_h^B(S_h, U_h; \pi) = \hat{\omega}_h^{B,1}(S_h, U_h; \pi) + \hat{\omega}_h^{B,2}(S_h, U_h; \pi) + \hat{\omega}_h^{B,3}(S_h, U_h; \pi) \\ & \quad + \hat{\gamma}_h^{B,1}(S_h, U_h; \pi) + \hat{\gamma}_h^{B,3}(S_h, U_h; \pi). \end{aligned} \quad (47)$$

Estimation of $Q_{h+1/2}^{B,\pi}$. For any $h \in [H]$, we estimate $Q_{h+1/2}^{B,\pi}(\cdot, \cdot, \cdot, \cdot)$ as follows. Since we have estimators $\hat{\theta}_{h+1}^B$, $\hat{\gamma}_{h+1}^B$, and $\hat{\omega}_{h+1}^B$ of $\bar{\theta}_{h+1}^B$, $\bar{\gamma}_{h+1}^B$, and $\bar{\omega}_{h+1}^B$, respectively from the estimator $\hat{Q}_{h+1}^B$, we first estimate the following terms,

$$\begin{aligned} & \mathbb{E} [R_{h+1/2}^B \mid S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2}], \\ & \mathbb{E} [\hat{\theta}_{h+1}^B(S_{h+1}, U_{h+1}^A; \pi) \cdot \pi_{h+1}^B(S_{h+1}, U_{h+1}^A) \mid S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2}], \\ & \mathbb{E} [\hat{\gamma}_{h+1}^B(S_{h+1}, U_{h+1}^A; \pi) \mid S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2}], \\ & \mathbb{E} [\hat{\omega}_{h+1}^B(S_{h+1}, U_{h+1}^A; \pi) \cdot \pi_{h+1}^B(S_{h+1}, U_{h+1}^A) \mid S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}, V_{h+1/2}] \end{aligned}$$

by

$$\begin{aligned} & \hat{\beta}_a^B(S_{h+1/2}, U_{h+1/2}) A_h + \hat{\beta}_z^B(S_{h+1/2}, U_{h+1/2}) B_{h+1/2} + \hat{\beta}_{az}^B(S_{h+1/2}, U_{h+1/2}) A_h B_{h+1/2}, \\ & \hat{\theta}_{h+1/2}^{B,1}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h + \hat{\gamma}_{h+1/2}^{B,1}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot B_{h+1/2} \\ & \quad + \hat{\omega}_{h+1/2}^{B,1}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h B_{h+1/2}, \\ & \hat{\theta}_{h+1/2}^{B,2}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h + \hat{\gamma}_{h+1/2}^{B,2}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot B_{h+1/2} \\ & \quad + \hat{\omega}_{h+1/2}^{B,2}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h B_{h+1/2}, \\ & \hat{\theta}_{h+1/2}^{B,3}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h + \hat{\gamma}_{h+1/2}^{B,3}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot B_{h+1/2} \\ & \quad + \hat{\omega}_{h+1/2}^{B,3}(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h B_{h+1/2}, \end{aligned} \quad (48)$$

respectively. We remark that the above estimators in (48) are constructed following from a similar idea as in (39). Similar as in (40), we obtain the following estimator of $Q_{h+1/2}^{B,\pi}$,

$$\begin{aligned} & \hat{Q}_{h+1/2}^B(S_{h+1/2}, A_h, B_{h+1/2}, U_{h+1/2}; \pi) \\ & = \hat{\theta}_{h+1/2}^B(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h + \hat{\gamma}_{h+1/2}^B(S_{h+1/2}, U_{h+1/2}; \pi) \cdot B_{h+1/2} \\ & \quad + \hat{\omega}_{h+1/2}^B(S_{h+1/2}, U_{h+1/2}; \pi) \cdot A_h B_{h+1/2}. \end{aligned} \quad (49)$$

where

$$\begin{aligned}
\hat{\theta}_{h+1/2}^B(S_{h+1/2}, U_{h+1/2}; \pi) &= \hat{\beta}_a^B(S_{h+1/2}, U_{h+1/2}) + \hat{\theta}_{h+1/2}^{B,1}(S_{h+1/2}, U_{h+1/2}; \pi), \\
\hat{\gamma}_{h+1/2}^B(S_{h+1/2}, U_{h+1/2}; \pi) &= \hat{\beta}_z^B(S_{h+1/2}, U_{h+1/2}) + \hat{\gamma}_{h+1/2}^{B,1}(S_{h+1/2}, U_{h+1/2}; \pi) \\
&\quad + \hat{\gamma}_{h+1/2}^{B,2}(S_{h+1/2}, U_{h+1/2}; \pi) + \hat{\gamma}_{h+1/2}^{B,3}(S_{h+1/2}, U_{h+1/2}; \pi), \\
\hat{\omega}_{h+1/2}^B(S_{h+1/2}, U_{h+1/2}; \pi) &= \hat{\beta}_{az}^B(S_{h+1/2}, U_{h+1/2}) + \hat{\omega}_{h+1/2}^{B,1}(S_{h+1/2}, U_{h+1/2}; \pi) \\
&\quad + \hat{\omega}_{h+1/2}^{B,2}(S_{h+1/2}, U_{h+1/2}; \pi) + \hat{\omega}_{h+1/2}^{B,3}(S_{h+1/2}, U_{h+1/2}; \pi) \\
&\quad + \hat{\theta}_{h+1/2}^{B,2}(S_{h+1/2}, U_{h+1/2}; \pi) + \hat{\theta}_{h+1/2}^{B,3}(S_{h+1/2}, U_{h+1/2}; \pi).
\end{aligned} \tag{50}$$

Estimation of J^A and J^B . We estimate $J^A(\pi^A, \pi^B)$ as

$$\hat{J}^A(\pi^A, \pi^B) = \mathbb{E}_\pi \left[\hat{Q}_1^A(S_1, A_1^A, A_{1/2}^B, U_1; \pi) \right],$$

where the expectation $\mathbb{E}_\pi[\cdot]$ is taken with respect to $A_{1/2}^B \sim \pi_{1/2}^B(\cdot)$, $S_1 \sim \nu(\cdot)$, U_1 , and $A_1^A \sim \pi_1^A(\cdot | S_1, U_1)$. Similarly, we construct an estimator $\hat{J}^B(\pi^A, \pi^B)$ for $J^B(\pi^A, \pi^B)$.

A.2. Construction of Confidence Regions

In a multi-stage game with $H \geq 1$, with a similar evaluation procedure proposed in §A.1, we present our method to find an optimal policy in this section. To begin with, in light of (57) in §B of the appendix, we have the following equations

$$\begin{aligned}
W^{\diamond, r}(\tilde{\mathbf{f}}^\diamond) &= \begin{pmatrix} \Phi^{\diamond, r}(\tilde{\mathbf{f}}^\diamond) \beta^\diamond \\ \varphi^{\diamond, r}(\tilde{\mathbf{f}}^\diamond) \end{pmatrix} + \alpha^{\diamond, r}(\tilde{\mathbf{f}}^\diamond), \\
W_h^{\diamond, j, \pi}(\tilde{\theta}_{h+1/2}^\diamond, \tilde{\mathbf{f}}_h^\diamond) &= \begin{pmatrix} \Phi_h^{\diamond, j, \pi}(\tilde{\theta}_{h+1/2}^\diamond, \tilde{\mathbf{f}}_h^\diamond) \theta_h^{\diamond, j, \pi}(\tilde{\theta}_{h+1/2}^\diamond) \\ \varphi_h^{\diamond, j, \pi}(\tilde{\theta}_{h+1/2}^\diamond, \tilde{\mathbf{f}}_h^\diamond) \end{pmatrix} + \alpha_h^{\diamond, j, \pi}(\tilde{\theta}_{h+1/2}^\diamond, \tilde{\mathbf{f}}_h^\diamond), \\
W_{h+1/2}^{\diamond, j, \pi}(\tilde{\theta}_{h+1}^\diamond, \tilde{\mathbf{f}}_{h+1/2}^\diamond) &= \begin{pmatrix} \Phi_{h+1/2}^{\diamond, j, \pi}(\tilde{\theta}_{h+1}^\diamond, \tilde{\mathbf{f}}_{h+1/2}^\diamond) \theta_{h+1/2}^{\diamond, j, \pi}(\tilde{\theta}_{h+1}^\diamond) \\ \varphi_{h+1/2}^{\diamond, j, \pi}(\tilde{\theta}_{h+1}^\diamond, \tilde{\mathbf{f}}_{h+1/2}^\diamond) \end{pmatrix} + \alpha_{h+1/2}^{\diamond, j, \pi}(\tilde{\theta}_{h+1}^\diamond, \tilde{\mathbf{f}}_{h+1/2}^\diamond),
\end{aligned} \tag{51}$$

where

$$\begin{aligned}
&\Phi^{\diamond, r}(\tilde{\mathbf{f}}^\diamond), \Phi_h^{\diamond, j, \pi}(\tilde{\theta}_{h+1/2}^\diamond, \tilde{\mathbf{f}}_h^\diamond), \Phi_{h+1/2}^{\diamond, j, \pi}(\tilde{\theta}_{h+1}^\diamond, \tilde{\mathbf{f}}_{h+1/2}^\diamond) \in \mathbb{R}^{3 \times 3}, \\
&\varphi^{\diamond, r}(\tilde{\mathbf{f}}^\diamond), \varphi_h^{\diamond, j, \pi}(\tilde{\theta}_{h+1/2}^\diamond, \tilde{\mathbf{f}}_h^\diamond), \varphi_{h+1/2}^{\diamond, j, \pi}(\tilde{\theta}_{h+1}^\diamond, \tilde{\mathbf{f}}_{h+1/2}^\diamond) \in \mathbb{R}^5, \\
&\alpha^{\diamond, r}(\tilde{\mathbf{f}}^\diamond), \alpha_h^{\diamond, j, \pi}(\tilde{\theta}_{h+1/2}^\diamond, \tilde{\mathbf{f}}_h^\diamond), \alpha_{h+1/2}^{\diamond, j, \pi}(\tilde{\theta}_{h+1}^\diamond, \tilde{\mathbf{f}}_{h+1/2}^\diamond) \in \mathbb{R}^3
\end{aligned}$$

for any $\tilde{\mathbf{f}}^\diamond, \tilde{\mathbf{f}}_h^\diamond, \tilde{\mathbf{f}}_{h+1/2}^\diamond \in \mathcal{H}^5$, $\tilde{\theta}_{h+1/2}^\diamond, \tilde{\theta}_{h+1}^\diamond \in \mathcal{H}^3$, $\diamond \in \{A, B\}$, $j \in [3]$, and $h \in [H]$. Here $W^{\diamond, r}(\tilde{\mathbf{f}}^\diamond)$, $W_h^{\diamond, j, \pi}(\tilde{\theta}_{h+1/2}^\diamond, \tilde{\mathbf{f}}_h^\diamond)$, and $W_{h+1/2}^{\diamond, j, \pi}(\tilde{\theta}_{h+1}^\diamond, \tilde{\mathbf{f}}_{h+1/2}^\diamond)$ are constructed via (55). With (51), we further define the loss functions for the construction of estimators in (40), (43), (46), and (49) as follows,

$$\begin{aligned}
\hat{L}^{\diamond, r}(\tilde{\beta}^\diamond; \tilde{\mathbf{f}}^\diamond) &= \frac{1}{n} \sum_{i=1}^n \hat{m}^{i, A, r}(\tilde{\beta}^\diamond; \tilde{\mathbf{f}}^\diamond)^\top \hat{m}^{i, A, r}(\tilde{\beta}^\diamond; \tilde{\mathbf{f}}^\diamond), \\
\hat{L}_h^{\diamond, j, \pi}(\tilde{\theta}_h^{\diamond, j}, \tilde{\theta}_{h+1/2}^\diamond; \tilde{\mathbf{f}}_h^\diamond) &= \frac{1}{n} \sum_{i=1}^n \hat{m}_h^{i, B, j, \pi}(\tilde{\theta}_h^{\diamond, j}, \tilde{\theta}_{h+1/2}^\diamond; \tilde{\mathbf{f}}_h^\diamond)^\top \hat{m}_h^{i, B, j, \pi}(\tilde{\theta}_h^{\diamond, j}, \tilde{\theta}_{h+1/2}^\diamond; \tilde{\mathbf{f}}_h^\diamond), \\
\hat{L}_{h+1/2}^{\diamond, j, \pi}(\tilde{\theta}_{h+1/2}^{\diamond, j}, \tilde{\theta}_{h+1}^\diamond; \tilde{\mathbf{f}}_{h+1/2}^\diamond) &= \frac{1}{n} \sum_{i=1}^n \hat{m}_{h+1/2}^{i, B, j, \pi}(\tilde{\theta}_{h+1/2}^{\diamond, j}, \tilde{\theta}_{h+1}^\diamond; \tilde{\mathbf{f}}_{h+1/2}^\diamond)^\top \hat{m}_{h+1/2}^{i, B, j, \pi}(\tilde{\theta}_{h+1/2}^{\diamond, j}, \tilde{\theta}_{h+1}^\diamond; \tilde{\mathbf{f}}_{h+1/2}^\diamond),
\end{aligned} \tag{52}$$

where $\widehat{m}^{i,A,r}(\widetilde{\beta}^\diamond; \widetilde{\mathbf{f}}^\diamond)$, $\widehat{m}_h^{i,B,j,\pi}(\widetilde{\theta}_h^{\diamond,j}, \widetilde{\theta}_{h+1/2}^\diamond; \widetilde{\mathbf{f}}_h^\diamond)$, and $\widehat{m}_{h+1/2}^{i,B,j,\pi}(\widetilde{\theta}_{h+1/2}^{\diamond,j}, \widetilde{\theta}_{h+1}^\diamond; \widetilde{\mathbf{f}}_{h+1/2}^\diamond)$ are consistent estimators of (51). By minimizing the loss functions in (52), for any policy pair π , we can obtain the following estimators needed to construct estimators of action-value functions defined in (40), (43), (46), and (49),

$$\widehat{\beta}^\diamond = (\widehat{\beta}_a^\diamond, \widehat{\beta}_z^\diamond, \widehat{\beta}_{az}^\diamond), \quad \widehat{\theta}_h^{\diamond,j,\pi} = (\widehat{\theta}_h^{\diamond,j,\pi}, \widehat{\gamma}_h^{\diamond,j,\pi}, \widehat{\omega}_h^{\diamond,j,\pi}), \quad (53)$$

for any $j \in [3]$, $h \in [H] \cup [H]_{1/2}$, and $\diamond \in \{A, B\}$. With (52) and (53), we define the following confidence regions,

$$\begin{aligned} \text{conf}^{\diamond,r} &= \left\{ (\widetilde{\beta}_a^\diamond, \widetilde{\beta}_z^\diamond, \widetilde{\beta}_{az}^\diamond) : \widehat{L}^{\diamond,r}(\widetilde{\beta}^\diamond; \widetilde{\mathbf{f}}^\diamond) - \widehat{L}^{\diamond,r}(\widehat{\beta}^\diamond; \widehat{\mathbf{f}}^\diamond) \leq \eta^{\diamond,r} \right\}, \\ \text{conf}_h^{\diamond,j,\pi}(\widetilde{\theta}_{h+1/2}^\diamond) &= \left\{ (\widetilde{\theta}_h^{\diamond,j}, \widetilde{\gamma}_h^{\diamond,j}, \widetilde{\omega}_h^{\diamond,j}) : \widehat{L}_h^{\diamond,j,\pi}(\widetilde{\theta}_h^{\diamond,j}, \widetilde{\theta}_{h+1/2}^\diamond; \widetilde{\mathbf{f}}_h^{\diamond,j}) - \widehat{L}_h^{\diamond,j,\pi}(\widehat{\theta}_h^{\diamond,j,\pi}, \widehat{\theta}_{h+1/2}^\diamond; \widehat{\mathbf{f}}_h^{\diamond,j}) \leq \eta_h^{\diamond,j} \right\}, \\ \text{conf}_h^{\diamond,\pi}(\widetilde{\theta}_{h+1/2}^\diamond) &= \left\{ (\widetilde{\theta}_h^{\diamond,j}, \widetilde{\gamma}_h^{\diamond,j}, \widetilde{\omega}_h^{\diamond,j}) : (\widetilde{\theta}_h^{\diamond,j}, \widetilde{\gamma}_h^{\diamond,j}, \widetilde{\omega}_h^{\diamond,j}) \in \text{conf}_h^{\diamond,j,\pi}(\widetilde{\theta}_{h+1/2}^\diamond) \text{ for any } j \in [3] \right\}, \end{aligned} \quad (54)$$

for any $j \in [3]$, $h \in [H] \cup [H]_{1/2}$, and $\diamond \in \{A, B\}$. Here we denote by $\widehat{\theta}_H^A = \widehat{\beta}^A$ and $\widehat{\theta}_{H+1/2}^B = \widehat{\beta}^B$ for notational convenience.

Appendix B: Estimation Procedure without Oracle

The estimators constructed in (40), (43), (46), and (49) follow a similar procedure. Specifically, for a given policy pair π and a function ϑ , we aim to estimate the conditional expectation of the form,

$$\mathbb{E}[Y | S, A, B, U] = \vartheta_a^*(S, U)A + \vartheta_z^*(S, U)B + \vartheta_{az}^*(S, U)AB,$$

where $Y = g(R, S', U')$ and the expectation $\mathbb{E}[\cdot]$ is taken following the policy pair π and the turn-based model introduced in §2. Note here $\mathbb{E}[\cdot]$, ϑ_a^* , ϑ_z^* , and ϑ_{az}^* depend on π but we drop the subscripts for notational convenience. Specifically, we want to construct estimators for ϑ_a^* , ϑ_z^* , and ϑ_{az}^* .

B.1. Identification Results

We rely on the following three lemmas for our identification results in Theorem 1.

LEMMA 3. *For any π , we have*

$$\begin{aligned} &\mathbb{E}[(B - \mathbb{E}[B | S, U]) \cdot (A - \mathbb{E}[A | B, S, U]) \cdot Y | S, U] \\ &= \mathbb{E}[(B - \mathbb{E}[B | S, U]) \cdot (A - \mathbb{E}[A | B, S, U]) \cdot A | S, U] \cdot \vartheta_a^*(S, U) \\ &\quad + \mathbb{E}[B \cdot (B - \mathbb{E}[B | S, U]) \cdot (A - \mathbb{E}[A | B, S, U]) \cdot A | S, U] \cdot \vartheta_{az}^*(S, U). \end{aligned}$$

LEMMA 4. *For any π , we have*

$$\begin{aligned} &\mathbb{E}[(B - \mathbb{E}[B | S, U]) \cdot Y | S, U] \\ &= \mathbb{E}[A \cdot (B - \mathbb{E}[B | S, U]) | S, U] \cdot \vartheta_a^*(S, U) \\ &\quad + \mathbb{E}[B \cdot (B - \mathbb{E}[B | S, U]) | S, U] \cdot \vartheta_z^*(S, U) \\ &\quad + \mathbb{E}[AB \cdot (B - \mathbb{E}[B | S, U]) | S, U] \cdot \vartheta_{az}^*(S, U). \end{aligned}$$

LEMMA 5. *For any π , we have*

$$\begin{aligned} &\text{Cov}(B \cdot (B - \mathbb{E}[B | S, U]), (A - \mathbb{E}[A | B, S, U]) \cdot Y | S, U) \\ &= \text{Cov}(B \cdot (B - \mathbb{E}[B | S, U]), A \cdot (A - \mathbb{E}[A | B, S, U]) | S, U) \cdot \vartheta_a^*(S, U) \\ &\quad + \text{Cov}(B \cdot (B - \mathbb{E}[B | S, U]), AB \cdot (A - \mathbb{E}[A | B, S, U]) | S, U) \cdot \vartheta_{az}^*(S, U). \end{aligned}$$

By Lemmas 3, 4, and 5, we can identify ϑ_a^* , ϑ_z^* , and ϑ_{az}^* non-parametrically.

B.2. Estimation

Recall that Y depends on some generic policy π . For a fixed policy π , we define the following nuisance parameters,

$$\begin{aligned} f_1(S, U) &= \mathbb{E}[B | S, U], \quad f_2(X, B) = \mathbb{E}[A | X, B], \quad f_3(S, U) = \mathbb{E}[(A - f_2(S, B))Y | S, U], \\ f_4(S, U) &= \mathbb{E}[A(A - f_2(S, B)) | S, U], \quad f_5(S, U) = \mathbb{E}[AB(A - f_2(S, B)) | S, U]. \end{aligned}$$

Note the nuisance parameters defined here depend on π but we drop the subscripts for notational convenience.

Further, we have

$$\begin{aligned} w_1(Y, A, B, S, U, f_1, f_2) &= \rho_1(Y, S, B, U, f_1, f_2) - \rho_2(S, B, U, f_1, f_2)\vartheta_a^*(S, U) \\ &\quad - \rho_3(S, B, U, f_1, f_2)\vartheta_{az}^*(S, U), \\ w_2(Y, A, B, S, U, f_1) &= \rho_4(Y, S, U, B, f_1) - \rho_2(S, B, U, f_1)\vartheta_a^*(S, U) \\ &\quad - \rho_6(S, B, U, f_1)\vartheta_z^*(S, U) - \rho_7(S, B, U, f_1)\vartheta_{az}^*(S, U), \\ w_3(Y, A, B, S, U, f_1, f_2, f_3) &= \rho_8(Y, S, B, U, f_1, f_2) - (1 - f_1(S, U))f_1(S, U)f_3(S, U) \\ &\quad - (\rho_9(S, B, U, f_1, f_2) - (1 - f_1(S, U))f_1(S, U)f_4(S, U))\vartheta_a^*(S, U) \\ &\quad - (\rho_{10}(S, B, U, f_1, f_2) - (1 - f_1(S, U))f_1(S, U)f_5(S, U))\vartheta_{az}^*(S, U), \\ w_4(S, B, U, f_1) &= B - f_1(S, U), \\ w_5(A, B, S, U, f_2) &= A - f_2(S, B, U), \\ w_6(Y, A, B, S, U, f_2, f_3) &= (A - f_2(S, B, U)) \cdot Y - f_3(S, U), \\ w_7(A, B, S, U, f_2, f_4) &= (A - f_2(S, B, U))A - f_4(S, U), \\ w_8(A, B, S, U, f_2, f_5) &= (A - f_2(S, B, U))AB - f_5(S), \end{aligned} \tag{55}$$

where

$$\begin{aligned} \rho_1(Y, S, A, B, U, f_1, f_2) &= (B - f_1(S, U)) \cdot (A - f_2(S, B, U)) \cdot Y, \\ \rho_2(S, A, B, U, f_1, f_2) &= (B - f_1(S, U)) \cdot (A - f_2(S, B, U)) \cdot A, \\ \rho_3(S, A, B, U, f_1, f_2) &= B(B - f_1(S, U)) \cdot (A - f_2(S, B, U)) \cdot A, \\ \rho_4(Y, S, B, U, f_1) &= (B - f_1(S, U)) \cdot Y, \\ \rho_5(S, A, B, U, f_1) &= (B - f_1(S, U)) \cdot A, \\ \rho_6(S, B, U, f_1) &= B(B - f_1(S, U)), \\ \rho_7(S, A, B, U, f_1) &= AB(B - f_1(S, U)), \\ \rho_8(Y, S, A, B, U, f_1, f_2) &= B(B - f_1(S, U)) \cdot (A - f_2(S, B, U)) \cdot Y, \\ \rho_9(S, A, B, U, f_1, f_2) &= A(B - f_1(S, U)) \cdot (A - f_2(S, B, U)), \\ \rho_{10}(S, A, B, U, f_1, f_2) &= AB(B - f_1(S, U)) \cdot (A - f_2(S, B, U)). \end{aligned} \tag{56}$$

Note again that the parameters defined here depend on π but we drop the subscripts for notational convenience. We denote by $W(Y, S, A, B, U, \varpi^*) = (w_1, w_2, \dots, w_8)$, where $\varpi^* = (\vartheta_a^*, \vartheta_z^*, \vartheta_{az}^*, f_1, f_2, \dots, f_5)$ depends on π . Note that we can write

$$W = \begin{pmatrix} \Phi(\mathbf{f})\boldsymbol{\vartheta} \\ \varphi(\mathbf{f}) \end{pmatrix} + \alpha(\mathbf{f}), \tag{57}$$

where $\Phi(\mathbf{f}) \in \mathbb{R}^{3 \times 3}$, $\varphi(\mathbf{f}) \in \mathbb{R}^5$, $\boldsymbol{\vartheta} = (\vartheta_a^*, \vartheta_z^*, \vartheta_{az}^*)^\top$, and $\mathbf{f} = (f_1, f_2, \dots, f_5)^\top$. Then by Lemmas 3, 4, and 5, we have the following conditional moment restriction for any policy π ,

$$\mathbb{E}[W(Y, S, A, B, U, \varpi^*) | S, U] = 0. \quad (58)$$

We aim to recover ϖ^* via such a conditional moment restriction. We consider the following sieve minimum distance (SMD) estimator,

$$\begin{aligned} \hat{\varpi}_n &\in \arg \min_{\varpi \in \mathcal{H}_n} \hat{L}(\varpi), \\ \text{where } \hat{L}(\varpi) &= \frac{1}{n} \sum_{i=1}^n \hat{m}(Y^i, S^i, U^i, A^i, B^i; \varpi)^\top \hat{m}(Y^i, S^i, U^i, A^i, B^i; \varpi). \end{aligned} \quad (59)$$

Here $\hat{m}(Y^i, S^i, U^i, A^i, B^i)$ is a consistent estimator of $\mathbb{E}[W(Y, S, A, B, U, \varpi) | S, U]$ and $\mathcal{H}_n$ is a sieve parameter space. Also note that here the estimator $\hat{\varpi}_n$ and the function $\hat{L}$ both depends on π . The following result characterizes the convergence of $\hat{\varpi}_n$ to ϖ^* . In the meanwhile, we denote by

$$L(\varpi) = \mathbb{E} [m(S, U, \varpi)^\top m(S, U, \varpi)],$$

where $m(S, U, \varpi) = \mathbb{E}[W(Y, S, A, B, U, \varpi) | S, U]$. We know that $L(\varpi^*) = 0$ by the conditional moment restriction.

We first introduce the definition of sieve space $\mathcal{H}_n$ as follows,

$$\mathcal{H}_n = \left\{ \varpi \in \mathcal{H}: \varpi(\cdot) = \sum_{k=1}^{k(n)} a_k q_k(\cdot) \right\},$$

where $k(n) < \infty$, $k(n) \rightarrow \infty$ as $n \rightarrow \infty$, and $\{q_k\}_{k=1}^\infty$ is a sequence of known basis functions of a Banach space.

We introduce the following assumptions.

ASSUMPTION 10 (Sieves). *The following statements hold uniformly for any policy $\pi \in \Pi$.*

- We have $\|\varpi^* - \varpi\| = 0$ for any $\varpi \in \mathcal{H}$ such that $\mathbb{E}[W(Y, S, A, B, U, \varpi) | S, U] = 0$.
- The sequence of sieve spaces $\{\mathcal{H}_k\}_{k \geq 1}$ is a sequence of nonempty closed subsets satisfying $\mathcal{H}_k \subseteq \mathcal{H}_{k+1} \subseteq \mathcal{H}$, and there exists $\Gamma_n \varpi^* \in \mathcal{H}_{k(n)}$ such that $\|\Gamma_n \varpi^* - \varpi^*\| = o(1)$.
- We have $\mathbb{E}[\|m(S, U, \Gamma_n \varpi^*)\|^2] = o(1)$.

ASSUMPTION 11 (Sample Criterion). *The following statements hold uniformly for any policy $\pi \in \Pi$.*

- We have $\frac{1}{n} \sum_{i=1}^n \|\hat{m}(Y^i, S^i, U^i, A^i, B^i; \Gamma_n \varpi^*)\|^2 \leq c_0 \cdot \mathbb{E}[\|m(S, U, \Gamma_n \varpi^*)\|^2] + O_p(\eta_n)$, where $c_0 > 0$ is a constant and $\eta_n = o(1)$.
- We have $\frac{1}{n} \sum_{i=1}^n \|\hat{m}(Y^i, S^i, U^i, A^i, B^i; \varpi)\|^2 \geq c \cdot \mathbb{E}[\|m(S, U, \varpi)\|^2] - O_p(\delta_n^2)$ for any $\varpi \in \mathcal{H}_{k(n)}$, where $c > 0$ is a constant and $\delta_n^2 = c \cdot k(n)/n = o(1)$ satisfying $\eta_n = O(\delta_n^2)$.

With Assumptions 10 and 11, we have the following result showing that $\hat{\varpi}_n$ is a consistent estimator.

THEOREM 5 (Consistency). *Suppose Assumptions 8, 10 and 11 hold. We have $\|\hat{\varpi}_n - \varpi^*\| = o_p(1)$ uniformly for any policy $\pi \in \Pi$, where $\hat{\varpi}_n$ is the SMD estimator defined in (59).*

Under Assumption 8, as long as we can show the consistency for a fixed $\pi \in \Pi$, then by using the union bound, we can conclude the proof of Theorem 5. The proof for fixed π follows that in [Chen and Pouzo \(2012\)](#).

Given the consistency result in Theorem 5, we can restrict our space $\mathcal{H}$ to a shrinking neighborhood around the ground truth ϖ^* . We define

$$\mathcal{H}_s = \{\varpi \in \mathcal{H} : \|\varpi - \varpi^*\| \leq \delta, \|\varpi\| \leq M\}, \quad \mathcal{H}_{sn} = \mathcal{H}_s \cap \mathcal{H}_n$$

for a positive constant M and a sufficiently small positive δ such that $\mathbb{P}(\widehat{\varpi}_n \notin \mathcal{H}_s) < \delta$. Meanwhile, we define the path-wise derivative in the direction $[\varpi - \varpi^*]$ evaluated at ϖ^* as follows,

$$\frac{dm(S, U, \varpi^*)}{d\varpi}[\varpi - \varpi^*] = \frac{d\mathbb{E}[W(Y, S, A, B, U, (1-\tau)\varpi + \tau\varpi^*) | S, U]}{d\tau} \Big|_{\tau=0}.$$

We define the following notation of pseudometric for any $\varpi_1, \varpi_2 \in \mathcal{H}_s$,

$$\|\varpi_1 - \varpi_2\|_{ps} = \sqrt{\mathbb{E} \left[\left\| \frac{dm(S, U, \varpi^*)}{d\varpi}[\varpi_1 - \varpi_2] \right\|^2 \right]}.$$

We introduce the following assumption.

ASSUMPTION 12 (Local Curvature). *The following statements hold for $\pi \in \Pi$.*

- $\mathcal{H}_s$ and $\mathcal{H}_{sn}$ are convex, and $m(S, U, \varpi)$ is continuously path-wise differentiable with respect to $\varpi \in \mathcal{H}_s$.
- There exists a constant $c > 0$ such that $\|\varpi - \varpi^*\|_{ps} \leq c \cdot \|\varpi - \varpi^*\|$ for any $\varpi \in \mathcal{H}_s$.
- There are finite constants $c_1, c_2 > 0$ such that $\|\varpi - \varpi^*\|_{ps}^2 \leq c_1 \mathbb{E}[\|m(S, U, \varpi)\|^2]$ for any $\varpi \in \mathcal{H}_{sn}$; and $c_2 \mathbb{E}[\|m(S, U, \Gamma_n \varpi^*)\|^2] \leq \|\Gamma_n \varpi^* - \varpi^*\|_{ps}^2$.

ASSUMPTION 13 (Sieve Approximation Error). *We have $\|\varpi^* - \sum_{j=1}^{k(n)} \langle \varpi^*, q_j \rangle q_j\| = O(\{\nu_{k(n)}^{-\alpha}\})$ for any $\pi \in \Pi$ with a finite $\alpha > 0$ and a positive sequence $\{\nu_j\}_{j=1}^{\infty}$ that strictly increases and $\nu_j = \Theta(j^{1/d})$.*

ASSUMPTION 14 (Sieve Link Condition). *There are finite constants $c, C > 0$ and a continuous increasing function $\varphi: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that for any $\pi \in \Pi$*

- $\varphi(\tau) = \tau^\varsigma$ for some $\varsigma \geq 0$;
- $\|\varpi\|_{ps}^2 \geq c \sum_{j=1}^{\infty} \varphi(\nu_j^{-2}) \cdot |\langle \varpi, q_j \rangle|^2$ for all $\varpi \in \mathcal{H}_{sn}$;
- $\|\Gamma_n \varpi^* - \varpi^*\|_{ps}^2 \leq C \sum_{j=1}^{\infty} \varphi(\nu_j^{-2}) \cdot |\langle \Gamma_n \varpi^* - \varpi^*, q_j \rangle|^2$.

THEOREM 6 ([Chen and Pouzo \(2012\)](#)). *Suppose that Assumptions 8, 10–14 hold. It holds for any $\pi \in \Pi$ that*

$$\|\widehat{\varpi}_n - \varpi^*\| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right),$$

where $\widehat{\varpi}_n$ is the SMD estimator defined in (59).

We can thus construct plug-in estimators of ϑ_a^* , ϑ_z^* , and ϑ_{az}^* as $\widehat{\vartheta}_a^*$, $\widehat{\vartheta}_z^*$, and $\widehat{\vartheta}_{az}^*$ and obtain the following convergence results directly from Theorem 6.

COROLLARY 3. *Suppose that Assumptions 8 10–14 hold. It holds for any $\pi \in \Pi$ that*

$$\left\| \widehat{\vartheta}_a^* - \vartheta_a^* \right\| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right), \quad \left\| \widehat{\vartheta}_z^* - \vartheta_z^* \right\| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right), \quad \left\| \widehat{\vartheta}_{az}^* - \vartheta_{az}^* \right\| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right).$$

Appendix C: Details of Numerical Simulation

In this section, we detail the parameters used in the experiments of Section 6. For the transition model defined in (3), we only need to characterize for $f_1(u, v, s) = u$, $f_2(u, v, s) = v$, and $f_3(u, v, s) = s$, i.e., we can re-write (3) in its corresponding vector form as follows,

$$\mathbb{E} \left[\begin{pmatrix} f_1(S_{h+1/2}, U_{h+1/2}, V_{h+1/2}) \\ f_2(S_{h+1/2}, U_{h+1/2}, V_{h+1/2}) \\ f_3(S_{h+1/2}, U_{h+1/2}, V_{h+1/2}) \end{pmatrix} \middle| S_h, A_h, B_{h-1/2}, U_h, V_h \right] \\ = \theta_h(U_h, V_h, S_h)A_h + \gamma_h(U_h, V_h, S_h)B_{h-1/2} + \omega_h(U_h, V_h, S_h)A_h B_{h-1/2} + \iota_h(U_h, V_h, S_h),$$

where $\theta_h, \gamma_h, \omega_h, \iota_h: \mathcal{U} \times \mathcal{V} \times \mathcal{S} \rightarrow \mathbb{R}^3$. Similar result holds for (5). In this way, we only need to specify $\alpha_z^\diamond, \alpha_{u,s}^\diamond, \beta_a^\diamond, \beta_z^\diamond, \beta_{a,z}^\diamond, \beta_{u,s}^\diamond, \theta_h, \gamma_h, \omega_h, \iota_h$ for any $(\diamond, h, f) \in \{\mathbf{A}, \mathbf{B}\} \times \{0, 0.5, 1, 1.5\} \times \mathbb{R}^{\mathcal{S} \times \mathcal{U} \times \mathcal{V}}$.

C.1. Parameter Set 1

Parameter p in the behavior policy is taken as $p = 0.2$.

Alice model:

$$\alpha_z^{\mathbf{A}}(u, v, s) = \begin{pmatrix} -1.255 \\ 4.507 \\ 2.32 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \alpha_{u,s}^{\mathbf{A}}(u, v, s) = \begin{pmatrix} 0.987 \\ -3.44 \\ -3.44 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \beta_a^{\mathbf{A}}(u, v, s) = \begin{pmatrix} -4.419 \\ 3.662 \\ 1.011 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_z^{\mathbf{A}}(u, v, s) = \begin{pmatrix} 2.081 \\ -4.794 \\ 4.699 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \beta_{a,z}^{\mathbf{A}}(u, v, s) = \begin{pmatrix} 3.324 \\ -2.877 \\ -3.182 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_{u,s}^{\mathbf{A}}(u, v, s) = \begin{pmatrix} 0.0 \\ 0.0 \\ 0.0 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}.$$

Bob model:

$$\alpha_z^{\mathbf{B}}(u, v, s) = \begin{pmatrix} -3.166 \\ -1.958 \\ 0.248 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \alpha_{u,s}^{\mathbf{B}}(u, v, s) = \begin{pmatrix} -0.681 \\ -2.088 \\ 1.119 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \beta_a^{\mathbf{B}}(u, v, s) = \begin{pmatrix} -3.605 \\ -2.079 \\ -1.336 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_z^{\mathbf{B}}(u, v, s) = \begin{pmatrix} -0.439 \\ 2.852 \\ -3.003 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \beta_{a,z}^{\mathbf{B}}(u, v, s) = \begin{pmatrix} 0.142 \\ 0.924 \\ -4.535 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_{u,s}^{\mathbf{B}}(u, v, s) = \begin{pmatrix} 0.0 \\ 0.0 \\ 0.0 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}.$$

Transition model at step 0:

$$\theta_0(u, v, s) = \begin{pmatrix} 1.075, -3.295, -4.349 \\ 4.489, 4.656, 3.084 \\ -1.954, -4.023, 1.842 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_0(u, v, s) = \begin{pmatrix} -1.89, -1.748, 2.296 \\ 1.376, 3.872, -0.278 \\ -3.804, 2.132, 2.608 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \omega_0(u, v, s) = \begin{pmatrix} -1.364, 4.718, 4.624 \\ -2.482, -0.028, -1.991 \\ -2.152, -4.631, 1.096 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_0(u, v, s) = \begin{pmatrix} 4.004, 1.331, -1.61 \\ -1.508, 2.26, 3.971 \\ 3.871, 2.799, 1.42 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}.$$

Transition model at step 0.5:

$$\theta_{0.5}(u, v, s) = \begin{pmatrix} -0.718, 4.667, 4.636 \\ 3.53, -2.056, -1.149 \\ 3.511, -1.831, -3.305 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_{0.5}(u, v, s) = \begin{pmatrix} -4.483, 0.314, 0.406 \\ 1.374, 2.261, 4.759 \\ 0.163, -1.77, 2.952 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \omega_{0.5}(u, v, s) = \begin{pmatrix} -4.294, 1.424, -4.735 \\ 0.858, 4.402, 0.755 \\ -1.118, 1.433, -0.417 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_{0.5}(u, v, s) = \begin{pmatrix} 0.0, 0.0, 0.0 \\ 0.0, 0.0, 0.0 \\ 0.0, 0.0, 0.0 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}.$$

Transition model at step 1:

$$\begin{aligned}\theta_1(u, v, s) &= \begin{pmatrix} -0.598, -3.78, -0.048 \\ -4.656, 4.093, -2.412 \\ 1.625, -1.883, 0.201 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_1(u, v, s) = \begin{pmatrix} 0.613, 2.71, -0.062 \\ 0.227, -0.725, -4.746 \\ -3.921, -4.686, 1.364 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \omega_1(u, v, s) &= \begin{pmatrix} 0.027, -4.485, -2.214 \\ 4.083, -2.604, -3.551 \\ -0.105, 4.857, -2.579 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_1(u, v, s) = \begin{pmatrix} -4.159, -3.384, 3.986 \\ 1.064, -4.908, -3.985 \\ 1.635, -4.949, -3.392 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}.\end{aligned}$$

Transition model at step 1.5:

$$\begin{aligned}\theta_{1.5}(u, v, s) &= \begin{pmatrix} 0.568, 4.362, 1.96 \\ 0.701, -4.028, 1.15 \\ 4.901, -3.599, 0.183 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_{1.5}(u, v, s) = \begin{pmatrix} -2.292, -0.61, -4.215 \\ -4.746, 4.626, 3.36 \\ 1.96, -0.91, -3.267 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \omega_{1.5}(u, v, s) &= \begin{pmatrix} 0.456, 4.415, -1.139 \\ 4.612, 4.054, -3.042 \\ -4.306, -3.992, -4.818 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_{1.5}(u, v, s) = \begin{pmatrix} 0.0, 0.0, 0.0 \\ 0.0, 0.0, 0.0 \\ 0.0, 0.0, 0.0 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}.\end{aligned}$$

C.2. Parameter Set 2

Parameter p in the behavior policy is taken as $p = 0.3$.

Alice model:

$$\begin{aligned}\alpha_z^A(u, v, s) &= \begin{pmatrix} -0.096 \\ -4.243 \\ 1.678 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \alpha_{u,s}^A(u, v, s) = \begin{pmatrix} 4.273 \\ 2.237 \\ -0.563 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \beta_a^A(u, v, s) &= \begin{pmatrix} 3.522 \\ -0.414 \\ -4.2 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_z^A(u, v, s) = \begin{pmatrix} 3.748 \\ 0.152 \\ 4.141 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \beta_{a,z}^A(u, v, s) &= \begin{pmatrix} -0.992 \\ -0.812 \\ 1.041 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_{u,s}^A(u, v, s) = \begin{pmatrix} 0.0 \\ 0.0 \\ 0.0 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}.\end{aligned}$$

Bob model:

$$\begin{aligned}\alpha_z^B(u, v, s) &= \begin{pmatrix} 3.874 \\ -0.067 \\ -2.421 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \alpha_{u,s}^B(u, v, s) = \begin{pmatrix} 0.909 \\ -3.541 \\ 3.221 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \beta_a^B(u, v, s) &= \begin{pmatrix} 3.108 \\ 4.352 \\ -4.736 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_z^B(u, v, s) = \begin{pmatrix} -0.661 \\ 2.308 \\ -0.775 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \beta_{a,z}^B(u, v, s) &= \begin{pmatrix} 2.415 \\ 4.164 \\ -2.743 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_{u,s}^B(u, v, s) = \begin{pmatrix} 0.0 \\ 0.0 \\ 0.0 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}.\end{aligned}$$

Transition model at step 0:

$$\begin{aligned}\theta_0(u, v, s) &= \begin{pmatrix} -0.353 & -2.33 & -1.517 \\ -2.711 & 1.565 & 1.06 \\ -3.714 & 1.985 & -2.725 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_0(u, v, s) = \begin{pmatrix} -4.126 & -2.392 & -0.429 \\ -3.975 & -3.109 & 4.213 \\ 4.008 & -3.514 & -4.831 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \omega_0(u, v, s) &= \begin{pmatrix} 2.31 & 0.064 & -2.114 \\ 0.706 & 1.302 & -3.59 \\ 2.477 & 3.132 & 3.185 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_0(u, v, s) = \begin{pmatrix} 0.53 & -2.978 & 3.362 \\ -3.236 & -1.366 & 1.872 \\ -2.148 & -1.293 & 0.903 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}.\end{aligned}$$

Transition model at step 0.5:

$$\theta_{0.5}(u, v, s) = \begin{pmatrix} -0.221 & -2.825 & 3.516 \\ -2.843 & 1.228 & -1.429 \\ -1.403 & -2.35 & 0.957 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_{0.5}(u, v, s) = \begin{pmatrix} -3.296 & 4.693 & 4.336 \\ 1.416 & -0.919 & 1.299 \\ -3.302 & -3.805 & 3.486 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix},$$

$$\omega_{0.5}(u, v, s) = \begin{pmatrix} -4.258 & 2.669 & -4.05 \\ 0.58 & 0.193 & 0.324 \\ -4.067 & 0.615 & -3.492 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_{0.5}(u, v, s) = \begin{pmatrix} 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}.$$

Transition model at step 1:

$$\theta_1(u, v, s) = \begin{pmatrix} 2.371 & -4.946 & 2.183 \\ 0.122 & 3.326 & -2.147 \\ 2.258 & 1.415 & 0.736 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_1(u, v, s) = \begin{pmatrix} 2.287 & -1.425 & 3.889 \\ -2.338 & -2.044 & -2.083 \\ 4.94 & -1.624 & -1.828 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix},$$

$$\omega_1(u, v, s) = \begin{pmatrix} -0.804 & -4.158 & 3.002 \\ -4.361 & 0.936 & -1.299 \\ -4.859 & -4.379 & -2.102 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_1(u, v, s) = \begin{pmatrix} 2.734 & -1.034 & -0.947 \\ -4.153 & -0.043 & 1.444 \\ 4.832 & 2.785 & -1.813 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}.$$

Transition model at step 1.5:

$$\theta_{1.5}(u, v, s) = \begin{pmatrix} -4.745 & 4.26 & 0.306 \\ 4.419 & 0.479 & -0.889 \\ 2.482 & 0.414 & -2.357 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_{1.5}(u, v, s) = \begin{pmatrix} 1.51 & -0.416 & -4.185 \\ 1.584 & -2.808 & 3.228 \\ 1.937 & -0.782 & 1.727 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix},$$

$$\omega_{1.5}(u, v, s) = \begin{pmatrix} 0.499 & -4.478 & -3.49 \\ -1.604 & -4.853 & -2.381 \\ 3.729 & -3.736 & 4.767 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_{1.5}(u, v, s) = \begin{pmatrix} 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}.$$

C.3. Parameter Set 3

Parameter p in the behavior policy is taken as $p = 0.4$.

Alice model:

$$\alpha_z^A(u, v, s) = \begin{pmatrix} 4.476 \\ -2.144 \\ -0.371 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \alpha_{u,s}^A(u, v, s) = \begin{pmatrix} 0.727 \\ -0.579 \\ 0.16 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix},$$

$$\beta_a^A(u, v, s) = \begin{pmatrix} -4.094 \\ 4.963 \\ -4.739 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_z^A(u, v, s) = \begin{pmatrix} -1.762 \\ -4.377 \\ -1.237 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix},$$

$$\beta_{a,z}^A(u, v, s) = \begin{pmatrix} -1.042 \\ -1.764 \\ -3.144 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_{u,s}^A(u, v, s) = \begin{pmatrix} 0.0 \\ 0.0 \\ 0.0 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}.$$

Bob model:

$$\alpha_z^B(u, v, s) = \begin{pmatrix} -4.383 \\ 4.633 \\ -2.526 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \alpha_{u,s}^B(u, v, s) = \begin{pmatrix} 3.152 \\ -0.936 \\ -0.646 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix},$$

$$\beta_a^B(u, v, s) = \begin{pmatrix} 1.94 \\ 3.509 \\ -0.456 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_z^B(u, v, s) = \begin{pmatrix} 2.806 \\ 2.249 \\ -1.083 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix},$$

$$\beta_{a,z}^B(u, v, s) = \begin{pmatrix} -1.596 \\ -3.111 \\ -3.195 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \beta_{u,s}^B(u, v, s) = \begin{pmatrix} 0.0 \\ 0.0 \\ 0.0 \end{pmatrix}^\top \begin{pmatrix} u \\ v \\ s \end{pmatrix}.$$

Transition model at step 0:

$$\theta_0(u, v, s) = \begin{pmatrix} 4.27 & -4.137 & -3.366 \\ 3.493 & -4.769 & 2.891 \\ 2.093 & -2.805 & 4.349 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_0(u, v, s) = \begin{pmatrix} -0.185 & -4.059 & 0.467 \\ 0.68 & 3.714 & -1.753 \\ -4.119 & -3.274 & 0.429 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix},$$

$$\omega_0(u, v, s) = \begin{pmatrix} 3.726 & 0.907 & -3.295 \\ 0.365 & -3.488 & -2.545 \\ 3.098 & 2.343 & -4.023 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_0(u, v, s) = \begin{pmatrix} -3.718 & -0.383 & 0.37 \\ 2.809 & -3.941 & -2.193 \\ 0.775 & -2.285 & 1.685 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}.$$

Transition model at step 0.5:

$$\begin{aligned}\theta_{0.5}(u, v, s) &= \begin{pmatrix} -3.529 & -4.707 & -1.9 \\ 0.886 & 4.245 & -1.691 \\ -2.799 & 2.241 & -0.185 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_{0.5}(u, v, s) = \begin{pmatrix} 0.594 & 1.484 & 3.239 \\ -0.118 & 1.726 & -1.679 \\ -2.523 & 0.932 & -4.053 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \omega_{0.5}(u, v, s) &= \begin{pmatrix} 2.027 & 2.967 & 0.758 \\ 3.396 & -1.235 & -1.527 \\ 3.809 & -2.775 & -0.625 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_{0.5}(u, v, s) = \begin{pmatrix} 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}.\end{aligned}$$

Transition model at step 1:

$$\begin{aligned}\theta_1(u, v, s) &= \begin{pmatrix} -3.303 & 0.568 & 3.013 \\ 3.759 & 4.69 & -3.623 \\ 1.095 & 4.655 & 4.444 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_1(u, v, s) = \begin{pmatrix} -3.138 & 3.43 & 4.061 \\ 3.81 & 4.963 & 3.584 \\ 3.495 & -4.68 & -3.763 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \omega_1(u, v, s) &= \begin{pmatrix} -0.002 & -3.209 & -1.807 \\ -3.813 & -3.817 & -0.042 \\ -4.564 & -0.379 & 4.923 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_1(u, v, s) = \begin{pmatrix} -1.163 & 3.088 & 3.99 \\ 0.995 & -0.977 & 2.982 \\ 4.955 & -4.946 & -3.789 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}.\end{aligned}$$

Transition model at step 1.5:

$$\begin{aligned}\theta_{1.5}(u, v, s) &= \begin{pmatrix} 0.485 & -0.363 & 3.965 \\ 4.862 & -2.63 & -2.086 \\ -1.768 & -0.435 & -1.389 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \gamma_{1.5}(u, v, s) = \begin{pmatrix} -2.278 & -4.0 & 2.825 \\ 1.118 & 0.34 & -4.091 \\ 4.013 & -2.415 & 4.516 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \\ \omega_{1.5}(u, v, s) &= \begin{pmatrix} -3.624 & 0.053 & -4.875 \\ 1.875 & 2.916 & -0.497 \\ -2.452 & 2.763 & 1.282 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}, \quad \iota_{1.5}(u, v, s) = \begin{pmatrix} 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 \end{pmatrix} \begin{pmatrix} u \\ v \\ s \end{pmatrix}.\end{aligned}$$

Appendix D: Proof of Identification Results

D.1. Proof of Lemma 3

Proof. Without loss of generality, we assume $(S, U) = \emptyset$. By Lemma 3.1 in [Tchetgen et al. \(2021\)](#), we can show that

$$\begin{aligned}& \mathbb{E}[(B - \mathbb{E}[B | S, U])(A - \mathbb{E}[A | B, S, U]) \{A\vartheta_a^*(V) + \vartheta_z^*(V)B + \vartheta_u^*(V)\}] \\ &= \mathbb{E}[(B - \mathbb{E}[B | S, U])(A - \mathbb{E}[A | B, S, U])A] \times \mathbb{E}[\vartheta_a^*(V)].\end{aligned}$$

By direct calculation, we can show that

$$\begin{aligned}& \mathbb{E}[(B - \mathbb{E}[B | S, U])(A - \mathbb{E}[A | B])Y] \\ &= \mathbb{E}[(B - \mathbb{E}[B | S, U])(A - \mathbb{E}[A | B]) \{A\vartheta_a^*(V) + \vartheta_z^*(V)B + \vartheta_u^*(V)\}] \\ &+ \mathbb{E}[B(B - \mathbb{E}[B | S, U])(A - \mathbb{E}[A | B])A\vartheta_{a,z}^*(V)] \\ &= \mathbb{E}[(B - \mathbb{E}[B | S, U])(A - \mathbb{E}[A | B])A] \times \mathbb{E}[\vartheta_a^*(V)] \\ &+ \mathbb{E}[B(B - \mathbb{E}[B | S, U])(1 - \mathbb{E}[A | B])\vartheta_{a,z}^*(V) \{\alpha_z^*(V)B + \alpha_u^*(V)\}] \\ &= \mathbb{E}[(B - \mathbb{E}[B | S, U])(A - \mathbb{E}[A | B])A] \times \mathbb{E}[\vartheta_a^*(V)] \\ &+ \mathbb{E}[B(B - \mathbb{E}[B | S, U])(1 - \mathbb{E}[A | B])A] \times \mathbb{E}[\vartheta_{a,z}^*(V)] \\ &+ \mathbb{E}[B(B - \mathbb{E}[B | S, U])(1 - \mathbb{E}[A | B]) (\vartheta_{a,z}^*(V) - \mathbb{E}[\vartheta_{a,z}^*(V)]) \{\alpha_z^*(V)B + \alpha_u^*(V)\}] \\ &= \mathbb{E}[(B - \mathbb{E}[B | S, U])(A - \mathbb{E}[A | B])A] \times \mathbb{E}[\vartheta_a^*(V)] \\ &+ \mathbb{E}[B(B - \mathbb{E}[B | S, U])(1 - \mathbb{E}[A | B])A] \times \mathbb{E}[\vartheta_{a,z}^*(V)],\end{aligned}$$

where we use $Cov(\alpha_z^*(S, U, V), \vartheta_{a,z}^*(S, U, V) | S, U) = Cov(\vartheta_{a,z}^*(S, U, V), \alpha_{u,s}^*(S, U, V) | S, U) = 0$ in the last equation. ■

D.2. Proof of Lemma 4

Proof. Again, without loss of generality, assume $(S, U) = \emptyset$. By direct calculation, we can show that

$$\begin{aligned}
& \mathbb{E}[(B - \mathbb{E}[B])Y] \\
&= \mathbb{E}[(B - \mathbb{E}[B]) \{ \vartheta_a^*(V)A + \vartheta_z^*(V)B + \vartheta_{a,z}^*(V)BA + \vartheta_u^*(V) \}] \\
&= \mathbb{E}[(B - \mathbb{E}[B])A] \times \mathbb{E}[\vartheta_a^*(V)] + \mathbb{E}[(B - \mathbb{E}[B]) (\alpha_z^*(V)B + \alpha_u^*(V)) (\vartheta_a^*(V) - \mathbb{E}[\vartheta_a^*(V)])] \\
&\quad + \mathbb{E}[(B - \mathbb{E}[B])B] \times \mathbb{E}[\vartheta_z^*(V)] + \mathbb{E}[(B - \mathbb{E}[B])B \vartheta_{a,z}^*(V)A] \\
&= \mathbb{E}[(B - \mathbb{E}[B])A] \times \mathbb{E}[\vartheta_a^*(V)] + \mathbb{E}[(B - \mathbb{E}[B])B] \times \mathbb{E}[\vartheta_z^*(V)] \\
&\quad + \mathbb{E}[(B - \mathbb{E}[B])BA] \times \mathbb{E}[\vartheta_{a,z}^*(V)] + \mathbb{E}[(B - \mathbb{E}[B])B (\vartheta_{a,z}^*(V) - \mathbb{E}[\vartheta_{a,z}^*(V)]) (\alpha_z^*(V)B + \alpha_u^*(V))] \\
&= \mathbb{E}[(B - \mathbb{E}[B])A] \times \mathbb{E}[\vartheta_a^*(V)] + \mathbb{E}[(B - \mathbb{E}[B])B] \times \mathbb{E}[\vartheta_z^*(V)] \\
&\quad + \mathbb{E}[(B - \mathbb{E}[B])BA] \times \mathbb{E}[\vartheta_{a,z}^*(V)]
\end{aligned}$$

■

D.3. Proof of Lemma 5

Proof. Again, we omit (S, U) . First, by similar arguments in the proof of Lemma (3), we can show that

$$\begin{aligned}
& \mathbb{E}[B(B - \mathbb{E}[B])(A - \mathbb{E}[A|B])Y] \\
&= \mathbb{E}[B(B - \mathbb{E}[B])(A - \mathbb{E}[A|B])A] \times \mathbb{E}[\vartheta_a^*(V)] + \mathbb{E}[B(B - \mathbb{E}[B])(A - \mathbb{E}[A|B])A] \times \mathbb{E}[\vartheta_{a,z}^*(V)] \\
&\quad + \mathbb{E}[B(B - \mathbb{E}[B])(A - \mathbb{E}[A|B])\vartheta_u^*(V)] \\
&= \mathbb{E}[B(B - \mathbb{E}[B])(A - \mathbb{E}[A|B])A] \times \mathbb{E}[\vartheta_a^*(V)] + \mathbb{E}[B(B - \mathbb{E}[B])(A - \mathbb{E}[A|B])A] \times \mathbb{E}[\vartheta_{a,z}^*(V)] \\
&\quad + \mathbb{E}[B(B - \mathbb{E}[B]) \text{Cov}(\alpha_u^*(V), \vartheta_u^*(V))].
\end{aligned}$$

Next, we derive that

$$\begin{aligned}
& \mathbb{E}[(A - \mathbb{E}[A|B])Y] \\
&= \mathbb{E}[(A - \mathbb{E}[A|B]) \{ \vartheta_a^*(V)A + \vartheta_z^*(V)B + \vartheta_{a,z}^*(V)BA + \vartheta_u^*(V) \}] \\
&= \mathbb{E}[A(A - \mathbb{E}[A|B])] \times \mathbb{E}[\vartheta_a^*(V)] + \mathbb{E}[(1 - \mathbb{E}[A|B]) (\alpha_z^*(V)B + \alpha_u^*(V)) (\vartheta_a^*(V) - \mathbb{E}[\vartheta_a^*(V)])] \\
&\quad + \mathbb{E}[BA(A - \mathbb{E}[A|B])] \times \mathbb{E}[\vartheta_{a,z}^*(V)] + \mathbb{E}[B(1 - \mathbb{E}[A|B]) (\alpha_z^*(V)B + \alpha_u^*(V)) (\vartheta_{a,z}^*(V) - \mathbb{E}[\vartheta_{a,z}^*(V)])] \\
&\quad + \text{Cov}(\alpha_u^*(V), \vartheta_u^*(V)) \\
&= \mathbb{E}[A(A - \mathbb{E}[A|B])] \times \mathbb{E}[\vartheta_a^*(V)] + \mathbb{E}[BA(A - \mathbb{E}[A|B])] \times \mathbb{E}[\vartheta_{a,z}^*(V)] \\
&\quad + \text{Cov}(\alpha_u^*(V), \vartheta_u^*(V)).
\end{aligned}$$

Summarizing together, we obtain that

$$\begin{aligned}
& \mathbb{E}[B(B - \mathbb{E}[B])(A - \mathbb{E}[A|B])Y] \\
&= \mathbb{E}[B(B - \mathbb{E}[B])(A - \mathbb{E}[A|B])A] \times \mathbb{E}[\vartheta_a^*(V)] + \mathbb{E}[B(B - \mathbb{E}[B])(A - \mathbb{E}[A|B])A] \times \mathbb{E}[\vartheta_{a,z}^*(V)]
\end{aligned}$$

$$\begin{aligned}
& +\mathbb{E}[B(B-\mathbb{E}[B])] \\
& \times \left\{ \mathbb{E}[(A-\mathbb{E}[A|B])Y] - \mathbb{E}[A(A-\mathbb{E}[A|B])] \times \mathbb{E}[\vartheta_a^*(V)] - \mathbb{E}[BA(A-\mathbb{E}[A|B])] \times \mathbb{E}[\vartheta_{a,z}^*(V)] \right\} \\
& = \text{Cov}(B(B-\mathbb{E}[B]), A(A-\mathbb{E}[A|B]))\mathbb{E}[\vartheta_a^*(V)] + \text{Cov}(B(B-\mathbb{E}[B]), BA(A-\mathbb{E}[A|B]))\mathbb{E}[\vartheta_{a,z}^*(V)] \\
& + \mathbb{E}[B(B-\mathbb{E}[B])] \times \mathbb{E}[(A-\mathbb{E}[A|B])Y],
\end{aligned}$$

which concludes our proof. $\blacksquare$

Appendix E: Proofs for Single-Stage Game

E.1. Proof of Lemma 1

Proof. To show that $Q_1^{A,\pi} \in \text{conf}_1^{A,Q,\pi}$, we only need to show that $(\beta_a^A, \beta_z^A, \beta_{az}^A) \in \text{conf}^{A,r}$, i.e.,

$$\widehat{L}^{A,r}(\beta^A) - \widehat{L}^{A,r}(\widehat{\beta}^A) \leq \eta^{A,r}. \quad (60)$$

Note that

$$\begin{aligned}
& \widehat{L}^{A,r}(\beta^A) - \widehat{L}^{A,r}(\widehat{\beta}^A) \\
& = \widehat{L}^{A,r}(\beta^A) - L^{A,r}(\beta^A) + L^{A,r}(\beta^A) - L^{A,r}(\widehat{\beta}^A) + L^{A,r}(\widehat{\beta}^A) - \widehat{L}^{A,r}(\widehat{\beta}^A) \\
& \leq \widehat{L}^{A,r}(\beta^A) - L^{A,r}(\beta^A) + L^{A,r}(\widehat{\beta}^A) - \widehat{L}^{A,r}(\widehat{\beta}^A) \\
& = \left(\widehat{L}^{A,r}(\beta^A) - \widehat{L}^{A,r}(\widehat{\beta}^A) \right) - \left(L^{A,r}(\beta^A) - L^{A,r}(\widehat{\beta}^A) \right)
\end{aligned} \quad (61)$$

where in the first inequality, we use the fact that $L^{A,r}(\beta^A) = 0$ while $L^{A,r}(\widehat{\beta}^A) \geq 0$. Now, by using Bernstein's inequality on the right-hand side of (61), we have

$$\begin{aligned}
\widehat{L}^{A,r}(\beta^A) - \widehat{L}^{A,r}(\widehat{\beta}^A) & = O_p \left(\frac{c}{n} + \sqrt{\frac{\text{Var} \left[\|\widehat{m}(\beta^A)\|^2 - \|\widehat{m}(\widehat{\beta}^A)\|^2 \right]}{n}} \right) \\
& \leq O_p \left(\frac{c}{n} + \sqrt{\frac{\mathbb{E} \left[\left(\|\widehat{m}(\beta^A)\|^2 - \|\widehat{m}(\widehat{\beta}^A)\|^2 \right)^2 \right]}{n}} \right),
\end{aligned} \quad (62)$$

where we write $\widehat{m}(\beta^A) = \widehat{m}(Y, S, U, A, B; \beta^A)$ for notational convenience. Note that

$$\begin{aligned}
\left| \|\widehat{m}(\beta^A)\|^2 - \|\widehat{m}(\widehat{\beta}^A)\|^2 \right| & = \left(\|\widehat{m}(\beta^A)\| + \|\widehat{m}(\widehat{\beta}^A)\| \right) \cdot \left| \|\widehat{m}(\beta^A)\| - \|\widehat{m}(\widehat{\beta}^A)\| \right| \\
& \leq c \cdot \left| \|\widehat{m}(\beta^A)\| - \|\widehat{m}(\widehat{\beta}^A)\| \right| \\
& \leq c \cdot \left\| \widehat{m}(\beta^A) - \widehat{m}(\widehat{\beta}^A) \right\| = O_p \left(n^{-\frac{\alpha}{2(\alpha+\varsigma)+d}} \right),
\end{aligned} \quad (63)$$

where we use triangle's inequality and Corollary 1 in the second and last inequalities, respectively. By combining (62) and (63), it holds that

$$\widehat{L}^{A,r}(\beta^A) - \widehat{L}^{A,r}(\widehat{\beta}^A) = O_p \left(n^{-\frac{2\alpha}{2(\alpha+\varsigma)+d}} \right),$$

which verifies (60). Thus, by taking the union bound for $\pi \in \Pi$, it holds that $Q_1^{A,\pi} \in \text{conf}_1^{A,Q,\pi}$ as $n \rightarrow \infty$.

To show that $Q_1^{B,\pi} \in \text{conf}_1^{B,Q,\pi}$, we need to show that there exists $\beta^{B,\dagger} \in \text{conf}^{B,r}$ such that $\theta_1^B = (\theta_1^B, \gamma_1^B, \omega_1^B) \in \text{conf}_1^{B,\pi}(\beta^{B,\dagger})$, where $(\theta_1^B, \gamma_1^B, \omega_1^B)$ is associated with $Q_1^{B,\pi}$. In other words, we need to show that there exists

$\beta^{B,\dagger} \in \text{conf}^{B,r}$ such that $\theta_1^{B,j} = (\theta_1^{B,j}, \gamma_1^{B,j}, \omega_1^{B,j}) \in \text{conf}_1^{B,j,\pi}(\beta^{B,\dagger})$ for any $j \in [3]$. Following from a similar argument as in (60), we can show that as $n \rightarrow \infty$, we have $\beta^B \in \text{conf}^{B,r}$, where β^B is associated with $Q_{3/2}^{B,\pi}$. Thus, we only need to show that $(\theta_1^{B,j}, \gamma_1^{B,j}, \omega_1^{B,j}) \in \text{conf}_1^{B,j,\pi}(\beta^B)$, i.e.,

$$\widehat{L}_1^{B,j,\pi}(\theta_1^{B,j}; \beta^B) - \widehat{L}_1^{B,j,\pi}(\widehat{\theta}_1^{B,j}; \beta^B) \leq \eta_1^{B,j} \quad (64)$$

for any $j \in [3]$. Following from a similar argument to prove (60), we have

$$\widehat{L}_1^{B,j,\pi}(\theta_1^{B,j}; \beta^B) - \widehat{L}_1^{B,j,\pi}(\widehat{\theta}_1^{B,j}; \beta^B) = O_p\left(n^{-\frac{2\alpha}{2(\alpha+\varsigma)+d}}\right),$$

which verifies (64) for any $j \in [3]$. Thus, we know that $Q_1^{B,\pi} \in \text{conf}_1^{B,Q,\pi}$ as $n \rightarrow \infty$ by again taking union bound over $\pi \in \Pi$. This concludes the proof of the lemma. $\blacksquare$

E.2. Proof of Lemma 2

Proof. By (17) and the definition of the confidence region in (28), we obtain that

$$\begin{aligned} \left| \mathbb{E}_{\pi^*} [Q_1^{\pi^*} - Q_1^A] \right| &= \mathbb{E}_{\pi^*} \left[(\beta_a^A - \widetilde{\beta}_a^A) A_1 + (\beta_z^A - \widetilde{\beta}_z^A) B_{1/2} + (\beta_{az}^A - \widetilde{\beta}_{az}^A) A_1 B_{1/2} \right] \\ &\leq \left\| \beta_a^A - \widetilde{\beta}_a^A \right\|_{\pi^*} + \left\| \beta_z^A - \widetilde{\beta}_z^A \right\|_{\pi^*} + \left\| \beta_{az}^A - \widetilde{\beta}_{az}^A \right\|_{\pi^*}, \end{aligned} \quad (65)$$

where $\widetilde{\beta}^A = (\widetilde{\beta}_a^A, \widetilde{\beta}_z^A, \widetilde{\beta}_{az}^A)^\top$ is associated with Q_1^A . Here we denote by $\|\beta - \widetilde{\beta}\|_\pi^2 = \mathbb{E}_\pi[(\beta - \widetilde{\beta})^2]$ for any functions β , $\widetilde{\beta}$, and policy $\pi \in \Pi$. Note that

$$\widehat{L}^{A,r}(\widetilde{\beta}^A) = \frac{1}{n} \sum_{i=1}^n \left\| \Phi_i \widetilde{\beta}_i^A + \alpha_i \right\|^2,$$

where $\Phi_i \in \mathbb{R}^{3 \times 3}$ and $\widetilde{\beta}_i^A = \widetilde{\beta}^A(S_1^i, U_1^{A,i})$. In the meanwhile, by Bernstein's inequality, we have

$$\begin{aligned} &\left(L^{A,r}(\widetilde{\beta}^A) - L^{A,r}(\beta^A) \right) - \left(\widehat{L}^{A,r}(\widetilde{\beta}^A) - \widehat{L}^{A,r}(\beta^A) \right) \\ &= O_p \left(\frac{1}{n} + \sqrt{\frac{\mathbb{E} \left[(\|\Phi \widetilde{\beta}^A + \alpha\|^2 - \|\Phi \beta^A + \alpha\|^2)^2 \right]}{n}} \right). \end{aligned} \quad (66)$$

Note that

$$\mathbb{E} \left[\left(\|\Phi \widetilde{\beta}^A + \alpha\|^2 - \|\Phi \beta^A + \alpha\|^2 \right)^2 \right] \leq c \cdot \mathbb{E} \left[\left\| \Phi(\widetilde{\beta}^A - \beta^A) \right\|^2 \right]. \quad (67)$$

By combining (66) and (67), we have

$$\begin{aligned} &\left(L^{A,r}(\widetilde{\beta}^A) - L^{A,r}(\beta^A) \right) - \left(\widehat{L}^{A,r}(\widetilde{\beta}^A) - \widehat{L}^{A,r}(\beta^A) \right) \\ &= O_p \left(n^{-1} + \sqrt{n^{-1} \cdot \mathbb{E}[\|\Phi(\widetilde{\beta}^A - \beta^A)\|^2]} \right). \end{aligned} \quad (68)$$

By (68), it holds for any $\widetilde{\beta}^A \in \text{conf}^{A,r}$ that

$$L^{A,r}(\widetilde{\beta}^A) - L^{A,r}(\beta^A) = O_p \left(n^{-1} + \sqrt{n^{-1} \cdot \mathbb{E}[\|\Phi(\widetilde{\beta}^A - \beta^A)\|^2]} + \eta^{A,r} \right), \quad (69)$$

In the meanwhile, since $L^{A,r}(\cdot)$ is a quadratic functional, it holds for any $\widetilde{\beta}^A$ that

$$\mathbb{E} \left[\left\| \Phi(\widetilde{\beta}^A - \beta^A) \right\|^2 \right] \leq L^{A,r}(\widetilde{\beta}^A) - L^{A,r}(\beta^A). \quad (70)$$

Now, by combining (69) and (70), we have

$$\mathbb{E} \left[\left\| \Phi(\tilde{\beta}^A - \beta^A) \right\|^2 \right] = O_p \left(n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right).$$

By Assumption 7, we have

$$\mathbb{E} \left[\left\| \Phi(\tilde{\beta}^A - \beta^A) \right\|^2 \right] = \mathbb{E} \left[\left(\tilde{\beta}^A - \beta^A \right)^\top \mathbb{E} [\Phi^\top \Phi] \left(\tilde{\beta}^A - \beta^A \right) \right] \geq c \cdot \mathbb{E} \left[\left\| \tilde{\beta}^A - \beta^A \right\|^2 \right].$$

Thus, we know that

$$\mathbb{E} \left[\left\| \tilde{\beta}^A - \beta^A \right\|^2 \right] = O_p \left(n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right). \quad (71)$$

Combining (65) and (71), we have

$$\left| \mathbb{E}_{\pi^*} \left[Q_1^{A,\pi^*} - Q_1^A \right] \right| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right).$$

Similarly, we can show for any $j \in [3]$ that

$$\mathbb{E} \left[\left\| \tilde{\beta}_1^{B,j} - \beta_1^{B,j} \right\|^2 \right] = O_p \left(n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right),$$

which implies that

$$\mathbb{E} \left[\left\| \tilde{\beta}_1^B - \beta_1^B \right\|^2 \right] = O_p \left(n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right).$$

Further, by the definition of confidence regions in (27) and (28), we know that

$$\left| \mathbb{E}_{\pi^*} \left[Q_1^{B,\pi^*} - Q_1^B \right] \right| = O_p \left(n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right),$$

which concludes the proof of the lemma. ■

Appendix F: Proof of Multi-Stage Game

F.1. Proof of Theorem 4

Proof. We introduce two supporting lemmas. The following lemma shows that with the sizes of the confidence regions in (34), the true action-value functions $Q_1^{A,\pi}$ and $Q_1^{B,\pi}$ lie in the confidence regions.

LEMMA 6. *Under the assumptions stated in Theorem 2, it holds that $\lim_{n \rightarrow \infty} \mathbb{P}(Q_1^{A,\pi} \in \text{conf}_1^{A,Q,\pi}, Q_1^{B,\pi} \in \text{conf}_1^{B,Q,\pi} \text{ for any } \pi \in \Pi) = 1$.*

Proof. See §F.2 for a detailed proof. ■

The following lemma shows that with the sizes of the confidence regions in (34), for any functions in the confidence regions, they are close to the true action-value functions $Q_1^{A,\pi}$ and $Q_1^{B,\pi}$.

LEMMA 7. *Under Assumption 9 and the assumptions stated in Theorem 2, for any policy $\pi \in \Pi$, $Q_1^A \in \text{conf}_1^{A,Q,\pi}$, and $Q_1^B \in \text{conf}_1^{B,Q,\pi}$, we have*

$$\left| \mathbb{E}_\pi \left[Q_1^{A,\pi} - Q_1^A \right] \right| = O_p \left(H^2 \cdot n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right), \quad \left| \mathbb{E}_\pi \left[Q_1^{B,\pi} - Q_1^B \right] \right| = O_p \left(H^2 \cdot n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right).$$

Proof. See §F.3 for a detailed proof. ■

Note that

$$\begin{aligned}
& J(\pi^{*,A}, \pi^{*,B}) - J(\hat{\pi}^A, \hat{\pi}^B) \\
&= \mathbb{E}_{\pi^*} \left[(Q_1^{A,\pi^*} + Q_1^{B,\pi^*})(S_1, A_1, B_{1/2}, U_1) \right] - \mathbb{E}_{\hat{\pi}} \left[(Q_1^{A,\hat{\pi}} + Q_1^{B,\hat{\pi}})(S_1, A_1, B_{1/2}, U_1) \right] \\
&\leq \mathbb{E}_{\pi^*} \left[(Q_1^{A,\pi^*} + Q_1^{B,\pi^*})(S_1, A_1, B_{1/2}, U_1) \right] - \min_{Q_1^A \in \text{conf}_1^{A,Q,\hat{\pi}}, Q_1^B \in \text{conf}_1^{B,Q,\hat{\pi}}} \mathbb{E}_{\hat{\pi}} \left[(Q_1^A + Q_1^B)(S_1, A_1, B_{1/2}, U_1) \right] \\
&\leq \mathbb{E}_{\pi^*} \left[(Q_1^{A,\pi^*} + Q_1^{B,\pi^*})(S_1, A_1, B_{1/2}, U_1) \right] - \min_{Q_1^A \in \text{conf}_1^{A,Q,\pi^*}, Q_1^B \in \text{conf}_1^{B,Q,\pi^*}} \mathbb{E}_{\pi^*} \left[(Q_1^A + Q_1^B)(S_1, A_1, B_{1/2}, U_1) \right] \\
&\leq \max_{Q_1^A \in \text{conf}_1^{A,Q,\pi^*}, Q_1^B \in \text{conf}_1^{B,Q,\pi^*}} \left| \mathbb{E}_{\pi^*} \left[(Q_1^{A,\pi^*} - Q_1^A + Q_1^{B,\pi^*} - Q_1^B)(S_1, A_1, B_{1/2}, U_1) \right] \right|,
\end{aligned}$$

where in the first and second inequalities, we use Lemma 6 and the optimality of $(\hat{\pi}^A, \hat{\pi}^B)$, respectively. Further, by Lemma 7, we have

$$J(\pi^{*,A}, \pi^{*,B}) - J(\hat{\pi}^A, \hat{\pi}^B) = O_p \left(H^2 \cdot n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right),$$

which concludes the proof of the theorem. ■

F.2. Proof of Lemma 6

Proof. It suffices to show for any h and $\diamond \in \{A, B\}$ and π that

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\theta_h^{\diamond,j} \in \bigcup_{\substack{\tilde{\theta}^\diamond \in \text{conf}_h^{\diamond,j,\pi}(\tilde{\theta}_{h+1/2}^\diamond), \\ \tilde{\theta}_j^\diamond \in \text{conf}_j^{\diamond,\pi}(\tilde{\theta}_{j+1/2}^\diamond), \\ \text{for any } j \in \mathcal{C}_h^\diamond}} \right) = 1, \quad (72)$$

where $\mathcal{C}_h^A = \{h+1/2, h+1, \dots, H-1/2\}$ and $\mathcal{C}_h^B = \{h+1/2, h+1, \dots, H\}$. In the meanwhile, we denote by $\tilde{\theta}_H^A = \tilde{\beta}^A$ and $\tilde{\theta}_{H+1/2}^B = \tilde{\beta}^B$ for notational convenience. For the simplicity of presentation, we only show that (72) holds for $\diamond = A$ as follows. We use mathematical induction as follows.

Base. We want to show that $\lim_{n \rightarrow \infty} \mathbb{P}(\beta^A \in \text{conf}^{A,r}) = 1$, i.e.,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\hat{L}^{A,r}(\beta^A, \hat{f}^A) - \hat{L}^{A,r}(\hat{\beta}^A, \hat{f}^A) \leq \eta^{A,r} \right) = 1. \quad (73)$$

To show that (73) holds, we note that

$$\begin{aligned}
& \hat{L}^{A,r}(\beta^A, \hat{f}^A) - \hat{L}^{A,r}(\hat{\beta}^A, \hat{f}^A) \\
&= \left(\hat{L}^{A,r}(\beta^A, \hat{f}^A) - L^{A,r}(\beta^A, \hat{f}^A) \right) + \left(L^{A,r}(\beta^A, \hat{f}^A) - L^{A,r}(\beta^A, f^A) \right) \\
&\quad + \left(L^{A,r}(\beta^A, f^A) - L^{A,r}(\hat{\beta}^A, \hat{f}^A) \right) + \left(L^{A,r}(\hat{\beta}^A, \hat{f}^A) - \hat{L}^{A,r}(\hat{\beta}^A, \hat{f}^A) \right) \\
&\leq \underbrace{\left(\hat{L}^{A,r}(\beta^A, \hat{f}^A) - \hat{L}^{A,r}(\hat{\beta}^A, \hat{f}^A) \right)}_{\text{term (A)}} - \underbrace{\left(L^{A,r}(\beta^A, \hat{f}^A) - L^{A,r}(\hat{\beta}^A, \hat{f}^A) \right)}_{\text{term (B)}} \\
&\quad + \underbrace{\left(L^{A,r}(\beta^A, \hat{f}^A) - L^{A,r}(\beta^A, f^A) \right)}_{\text{term (B)}},
\end{aligned} \quad (74)$$

where we use the fact that $L^{A,r}(\beta^A, f^A) \leq L^{A,r}(\hat{\beta}^A, \hat{f}^A)$ in the last inequality. Similar to the proof of Lemma 1 in §E.1, we upper bound term (A) using Bernstein's inequality as follows,

$$\text{term (A)} = O_p(n^{-\frac{2\alpha}{2\alpha+2\zeta+d}}). \quad (75)$$

In the meanwhile, note that by Corollary 3, we have $\|\hat{f}^A - f^A\| = O_p(n^{-\frac{\alpha}{2\alpha+2\zeta+d}})$. Thus, it holds that

$$\text{term (B)} = O_p(n^{-\frac{2\alpha}{2\alpha+2\zeta+d}}). \quad (76)$$

Now, combining (74), (75), and (76), we obtain that

$$\hat{L}^{A,r}(\beta^A, \hat{f}^A) - \hat{L}^{A,r}(\hat{\beta}^A, \hat{f}^A) = O_p(n^{-\frac{2\alpha}{2\alpha+2\zeta+d}}),$$

which implies that (73) holds.

Hypothesis. We assume that it holds for any $i \in \{H-1/2, H-1, \dots, h+1/2\}$ that

$$\lim_{n \rightarrow \infty} \mathbb{P}(\theta_i^A \in \text{conf}_i^{A,\pi}(\theta_{i+1/2}^A) \forall \pi \in \Pi) = 1. \quad (77)$$

Induction. With the hypothesis in (77), it only suffices to show that $\lim_{n \rightarrow \infty} \mathbb{P}(\theta_h^A \in \text{conf}_h^{A,\pi}(\theta_{h+1/2}^A) \forall \pi \in \Pi) = 1$, i.e.,

$$\hat{L}_h^{A,j,\pi}(\theta_h^{A,j}, \hat{f}_h^{A,j}; \theta_{h+1/2}^A) - \hat{L}_h^{A,j,\pi}(\hat{\theta}_h^{A,j}, \hat{f}_h^{A,j}; \theta_{h+1/2}^A) \leq \eta_h^{A,j} \quad \text{for any } j \in [3] \quad (78)$$

for any π . Similar to (74), we obtain the following upper bound for any $j \in [3]$,

$$\begin{aligned} & \hat{L}_h^{A,j,\pi}(\theta_h^{A,j}, \hat{f}_h^{A,j}; \theta_{h+1/2}^A) - \hat{L}_h^{A,j,\pi}(\hat{\theta}_h^{A,j}, \hat{f}_h^{A,j}; \theta_{h+1/2}^A) \\ & \leq \left(\hat{L}_h^{A,j,\pi}(\theta_h^{A,j}, \hat{f}_h^{A,j}; \theta_{h+1/2}^A) - \hat{L}_h^{A,j,\pi}(\hat{\theta}_h^{A,j}, \hat{f}_h^{A,j}; \theta_{h+1/2}^A) \right) \\ & \quad - \left(L_h^{A,j,\pi}(\theta_h^{A,j}, \hat{f}_h^{A,j}; \theta_{h+1/2}^A) - L_h^{A,j,\pi}(\hat{\theta}_h^{A,j}, \hat{f}_h^{A,j}; \theta_{h+1/2}^A) \right) \\ & \quad + \left(L_h^{A,j,\pi}(\theta_h^{A,j}, \hat{f}_h^{A,j}; \theta_{h+1/2}^A) - L_h^{A,j,\pi}(\theta_h^{A,j}, f_h^{A,j}; \theta_{h+1/2}^A) \right), \end{aligned}$$

which can be further upper bounded following similar arguments as in (75) and (76) as follows,

$$\hat{L}_h^{A,j,\pi}(\theta_h^{A,j}, \hat{f}_h^{A,j}; \theta_{h+1/2}^A) - \hat{L}_h^{A,j,\pi}(\hat{\theta}_h^{A,j}, \hat{f}_h^{A,j}; \theta_{h+1/2}^A) \leq \eta_h^{A,j}.$$

Recall that Assumption 8 holds. This justifies (78), which also justifies (72). Thus we finish the mathematical induction, which concludes the proof of the lemma. $\blacksquare$

F.3. Proof of Lemma 7

Proof. Note that

$$\begin{aligned} \left| \mathbb{E}_{\pi^*} \left[Q_1^{A,\pi^*} - Q_1^A \right] \right| &= \mathbb{E}_{\pi^*} \left[(\theta_1^A - \tilde{\theta}_1^A) A_1 + (\gamma_h^A - \tilde{\gamma}_h^A) B_{1/2} + (\omega_h^A - \tilde{\omega}_h^A) A_1 B_{1/2} \right] \\ &\leq \left\| \theta_1^A - \tilde{\theta}_1^A \right\|_{\pi^*} + \left\| \gamma_1^A - \tilde{\gamma}_1^A \right\|_{\pi^*} + \left\| \omega_1^A - \tilde{\omega}_1^A \right\|_{\pi^*}, \end{aligned} \quad (79)$$

where $\tilde{\theta}_1^A = (\tilde{\theta}_1^A, \tilde{\gamma}_1^A, \tilde{\omega}_1^A)^\top$ is associated with Q_1^A . We use mathematical induction to upper bound (79) as follows.

Base. We upper bound $\mathbb{E}[\|\tilde{\beta}^A - \beta^A\|^2]$ as follows. Note that

$$\widehat{L}^{A,r}(\tilde{\beta}^A, \mathbf{f}^A) = \frac{1}{n} \sum_{i=1}^n \left\| \Phi_i(\mathbf{f}^A) \tilde{\beta}_i^A + \alpha_i(\mathbf{f}^A) \right\|^2,$$

where $\Phi_i(\mathbf{f}^A) \in \mathbb{R}^{3 \times 3}$ and $\tilde{\beta}_i^A = \tilde{\beta}^A(S_1^i, U_1^{A,i})$. In the meanwhile, by Bernstein's inequality, we have

$$\begin{aligned} & \left(L^{A,r}(\tilde{\beta}^A, \mathbf{f}^A) - L^{A,r}(\beta^A, \mathbf{f}^A) \right) - \left(\widehat{L}^{A,r}(\tilde{\beta}^A, \mathbf{f}^A) - \widehat{L}^{A,r}(\beta^A, \mathbf{f}^A) \right) \\ &= O_p \left(\frac{1}{n} + \sqrt{\frac{\mathbb{E} \left[\left(\|\Phi(\mathbf{f}^A) \tilde{\beta}^A + \alpha(\mathbf{f}^A)\|^2 - \|\Phi(\mathbf{f}^A) \beta^A + \alpha(\mathbf{f}^A)\|^2 \right)^2 \right]}{n}} \right). \end{aligned} \quad (80)$$

Note that

$$\mathbb{E} \left[\left(\left\| \Phi(\mathbf{f}^A) \tilde{\beta}^A + \alpha(\mathbf{f}^A) \right\|^2 - \left\| \Phi(\mathbf{f}^A) \beta^A + \alpha(\mathbf{f}^A) \right\|^2 \right)^2 \right] \leq c \cdot \mathbb{E} \left[\left\| \Phi(\mathbf{f}^A) (\tilde{\beta}^A - \beta^A) \right\|^2 \right]. \quad (81)$$

By combining (80) and (81), we have

$$\begin{aligned} & \left(L^{A,r}(\tilde{\beta}^A, \mathbf{f}^A) - L^{A,r}(\beta^A, \mathbf{f}^A) \right) - \left(\widehat{L}^{A,r}(\tilde{\beta}^A, \mathbf{f}^A) - \widehat{L}^{A,r}(\beta^A, \mathbf{f}^A) \right) \\ &= O_p \left(n^{-1} + \sqrt{n^{-1} \cdot \mathbb{E}[\|\Phi(\mathbf{f}^A)(\tilde{\beta}^A - \beta^A)\|^2]} \right). \end{aligned} \quad (82)$$

In the meanwhile, we have

$$\left(\widehat{L}^{A,r}(\tilde{\beta}^A, \mathbf{f}^A) - \widehat{L}^{A,r}(\beta^A, \mathbf{f}^A) \right) - \left(\widehat{L}^{A,r}(\tilde{\beta}^A, \widehat{\mathbf{f}}^A) - \widehat{L}^{A,r}(\beta^A, \widehat{\mathbf{f}}^A) \right) = O_p \left(n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right). \quad (83)$$

Combining (82) and (83), it holds for any $\tilde{\beta}^A \in \text{conf}^{A,r}$ that

$$L^{A,r}(\tilde{\beta}^A, \mathbf{f}^A) - L^{A,r}(\beta^A, \mathbf{f}^A) = O_p \left(n^{-1} + \sqrt{n^{-1} \cdot \mathbb{E}[\|\Phi(\mathbf{f}^A)(\tilde{\beta}^A - \beta^A)\|^2]} + n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right), \quad (84)$$

In the meanwhile, since $L^{A,r}(\cdot)$ is a quadratic functional, it holds that

$$\mathbb{E} \left[\left\| \Phi(\mathbf{f}^A) (\tilde{\beta}^A - \beta^A) \right\|^2 \right] \leq L^{A,r}(\tilde{\beta}^A, \mathbf{f}^A) - L^{A,r}(\beta^A, \mathbf{f}^A). \quad (85)$$

Now, by combining (84) and (85), we have

$$\mathbb{E} \left[\left\| \Phi(\mathbf{f}^A) (\tilde{\beta}^A - \beta^A) \right\|^2 \right] = O_p \left(n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right).$$

By Assumption 9, we know that

$$\mathbb{E} \left[\|\tilde{\beta}^A - \beta^A\|^2 \right] = O_p \left(n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right).$$

Hypothesis. We assume that it holds for any $i \in \{H-1/2, H-1, \dots, h+1/2\}$ that

$$\sup_{\tilde{\theta}_i^A} \mathbb{E} \left[\|\tilde{\theta}_i^A - \theta_i^A\|^2 \right] = O_p \left((H-i)^4 \cdot n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right), \quad (86)$$

where the sup is taken over $\tilde{\theta}_i^A \in \bigcup_{\tilde{\beta}^A \in \text{conf}^{A,r}, \tilde{\theta}_k^A \in \text{conf}_k^{A,\pi^*}(\tilde{\theta}_{k+1/2}^A)} \text{conf}_i^{A,k,\pi^*}(\tilde{\theta}_{i+1/2}^A)$.

Induction. We want to upper bound $\mathbb{E}[\|\tilde{\theta}_h^{A,j} - \theta_h^{A,j}\|^2]$ for any $j \in [3]$ as follows. By Bernstein's inequality, we have

$$\begin{aligned} & \left(L_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \mathbf{f}_h^{A,j}; \theta_{h+1/2}^A) - L_h^{A,j,\pi^*}(\theta_h^{A,j}, \mathbf{f}_h^{A,j}; \theta_{h+1/2}^A) \right) \\ & - \left(\widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \mathbf{f}_h^{A,j}; \theta_{h+1/2}^A) - \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \mathbf{f}_h^{A,j}; \theta_{h+1/2}^A) \right) \\ & = O_p \left(\frac{H(H-h)^2}{n} + \sqrt{\frac{H \mathbb{E} \left[\left(\|\Phi_h(\mathbf{f}^A) \tilde{\theta}_h^{A,j} + \alpha_h(\mathbf{f}^A)\|^2 - \|\Phi_h(\mathbf{f}^A) \theta_h^{A,j} + \alpha_h(\mathbf{f}^A)\|^2 \right)^2 \right]}{n}} \right). \end{aligned} \quad (87)$$

We upper bound the term $\widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \mathbf{f}_h^{A,j}; \theta_{h+1/2}^A) - \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \mathbf{f}_h^{A,j}; \theta_{h+1/2}^A)$ as follows. We assume that $\tilde{\theta}_h^{A,j} \in \text{conf}_h^{A,j,\pi^*}(\theta_{h+1/2}^{A,\dagger})$ for some $\theta_{h+1/2}^{A,\dagger} \in \bigcup_{\tilde{\theta}^A \in \text{conf}_h^{A,r}, \tilde{\theta}_k^A \in \text{conf}_k^{A,\pi^*}(\tilde{\theta}_{k+1/2}^A)} \text{conf}_{h+1/2}^{A,k,\pi^*}(\tilde{\theta}_{h+1}^A)$ and $\theta_h^{A,j} \in \text{conf}_h^{A,j,\pi^*}(\theta_{h+1/2}^{A,\ddagger})$ for some $\theta_{h+1/2}^{A,\ddagger} \in \bigcup_{\tilde{\theta}^A \in \text{conf}_h^{A,r}, \tilde{\theta}_k^A \in \text{conf}_k^{A,\pi^*}(\tilde{\theta}_{k+1/2}^A)} \text{conf}_{h+1/2}^{A,k,\pi^*}(\tilde{\theta}_{h+1}^A)$.

Note that by Corollary 3, we have

$$\begin{aligned} & \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \mathbf{f}_h^{A,j}; \theta_{h+1/2}^A) - \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \mathbf{f}_h^{A,j}; \theta_{h+1/2}^A) \\ & = \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^A) - \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^A) + O_p \left((H-h)^2 n^{-\frac{2\alpha}{2\alpha+2\zeta+d}} \right). \end{aligned} \quad (88)$$

In the meanwhile, we have

$$\begin{aligned} & \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^A) - \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^A) \\ & \leq \left| \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^A) - \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\dagger}) \right| \\ & \quad + \left| \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\dagger}) - \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\ddagger}) \right| \\ & \quad + \left| \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\ddagger}) - \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\ddagger}) \right| \\ & \quad + \left| \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\ddagger}) - \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^A) \right|. \end{aligned} \quad (89)$$

The first term on the RHS of (89) can be upper bounded by (86) as follows,

$$\left| \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^A) - \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\dagger}) \right| = O_p \left(H(H-h-1/2)^3 \cdot n^{-\frac{2\alpha}{2\alpha+2\zeta+d}} \right). \quad (90)$$

The second term on the RHS of (89) can be upper bounded by the definition of the confidence regions as follows,

$$\left| \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\dagger}) - \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\ddagger}) \right| \leq \eta_h^{A,j}. \quad (91)$$

The third term on the RHS of (89) can be upper bounded by (86) as follows,

$$\left| \widehat{L}_h^{A,j,\pi^*}(\tilde{\theta}_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\ddagger}) - \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\ddagger}) \right| = O_p \left(H(H-h-1/2)^3 \cdot n^{-\frac{2\alpha}{2\alpha+2\zeta+d}} \right). \quad (92)$$

The forth term on the RHS of (89) can be upper bounded by definition of the confidence regions and Lemma 6 as follows,

$$\left| \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^{A,\ddagger}) - \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^A) \right| \leq \eta_h^{A,j}. \quad (93)$$

The last term on the RHS of (89) can be upper bounded by (86) as follows,

$$\left| \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^A) - \widehat{L}_h^{A,j,\pi^*}(\theta_h^{A,j}, \widehat{\mathbf{f}}_h^{A,j}; \theta_{h+1/2}^A) \right| = O_p \left(H(H-h-1/2)^3 \cdot n^{-\frac{2\alpha}{2\alpha+2\zeta+d}} \right). \quad (94)$$

Now, by plugging (89), (90), (91), (92), (93), and (94) into (88), we have

$$\widehat{L}_h^{A,j,\pi^*}(\widetilde{\theta}_h^{A,j}, \mathbf{f}_h^{A,j}; \boldsymbol{\theta}_{h+1/2}^A) - \widehat{L}_h^{A,j,\pi^*}(\boldsymbol{\theta}_h^{A,j}, \mathbf{f}_h^{A,j}; \boldsymbol{\theta}_{h+1/2}^A) = O_p\left(H(H-h)^3 \cdot n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}}\right). \quad (95)$$

Note that

$$\begin{aligned} & \mathbb{E} \left[\left(\|\Phi_h(\mathbf{f}^A) \widetilde{\theta}_h^{A,j} + \alpha_h(\mathbf{f}^A)\|^2 - \|\Phi_h(\mathbf{f}^A) \boldsymbol{\theta}_h^{A,j} + \alpha_h(\mathbf{f}^A)\|^2 \right)^2 \right] \\ & \leq c(H-h)^2 \cdot \mathbb{E} \left[\left\| \Phi_h(\mathbf{f}^A) (\widetilde{\theta}_h^{A,j} - \boldsymbol{\theta}_h^{A,j}) \right\|^2 \right]. \end{aligned} \quad (96)$$

Combining (87), (95), and (96), we have

$$\begin{aligned} & \left(L_h^{A,j,\pi^*}(\widetilde{\theta}_h^{A,j}, \mathbf{f}_h^{A,j}; \boldsymbol{\theta}_{h+1/2}^A) - L_h^{A,j,\pi^*}(\boldsymbol{\theta}_h^{A,j}, \mathbf{f}_h^{A,j}; \boldsymbol{\theta}_{h+1/2}^A) \right) \\ & = O_p \left(\frac{H(H-h)^2}{n} + (H-h) \sqrt{\frac{H}{n} \cdot \mathbb{E} \left[\left\| \Phi_h(\mathbf{f}^A) (\widetilde{\theta}_h^{A,j} - \boldsymbol{\theta}_h^{A,j}) \right\|^2 \right]} + H(H-h)^3 \cdot n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right), \end{aligned}$$

which implies that

$$\mathbb{E}^{\pi^*} \left[\left\| \Phi_h(\mathbf{f}^A) (\widetilde{\theta}_h^{A,j} - \boldsymbol{\theta}_h^{A,j}) \right\|^2 \right] = O_p \left(H(H-h)^3 \cdot n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right)$$

following a similar argument as in the base case. Since the matrix $\Phi_h(\mathbf{f}^A; \boldsymbol{\theta}_{h+1/2}^A)$ is lower bounded, we know that

$$\mathbb{E} \left[\left\| \widetilde{\theta}_h^{A,j} - \boldsymbol{\theta}_h^{A,j} \right\|^2 \right] = O_p \left(H(H-h)^3 \cdot n^{-\frac{2\alpha}{2\alpha+2\varsigma+d}} \right),$$

which implies that

$$\sqrt{\mathbb{E} \left[\left\| \widetilde{\theta}_h^A - \boldsymbol{\theta}_h^A \right\|^2 \right]} = O_p \left(H^{1/2} (H-h)^{3/2} \cdot n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right).$$

By a union covering argument over $\widetilde{\theta}_h^A \in \bigcup_{\widetilde{\beta}^A \in \text{conf}^{A,\tau}, \widetilde{\theta}_k^A \in \text{conf}_k^{A,\pi^*}(\widetilde{\theta}_{k+1/2}^A) \text{ for any } k \in \mathcal{C}_h^A} \text{conf}_h^{A,k,\pi^*}(\widetilde{\theta}_{h+1/2}^A)$, we have

$$\sup_{\widetilde{\theta}_h^A} \sqrt{\mathbb{E} \left[\left\| \widetilde{\theta}_h^A - \boldsymbol{\theta}_h^A \right\|^2 \right]} = O_p \left(H^{1/2} (H-h)^2 \cdot n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right),$$

which concludes the proof of the induction part. Thus, we have

$$\left| \mathbb{E}_{\pi^*} \left[Q_1^{A,\pi^*} - Q_1^A \right] \right| = O_p \left(H^{5/2} \cdot n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right), \quad \left| \mathbb{E}_{\pi^*} \left[Q_1^{B,\pi^*} - Q_1^B \right] \right| = O_p \left(H^{5/2} \cdot n^{-\frac{\alpha}{2\alpha+2\varsigma+d}} \right),$$

where $H^{5/2}$ is obtained by combining $H^{1/2}$ and $(H-1)^2$. This concludes the proof of the lemma. $\blacksquare$

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